IIMB
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PM Case Book
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SIGMA
PM & TECH CLUB



About this case book

Foreword

We are proud to present the new edition of the PM Case Book by Sigma, the Product Management & Technology Club of IIM Bangalore. Leveraging interview experiences from alumni, the PM Case Book aims to give the readers a comprehensive view of the type of cases that form a major part of Product Management interviews. This case book covers Product Design, Root Cause Analysis, Pricing, App Improvement, Guesstimates, and Miscellaneous cases. Throughout the PM Case Book, we have tried to ensure enhanced readability while retaining the comprehensiveness of the cases.

We have a heartfelt gratitude for stalwarts such as Lewis C Lin, Gayle Laakmann McDowell and other authors in the domain of Product Management. Most of the frameworks used in this casebook have been derived from their extensive work.

How to make the most of it?

While reading this Case Book, we would suggest that the reader use the interview transcripts to set up a case between 2 people (or groups). The group should engage in active discussion after solving the case to provide feedback to each other and suggest areas of improvement. The transcripts given in the casebook should be seen as a template to gain a broader understanding of the approach and not be taken as the solution per se.

Acknowledgements

We are grateful to all the people that have contributed by sharing their cases and interview experiences that has enabled us to put together a comprehensive preparation resource for the future batches. We would like to thank Aman Jaswani (Casebook PoC), some major contributors (Swastika Nayak, Souparno Bhattacharyya, Ashika Yadav, Mehar Kaur, Ayushi Saini, Anshuman Deka), and the Sigma Cohort 2023-25 (Ojas Khewale, Ashish Mishra, Deepanshu Vats, Shashi Tomar, Bhanujya Saikia, Vikas Malavath, Sriraam E, Aviral Verma) for taking the initiative and putting together this edition of the Sigma PM Case Book.

We hope to get inputs and feedback from the students of IIM Bangalore on existing and new cases to further enhance the quality of the casebook in terms of breadth and depth.

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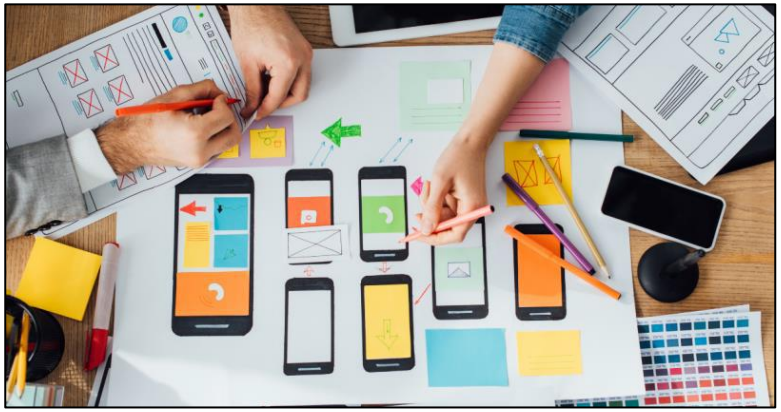
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Product Design

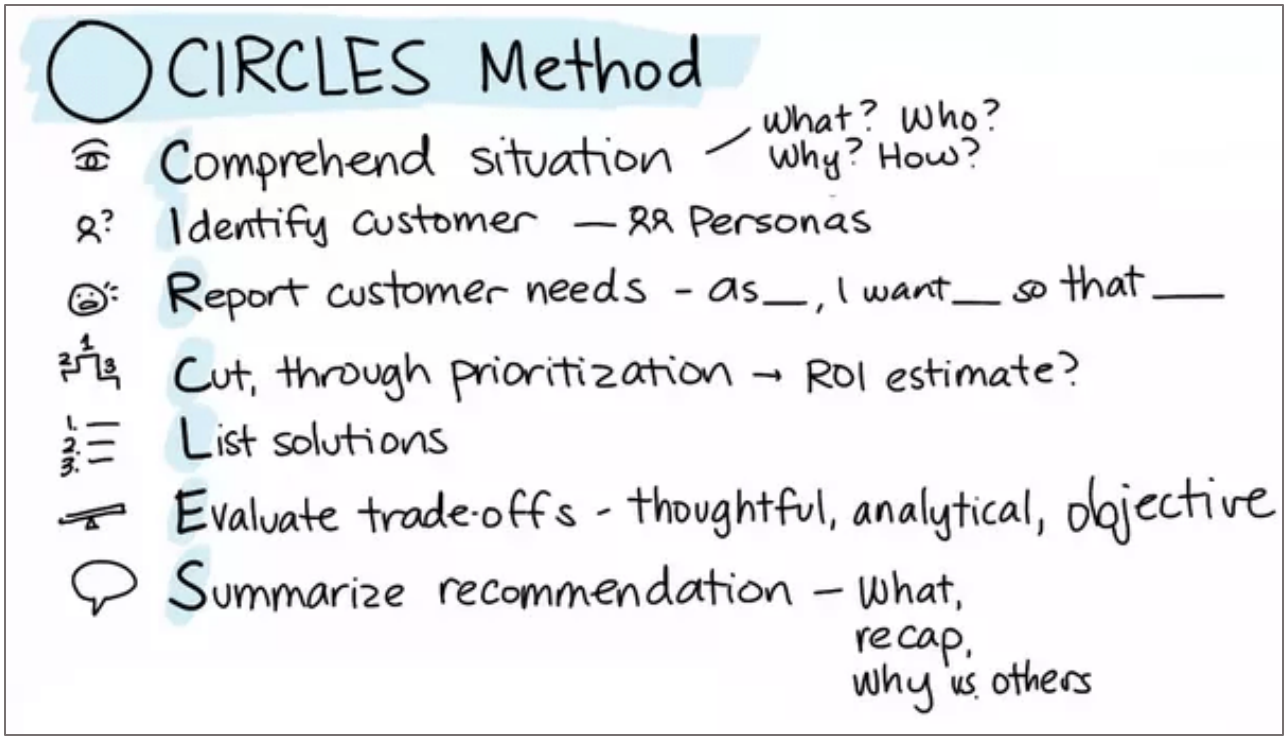


Product Design – Approach (1/2)

How would you build a product for ABC use case?

How would you improve ABC product?

- Product design is generally asked in 2 ways – building a new product or improving an existing one. The approach for either type of question is the same – the **CIRCLES** method.
- Wherever possible it is better to mention user journey before problem identification (e.g. Theme Park Application)



Source: Decode & Conquer

Product Design – Approach (2/2)

How would you build a product for ABC requirements?

How would you improve ABC product?

- **Context**: Start off by clarifying the context from the interviewer. What is the product? What is the objective? Which platform (website/mobile app) should we be targeting?
- **Target Segment**: Identify at least 3 user personas which could use your product. Tell the interviewer something about each persona – their demographics, psychographics and any other basic characteristics you would like to let the interviewer know. Select a persona you would like to build your product for. Give your reasons for selecting this persona to the interviewer. You could be asked to size the market at this stage. Explore more here.
- **User Needs**: You need to identify the needs and pain points for your selected persona. A great way of doing this is to build the user journey of how this persona currently carries out his or her requirements for ABC given in the question. This can also include points of the journey which currently happen outside the product. 5Es framework is a great way to build a user journey. Remember that all stages might not be valid for every question. Don't force fit!

Entice -> Enter -> Engage -> Exit -> Extend

- **Prioritize**: You would likely find a lot of needs and pain points across the user journey. Prioritize a few needs and pain points you would like to solve right now. Give your reasoning to the interviewer. The reasons could be reached, revenue, impact, cost, effort, and complexity of potentially solving this problem. You can get started here.
- **Solution**: Once you have finalized the needs and pain points you want to solve, come up with features. Try to be detailed with your solution. Clearly articulate each use case that would be solved. You can also build rough UI mocks to explain how your solution would look. Get started with defining features here.

(**Note**: Another valid approach for the prior two steps is using the pain points to develop features and then prioritizing the features themselves. In this case, you can define your features roughly, and once you have decided which ones to prioritize, you can then flesh them out)

- **KPIs**: Come up with metrics to measure your solution. There should be one north star metric to measure the overall objective of the product and a few metrics to measure the success of your features. You can also evaluate trade-offs by thinking of metrics that might go down due to this and whether the new feature is still worth it. Know more about KPIs here.

Refrigerator for Kids

You are a PM at Samsung, design a refrigerator for kids.

I have some initial clarifying questions. First, which age segment of kids are we targeting?

You are free to choose

Is there any competitor product in the market?

There is no competing product. You have to design a new product.

What is the end goal for the company? Is it user adoption or generating revenue?

You are free to decide

As this is a new product, I feel user adoption will be more necessary rather than generating revenue. I would like to first identify different customer segments and their pain points before I select a particular segment and recommend features. Is that alright?

Sure, please proceed.

I can identify 4 potential customer segments –

1. **Infants and young kids (0-5 year olds):** These kids majorly survive on breast milk or some formula or are in the early stages of eating solid foods. They do not have much need of storing or retrieving items from the fridge.
2. **Young school kids (5-12 year olds):** These kids do access the fridge quite regularly and are mostly looking for snacks, chocolates, juices, or cold drinks. Many kids in this age group run away from nutritional foods that are essential for their growth.
3. **Teenagers (13-18 year olds):** The kids in this segment are quite selective about what they eat. They want to try out new items and recipes and are also adequately tech savvy.
4. **Parents:** These are also stakeholders as they will be the ones purchasing the fridge for their kids. Their major concern is ensuring that their kids eat healthy and limit the intake of unhealthy junk foods.

I would like to proceed with the Young kids and Teenagers segment as I feel these will be the regular users of this new fridge. If that sounds reasonable I can proceed with listing pain points?

Yes, that is a fair segmentation. You can proceed.

1. Rigid positioning of different segments - Fridges have fixed positioning of slabs and kids of different heights may have difficulty reaching to all segments.
2. Lack of tasty yet nutritional snacking options - Often kids just reach out for the readymade junk option when they are hungry instead of scouting for items and making something healthy to eat. Alternatively, even if a bowl of salad is readymade in the fridge, the kids still might choose a junk option over it.
3. Tracking - Teenagers especially would be more interested in tracking the calories and nutritional value of what they are eating.

Coming to the features for these pain points –

1. **Compartmentalized design**: Different slabs to be adjusted at various heights for easier access. This includes being able to change the placement of beverage tray, food item slabs, and even the door handle.
2. **Gamification**: Points system for when kids select a healthy snacking option. A punishment (reward deduction) if kids cross a parent defined threshold for unhealthy snacking items. Parents can check these points and incentivize the kid that they'll allow him/her more play time if he/she has enough points.
3. **Healthy Recommendations**: The fridge can track what all ingredients are present within it and recommend some healthy options that can be made using those ingredients.
4. **Health Tracker**: On the fridge door itself we can have a screen that tracks the daily nutritional content consumed (including calories, protein, fats etc.). Different users can be tracked by perhaps a fingerprint sensor on the door handle.

How will you prioritize among these?

For prioritizing features, I will look into the value added by the feature and the complexity of implementation for the company.

Feature	Value	Complexity
Compartmentalization	Medium	Low
Gamification	High	Medium
Recommendations	High	Medium
Health Tracking	Medium	Medium

Refrigerator for Kids

On the basis of this, I would prioritize the Gamification and Recommendations feature.

Can you define what would the success metrics be?

One metric can be the sales number, i.e. the number of refrigerators sold (also considering returns in the same).
Another can be the number of kids profiles created.
We can also track certain activities like the number of times a kid chooses a healthy option over an unhealthy one.

That will be all for now, thank you.

Remote Worker Productivity

Design a product to enhance the work life balance & productivity of remote workers using AI (Microsoft)

Before I begin with the solution, I have some clarifying questions

Sure, go ahead

What do we mean by productivity here? Also, is Microsoft itself designing this product or is it a generic one by any other company? Do we have to create a brand new product or make some changes to any additional product? Also, is this is B2B or B2C product?

Great questions! To start with, you can define what productivity means to you here, there are no restrictions. Yes, Microsoft is the one launching this product. You are again free to devise a new product or make changes to an existing one. The choice of B2B or B2C is also up to you.

Alright, thanks! To start with, I'd define productivity as improving the time taken to complete a particular task and the overall satisfaction the worker has with his work. I'd start with a product from scratch and would create a B2C product to begin with.

Sounds good, please go ahead.

First of all, I'd talk about Microsoft's mission. Its mission is to develop tools to enhance workers' productivity. Developing such a tool aligns with Microsoft's overall strategic goal. Next, our objective is to enhance the worker productivity by leveraging AI, if this seems okay, I'd go ahead by defining some user personas.

Sure

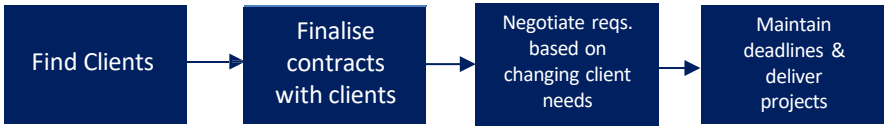
1. Raj is a consultant that struggles with long working hours as he deals with clients in the US and foreign countries, and he resides in India. He gets physically exhausted by the end of the day.
2. Rashmi is a software engineer, and she deals with complex coding projects that require a lot of her mental strength. She gets mentally exhausted and doesn't have time for any other chores.
3. Rohan is a freelancer, and he deals with multiple projects, does multiple context switching and needs to maintain track of multiple clients.

Great! Which one among these would you prioritize?

Since a freelancer usually works from home and deals with multiple clients and deadlines at the same time, enhancing their productivity would help them manage their time well and maximize their potential. Since, they also work alone, they tend to get bored too. Hence, I would prioritize the freelancer persona. If this is okay, I'll go ahead and create the user journey now.

Yes, go ahead

User Journey



What would be some pain points based on this user journey?

1. Difficulty in finding the right type of clients that fit the needs of both the parties i.e. the freelancer and the client
2. Difficulty in handling and managing deadlines, there could be a cyclic dependency among deadlines.
3. Difficulty in maintaining focus while long working hours.
4. Feeling lethargic at the end of the day, sitting at one place.

I would now create features catering to every pain point, post which I would prioritize 1-2 features

Sure, please go ahead

1. Client Meeting Sentiment Analysis → Analyse how our pitch is being received by the client to improve our future success rate.
2. GenAI to curate personal content pitch development based on different client backgrounds and needs.

Remote Worker Productivity

3. Deadline Analyser → Analyse our deadlines and accordingly schedule tasks at the beginning of every day.
4. Music Player → Analyzing the mood of the worker to recommend and play music during work.
5. Activity Analyser → Analyse what activities would be ideal for the employee to do at the end of the day based on how the day was. Eg: Running if it was a relatively free day, or watch a movie with in case of a hectic day

Which among these would you prioritise?

Feature	Reach	Impact	Confidence	Effort
Client Sentiment Analysis	L	L	S	L
GenAI	L	L	L	M
Music Player	M	M	M	L
Deadline Analyser	L	L	L	M
Activity Analyser	L	M	L	M

I would prioritise the following features based on the RICE score:

1. GenAI to curate personal content pitch development based on different client backgrounds and needs.
2. Activity Analyser → Analyse what activities would be ideal for the employee to do at the end of the day based on how the day was.
3. Deadline Analyser → Analyse our deadlines and accordingly schedule tasks at the beginning of every day.

Awesome! Can you quickly give some success metrics for this?

Sure.
1. # of projects successfully delivered
2. Avg. time to deliver
3. Client conversion rate
4. NPS

Great! I had a question, if you had to decide the platform to release this on? Mobile or Web, what would you choose and why?

I would go ahead with the Web version first since most of the work is done via laptop / PCs, hence developing an app for the web initially would make more sense. Once we find a product market fit for the product, we can then maybe extend it to the mobile version as well.

Okay! And what metric would you observe to check if your product has gained a product market fit?

We can see the NPS for that

Great, we can close the case now

Thank you

Classroom in the Metaverse

Design a classroom in the Metaverse (Google PM)

Before we start, I want to ask a few clarifying questions. Do we have a goal in mind while building this?

It should tackle the issue of missing human connection faced by users of video-based solutions.

I'll now go ahead with listing the user personas that I can think of.

1. Teachers: Age 25-45 years, familiar with technology but has concerns about maintaining student engagement.

Pain points:

• Find it challenging to maintain student engagement during virtual lectures, as the personal connection and interactive aspects are reduced

• Providing one-on-one assistance to students during virtual office hours can be challenging, leading to reduced interaction with students who need extra help

• Can be difficult to check instances of proxy attendance

2. Students: Age 13-21 years, tech-savvy and enjoy exploring new technologies and virtual experiences, often attend online lectures and collaborate with peers on group projects.

Pain points:

• Unable to access school or college infrastructure from home like labs etc

• Online classes sometimes lack the engagement and interactivity of traditional classroom settings, making it harder for her to stay focused

• Coordinating with team members for group projects is challenging, as current tools don't fully capture the essence of face-to-face collaboration

• Concerns about privacy

Would you like me to consider any other persona or prioritize a specific one?

Take both of them.

Sure! Let me describe the features which can solve these pain points.

Teachers:

- Interactive tools with gamified elements such as quizzes and polls
- Virtual whiteboards and breakout rooms for group activities and discussions
- Dashboards to monitor student engagement level
- Authenticate entry and exit

Students:

- Collaborative note-making
- Personalized virtual avatar and customizable desk
- Virtual lab and playground
- Avenues where students can interact informally with other students such as chatrooms, games & music rooms
- Privacy should be airtight, no one else should be able to log in to the classroom without proper credentials, and students' data should be secure

How can you ensure that some other large company won't copy your feature?

Large companies have a lot of resources at their disposal & can easily copy the feature. We could try creating hardware dependencies and patent the IP.

In the case of HW, It cannot be quickly modified, how will you decide what should go into the HW?

Most features should be allowed to be upgraded via software. Even at the hardware level, we can try pushing firmware updates that will improve the device.

What would be the North Star metric for this feature?

I think no. of hours spent learning could be a good North Star metric. It also takes into account the number of users indirectly.

Cooking App for Urban Users

Design an App to Address the Cooking Requirements of People Living in Urban Areas

That sounds interesting. Can I ask a few clarifying questions before moving forward?

Sure. Please go ahead

What specific cooking needs are we addressing for urban residents? Are we focusing on a digital application, a physical product, or a combination of both?

You can make your assumptions.

We can assume we are designing a digital application focused on simplifying and enhancing the cooking experience for urban residents. The specific needs would be to provide convenient, time-saving cooking solutions, promote healthy eating habits, and consider sustainability and reducing food waste. Does that sound good to you?

Yes, that sounds good. Let's proceed.

Here are four user personas I came up with:

1. **Busy Professionals** – Individuals with demanding jobs who need quick and easy meal solutions.
2. **Health-Conscious Individuals** – People who prioritize healthy eating but may lack time to cook elaborate meals.
3. **Families with Children** – Households that require meal planning and recipes suitable for both adults and children.
4. **Beginner Cooks** – Individuals new to cooking who need simple, step-by-step guidance.

That is comprehensive. How do you want to proceed?

Since we've identified the primary user personas, let's move on to selecting one and understanding their specific pain points. Here, I'll focus on the **Busy Professionals** persona. This group represents a significant portion of urban residents who often struggle to balance their work and personal lives, making them an ideal target for our application.

That makes sense. Focusing on busy professionals seems like a strategic choice. What are the pain points for this persona?

Here are the pain points for busy professionals:
1.Limited Time to Prepare Meals: Busy schedules leave little time for cooking.
2.Difficulty in Planning Meals in Advance: Lack of time to think about and plan meals.
3.Need for Quick and Healthy Recipes: Struggle to find recipes that are both quick to prepare and healthy.
Now I will prioritize them.

Ok. Go ahead

I'll prioritize the pain points based on their frequency and impact on the user's daily life:
1.Limited Time to Prepare Meals: This is the most acute problem as it directly affects their ability to cook regularly.
2.Need for Quick and Healthy Recipes: Providing quick and healthy options can significantly improve their cooking habits and health.
3.Difficulty in Planning Meals in Advance: This is important but can be addressed through planning tools once the first two pain points are tackled.

The prioritization seems logical. What solutions do you propose to address these pain points?

Based on the prioritized pain points, here are some potential solutions:
1.Quick Meal Recipes: A library of recipes that can be prepared in under 30 minutes.
2.Personalized Meal Plans: Customizable weekly meal plans based on user preferences and dietary needs.
3.Grocery Integration: Integration with grocery delivery services to streamline shopping for meal ingredients..

These solutions are well thought out. Why don't you prioritize those?

Cooking App for Urban Users

I'll prioritize the solutions based on their impact and ease of implementation:
1.Quick Meal Recipes: High impact and relatively easy to implement.
2.Personalized Meal Plans: High impact but requires more effort to develop.
3.Grocery Integration: Medium impact and medium effort required.

Okay. How would you measure the success of this application?

To gauge the success of the application, I will use the following metrics:
1.Adoption Metrics:
1. Number of new signups per month.

2.Engagement Metrics:
1. Average session duration.
2. Number of recipes viewed and saved.
3. Frequency of meal plan usage.

3.Retention Metrics:
1. Churn rate (percentage of users who stop using the app).

4.User Satisfaction Metrics:
1. Net Promoter Score (NPS).
2. User ratings and reviews.

Makes sense. Thank you for your time. We can close this case.

Tablet For Kids

Design a tablet for kids. (Google)

Thank you for the question. Can I ask you some questions to set the context?

Yeah sure.

What is the age range of kids that we are looking to cater to?

Let's say it's for 4 to 6 years

Ok, great, should I further segment this group into different age ranges to better understand their needs, preferences, and developmental stages. For instance, younger kids might need simpler interfaces, while older ones could handle more complex features.

For this example, consider this 4-6 age group having similar capabilities. Can you identify some more stakeholders?

There are other stakeholders involved like the parents and teachers considering it might be used for learning content. Essentially, these stakeholders have a say in the purchase of the product. So, their needs are to be identified too.

Which among the stakeholder you mentioned you would want to cater more to?

Majorly it's the parents and the kids. We can keep educators out for now as we plan to develop a more general-purpose yet customized experience for kids.

Ok what would be the product requirements you think?

Before jumping into it, I would like to categorize the pain points from kids' POV and parents' POV.

Kids:

- Wants a good display and sound to watch content and to interact with it.
- Might get overwhelmed with a lot of apps that they do not require.
- Is not able to write/ type properly with all the correct spellings as might be at initial stage of learning and development.
- Tired eyes

Parents:

- Privacy and safety concerns
- Screen time
- Control over the content

List out a few solutions

Based on the user journey, we need to think of hardware as well as software solutions. These might include a durable and child friendly design, like tactile buttons, simplified UI with minimalistic bloatware. The screen should be of high quality and have a good speaker as content for kids is mostly visual based. We can also bundle in YouTube Kids and make Google Assistant one of our major modes of interaction with the app.
We also require a safe browsing environment, some educational content, parental controls for managing usage, and robust security measures to protect kids' privacy. The security features might have certain restrictions on opening and usage of certain apps, while denying the installation of age-restricted apps altogether. We can also prioritize our screen timing options as well as blue-eye filter for kids' eyes.

Tell me one priority feature from both POVs.

For kids that would be simplified UI with durable hardware. And for parents' concerns, it would be parental control.

Any potential drawbacks or constraints?

Considering potential constraints, I'd think about factors such as budget, technical feasibility, and legal regulations around data privacy for children. These constraints will play a role in shaping the final product.

How would you test these features out?

With the requirements and constraints in mind, I'd start creating prototypes. These could be simple sketches or interactive wireframes that illustrate the user interface, navigation, and key features. This step helps visualize the concept and gather early feedback.

Application for Pet-Owners

Design a product for the pet industry

That sounds interesting. Can I ask a few clarifying questions before moving forward?

Sure. Please go ahead

Are we focusing on a specific pet segment? Could the product be a physical device? Which geographical area are we looking to target? What is the primary aim of this product?

We can assume that the application will cater to dogs and cats. The product should be a digital application. What are your thoughts on the target geography? Also, what do you see as the primary mission of this product?

I believe we should target the urban population as they typically have higher disposable income and a strong affinity for pet-related products and services. The primary objective of the application should be to generate revenue for the company.

Correct. Yes, we can assume the target segment to be Urban areas, let us move ahead.

Thanks for confirming the same. Next, I want to create the relevant user personas after which we can figure out their pain points by going through their user journey.

Sure. Let us get into creating the user personas.

I have identified four user personas to focus on:
1.Pet Owners – Individuals who own pets and are responsible for their food, hygiene, shelter, etc.
2.Veterinarians – Veterinary doctors who provide medical care to pets.
3.Pet Care Centers – Facilities that take care of pets when the owners are away for a short period.
4.Trainers – Professionals who take dogs out for walks or train them with commands and hand signals.

I think that should be enough. How do you want to proceed?

Since we have identified all the stakeholders, I'd like to proceed by understanding their pain points, with a primary focus on the pet owners.
Pet owners represent the largest user base and directly influence the success metrics such as user adoption, engagement, and retention. By addressing their primary pain points, we can significantly enhance the value proposition of the application, driving higher user satisfaction and loyalty.

Sure. Go ahead.

- Ensuring timely vaccinations is crucial for pet health and safety.
- Different breeds and ages of pets require specific nutrition and supplements, which many pet owners are not aware of.
- Pets are often fed human food due to the unavailability of pet-specific products.
- Emergency situations require immediate access to vets or care centers.
- Hiring and managing trainers for dogs is not streamlined.

This looks comprehensive. Let us move on to the solutioning part.

Sure. These are a few solutions that I have come up with -

1. Implement regular reminders for vaccinations and provide information on nearby locations where pet owners can get their pets vaccinated.
2. Offer nutrition recommendations tailored to the pet's breed, age, and weight through the application.
3. Sell food, nutrition, hygiene products, and accessories on the platform to complete the user journey.
4. Develop an AI tool to assist pet owners in diagnosing issues during emergency situations.
5. Facilitate hiring and negotiating with trainers.
6. Provide contact information for the nearest vet and care center facilities.
7. Organize community activities such as pet shows and dog walks in societies to promote social interaction.

These solutions are well thought out and cover various aspects of pet care. What are some metrics that you can use to gauge the success of the app?

Application for Pet-Owners

In my opinion, the North Star metric for the application should be the number of pets onboarded and the activation rate of their owners (measured by completion of key activities such as signup, purchase, hiring a trainer, etc.).

- **Adoption Metrics:** Number of new signups per month, Monthly growth rate of new users.
- **Engagement Metrics**: Average session duration per user, Daily and monthly active users (DAU/MAU), Number of interactions with key features (e.g., vaccination reminders, nutrition recommendations)
- **Retention Metrics:** Monthly retention rate, Churn rate (percentage of users who stop using the app), Frequency of repeat purchases

Thanks. We can close the case now!

Theme Park Application

How would you design an application to delight customers of a theme park?

Since we are talking about a theme park with no other details mentioned, I would first like to state a few assumptions. The theme park has an entry gate where they need to purchase tickets and rides with individual gates. The mobile app being designed would be used for entry as well as all the rides. The main aim of the park authorities with the introduction of this app is to maximize user satisfaction and ensure smooth operations.

These assumptions sound fair. Please proceed.

I would now like to identify the different customers that visit such parks.

- **Couples/Family:** 2-5 people.
- **Group Bookings:** 5+ people, non-corporate customers.
- **School/College Trip or Office Offsite:** 10+ people, corporate customers.
- **Events:** 50+ people (For example: Birthday parties)

Out of these customer segments, the first 2 would constitute the majority of the bookings. Hence, I will focus on them and try to figure out their pain points. Are the customer segments fine?

You have covered all the segments. Go ahead.

For both the segments, some issues are common. These include:

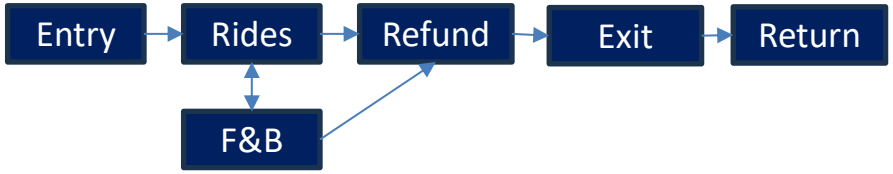
- Large queue times in individual rides and F&B counters.
- ~30 mins of waiting time at the entry gate to procure tickets & claim refunds.
- In case of lost items, the customer has to travel to the lost & found department at the opposite end of the park.

For group bookings, some additional pain points could be noticed including:

- Lack of SUVs/large vehicles for 5+ people while returning home.
- Individual payment for every person at ride counters, adding up to the queue time for all the waiting customers.

These seem to be true for most parks. How would you go about solving these?

I would first like to chalk out the user journey flow of a visitor in the theme park, and then go about solving the problems one by one.



Based on this user journey, we propose the following features:

- **Digital Wallet**: A user can top-up the digital wallet and purchase all services including tickets, add-on rides and F&B services inside the park. The remaining balance can be transferred back to bank account automatically after 7 days of inactivity.
- **Unified Tags:** Every user's application would be connected to a digital tag that they would receive physically in the park. Using these tags, all services can be purchased seamlessly using tap and pay. For group bookings, a unified tag for all the group members would be provided.
- **Estimated Wait Time:** The application should show the estimated waiting time for each individual ride at the park.
- **Report lost item:** Lost items can be reported in real-time in the application.
- **Pre-book cabs**: We would partner with Uber/Ola and offer guaranteed rides back home, including SUVs for ease of travel for the customers.

You have listed several features. Which of these would you prioritize and why?

I would prioritize the digital wallet, estimated wait time, and the unified tags features initially, as these are low effort and the reach and impact would be much more. Since only 2-3% of people in the park lose an item, we can implement the report lost item feature later. The pre-booking of cabs feature would require serious discussions with our cab partners and only then we can launch it, so it should be parked for later.

What metrics would you be monitoring and what are some risks

Average waiting time at the different checkpoints along the user journey will be our north star metric. We will also monitor the number of transactions done through the digital wallet and compare it with the offline channel.

A significant risk would be downtime in our application can result in a sudden spike in offline counters. Also, the park must ensure good WiFi connection spots throughout, otherwise the app would not be much of a benefit.

Sports App for India

Design an app to increase sports adoption in India

Sure. I would first like to ask a couple of clarifying questions. Firstly, is there any specific geography in India which this app is targeting? Second, what is the overall goal of the app – user engagement, monetization, or something else?

The app is supposed to be pan-India. You are free to choose the objective.

Absolutely. As this is a new app, I would like to focus on user acquisition and engagement initially. The key to designing an effective app for this purpose is understanding our user personas and their pain points. I've identified four primary user personas:

1.Young Adults (18-25 years)
2.Parents (30-45 years)
3.Working Professionals (25-40 years)
4.Sports Enthusiasts and Athletes (15-35 years)

That makes sense. Can you elaborate on the pain points of these personas?

Sure. Here are the pain points for each persona:

1.Young Adults:

1. Lack of awareness about local sports events and facilities.
2. Limited access to affordable sports coaching and facilities.
3. Lack of motivation and community support.

2.Parents:

1. Concern about safety and quality of sports facilities.
2. Difficulty in finding reliable information on sports programs for their children.
3. Time constraints in managing their children's sports activities.

3.Working Professionals:

1. Limited time to engage in sports activities.
2. Difficulty in finding convenient locations and schedules for sports.
3. Lack of social connections for group sports activities.

4.Sports Enthusiasts and Athletes:

1. Difficulty in finding high-quality training and coaching.
2. Lack of information on competitions and events.
3. Need for a platform to showcase their achievements and track progress.

Great. What features would you suggest to address these pain points?

1. Event and Facility Locator: A map-based feature to find nearby sports events, facilities, and coaching centres. Filters for different sports, age groups, and skill levels.
2. Scheduling and Booking: Easy scheduling and booking for sports activities. Integration with calendars for reminders and notifications.
3. Community and Social Features: Forums and groups for different sports and interests. Options to join or create sports groups and teams.
4. Safety and Quality Assurance: Verified reviews and ratings for sports facilities and coaches. Information on safety protocols and quality standards.
5. Progress Tracking and Achievements: Tools for tracking personal progress and setting goals. Leaderboards and achievements to motivate users.

How would you prioritize these features?

I would use the RICE framework to prioritize these features. Here's the evaluation:

Feature	Reach	Impact	Confidence	Effort
Event and Facility Locator	High	High	Medium	Medium
Scheduling and Booking	High	High	Medium	Medium
Community and Social Features	Medium	Medium	High	Medium
Safety and Quality Assurance	Medium	Medium	High	Medium
Progress Tracking and Achievements	Medium	High	Medium	High

Based on this evaluation, the prioritized features are:
1.Event and Facility Locator
2.Scheduling and Booking

Sports App for India

Excellent. What key metrics would you use to track the measure of app's success?

I would track several key metrics:

1.User Acquisition:

1. Number of downloads and active users.
2. Referral rates.

2.User Engagement:

1. Daily and monthly active users.
2. Time spent on the app.
3. Number of events booked and facilities used.

3.User Retention:

1. Retention rate (weekly/monthly).
2. Churn rate.

4.User Satisfaction:

1. Ratings and reviews.
2. Net Promoter Score (NPS).

5.Monetization:

1. Revenue from bookings and premium features.
2. Conversion rate of free to paid users.

The North Star Metric would be the **number of events booked and facilities used**. This metric captures user engagement and directly correlates with the app's goal of increasing sports adoption. It reflects how well the app is connecting users with sports activities and facilities.

Great! Thank you for walking me through your thought process and design strategy.

Social Media App for the Elderly

You are a PM, design a social media app for elderly people.

I have some initial clarifying questions. Are we focusing on any specific age group within the elderly population?

You are free to choose

Is there any competitor product in the market?

There is no competing product. You have to design a new product.

Finally, is there any specific goal such as driving user adoption or generating? revenue?

You are free to decide

As this is a new product, I feel focusing on user adoption is necessary rather than generating revenue. I would like to first identify different customer segments amongst elderly people above the age of 60 and identify their pain points before I select a particular segment and recommend features. Is that alright?

Sure, please proceed.

I can identify 3 potential customer segments –

1. **Tech-Savvy Seniors**: These are individuals who are already using some form of social media. They have basic technical proficiency but face concerns such as privacy, managing multiple apps, finding relevant content.
2. **Family Connectors**: These seniors love staying connected with family and friends, want to share photos, and updates. They often struggle with navigating complex interfaces and fear making mistakes.
3. **Lonely Seniors**: These seniors experience loneliness and look for activities, apps to keep them engaged and alleviate their sense of isolation. They face challenges such as a lack of digital literacy, fear of technology, and accessibility issues etc.

I would like to proceed with the Family Connectors segment as I feel these will be the regular users of this social media app. They have a strong desire to stay connected and can significantly benefit from a well-designed social media app, which aligns well with our goals. If that sounds reasonable, should I proceed with listing the pain points?

Yes, that is a fair segmentation. You can proceed with outlining the pain points and features. Focus on incorporating the latest technologies, such as AI.

1. Difficulty Navigating User Interfaces: The complexity of apps can be intimidating, making it challenging for elderly to use the app.
2. Using Features: These individuals often struggle with understanding and using advanced features, which can prevent them from fully engaging with the app and their family members.
3. Data Security and Privacy: Protection of shared memories and personal information is a worry hence they are skeptical of using the app.
4. Finding Relevant Content: Elderly people find it difficult to discover and share engaging content and topics that would interest and involve their relatives, leading to missed opportunities for connection.

Coming to the features for these pain points –

1. **Simplified UI and Features for Elderly**: An interface designed to be user-friendly for elderly users, featuring larger text, simplified navigation, and easy-to-understand options to enhance accessibility and usability.
2. **AI-Powered Event Reminders and Suggestions**: AI-driven reminders that suggest events based on user interests and family input.
3. **Adaptive Privacy Settings**: Enhanced privacy settings that adapt based on user behavior and content sharing patterns
4. **Personalized Content Recommendations**: Suggest relevant content, groups, and events based on user interests and interactions.
5. **Emotional Analysis:** Analyze user interactions and feedback to gauge mood and emotional state, offering personalized support and suggestions.

How will you prioritize among these?

Feature	Value	Complexity
Simplified UI and Features for Elderly	High	Medium
Adaptive Privacy Settings	High	Medium
AI-Powered Event Reminders & Suggestions	Medium	Medium
Personalized Content Recommendations	Medium	Medium
Emotional Analysis	Low	High

This was comprehensive. How would you monetize this app?

Social Media App for the Elderly

For monetization, I would focus on the following revenue models:

1. **Freemium Model**: Offer basic features for free while charging for premium. This will attract a large user base with free access and generate revenue from those seeking advanced features.
2. **Advertising and Partnerships**: Incorporate targeted ads and form partnerships with brands relevant to seniors, like health products and travel services.
3. **Subscription Model**: A subscription service for exclusive features, regular updates, and enhanced privacy settings.

This was comprehensive solution.

Thank you.

Supermarket Application

Design a product to solve the problem of long queues at Supermarkets (DP World)

Before we start, I want to ask a few clarifying questions. Are we looking to design a physical product or digital to solve this problem?

You can choose either.

I'll design a digital product and list the user personas that I can think of.

1. Cashier : 25-40. years old. Would like to process customer orders quickly.

2. Bulk buyer: 30-55 years old. Purchases groceries for a couple of weeks at a time.

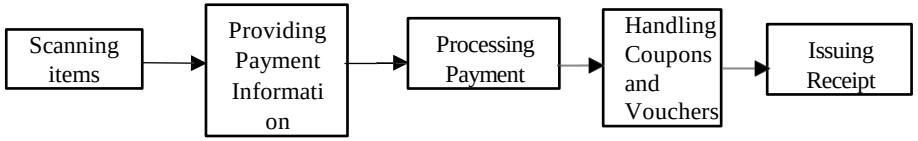
Out of these do you want me to prioritize any one persona?

No, you can consider both

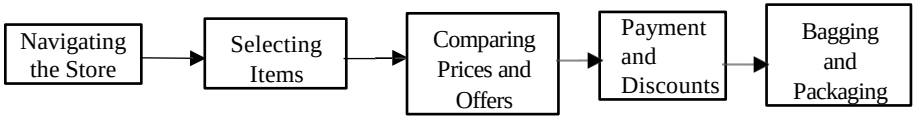
Great! Just give me a couple of minutes to gather my thoughts. I will describe the user journeys of the cashier and bulk buyer.

Sure.

Journey of a cashier in a supermarket.



Journey of a bulk buyer in a supermarket.



For these two personas I have identified a few pain points in their journey –

1. **Cash Handling (for cashier):** Often customers pay with cash. The cashier must find the exact change, which can increase waiting times.
2. **Payment failure:** Sometimes payments via digital channels enter a pending state for an indefinite amount of time, in which case the customer waits for 5-10 min before making another payment.
3. **Finding Items:** The buyer has to spend a lot of time to search all the different items that he/she needs.
4. **Long Checkout Lines:** Waiting in long checkout lines can be time-consuming and frustrating.

I think these look good. You can go ahead with the solution to solve these.

Sure! Let me describe the features which can solve these pain points.

1. **Smart wallet linked to the payments feature in the app**. Users can directly top up the wallet and use it for shopping. They can also keep a reserve balance in the wallet which will be used in case of a pending transaction. If the pending transaction was successful after some time the reserve balance will be refunded.
2. **Virtual map for supermarkets** which gives users the directions for any product that they want.
3. **QR Code Scanning:** Integrate QR codes on product labels. Buyers can scan these codes with their smartphones to access detailed product information, including manufacturing dates, expiry dates, and quality assurance details.
4. **AR Product Information:** Use Augmented Reality and display detailed product information, reviews, and usage tips when users point their phones at products.
5. **AR Navigation**: Utilize Augmented Reality to guide users through the supermarket. By using their smartphone camera, users can follow AR markers and arrows overlaid on their screen to quickly find aisles and specific products, making the shopping experience more efficient and enjoyable.
6. **AI-Powered Shopping Assistant**: An AI assistant that learns user's preferences and habits to suggest the fastest shopping routes and the best times to visit the store.

Sounds good. Thanks

ATM for Airport

You are a PM, you have been assigned to design an ATM for the departure terminal at an Airport

I would like to begin by asking a few clarifying questions. What is the business objective we are trying to achieve here?

Adoption and user engagement is the primary goal, we want the user to have hassle free experience.

Great, next I would like to move to user personas, different type of people who will be using this product and mention their pain points

Persona	Pain points
Frequent Business Travelers	• Inability to quickly print receipts for expense tracking. • Long wait times at the ATM. • Limited multi-currency options, leading to inconvenience for international travel.
International Tourists	• Language barriers making it difficult to understand instructions. • Unfamiliarity with local banking processes causing confusion. • Lack of multilingual support. • Currency Exchange Rates is an issue
Senior Citizens	• Small or unclear text making it hard to read. • Lack of assistance options (help button, voice guidance).
Tech-Savvy Millennials	• Lack of integration with digital wallets and payment apps. • Slow or outdated technology causing frustration
People with Disabilities	• Difficulty reaching and using the ATM due to poor design. • Absence of Braille support and audio instructions. • Lack of support for visual or physical impairments.

Okay, would you prioritize one of these personas, or are you going to solve for all?

I think we should prioritize all personas because the needs don't cannibalize. The base of the product will remain largely the same, with minor features adjusted for each persona, ensuring a comprehensive and user-friendly solution. This approach increases usage and revenue while strengthening the airport's reputation for accessibility and customer focus.

Okay, go on.

Now, I will be defining the features for the ATM

1. **Design:** Modern, Sleek, High-Resolution Touchscreen, Large Text, LED Lighting, Audio Instructions, Privacy Screens, CCTV Cameras, Anti-Skimming.
2. **Accessibility:** Near Check-In Counters, Departure Gates, Currency Exchange Desks
3. **ATM Machine features**
    1. **Fast Transaction Processing**: Implement a high-speed interface to minimize wait times.
    2. **Assistance Options**: Include help buttons and voice guidance features to assist with transactions.
    3. **Integration with Digital Wallets**: Support for digital wallets and payment apps (e.g., Apple Pay, Google Wallet) for contactless transactions.
    4. **Braille Support and Audio Instructions**: Provide Braille on all keypads and offer audio instructions for visually impaired users.
    5. **Multilingual Support**: Offer multiple language options for instructions and transaction prompts.
    6. **Clear Fee Information**: Clearly display any fees associated with transactions to avoid confusion.
    7. **Multi-Currency Options**: Provide a wide range of currency options for international travel, with the ability to quickly switch between them.
    8. **Quick Print Receipts**: Ensure the ATM can quickly print receipts with detailed transaction information for easy expense tracking.

What will you prioritize among them?

For prioritization, design and accessibility is a must because it bring value to a physical product, rest of the in-ATM feature I would like to prioritize based on value vs effort framework (on a scale of 1-10).

ATM for Airport

Feature	Value	Complexity	Score
Fast Transaction Processing	10	8	1.25
Assistance Options	7	8	0.85
Integration with Digital Wallets	10	6	1.66
Braille Support	5	2	2.25
Multilingual Support	8	5	1.6
Clear Fee Information	7	2	3.5
Multi-Currency Options	10	10	1
Quick Print Receipts	6	2	3

Therefore, the features are
1. Design: Modern, Sleek, High-Resolution Touchscreen, Large Text, LED Lighting, Privacy Screens, CCTV Cameras, Anti-Skimming.
2. Accessibility: Near Check-In Counters, Departure Gates, Currency Exchange Desks
3. ATM Machine: Integration with Digital Wallets, Clean Fee Information, Quick Print Receipts, Multilingual Support, Braille Support

Okay, and can you give me some success metrics for your app?

1. No. of Transactions processed/week
2. Time to Success

Finally, do you see some issues with your solution?

Cost Implications: High-end features could significantly increase costs, requiring a cost-benefit analysis to justify the expense.
Implementation Complexity vs. Prioritization: Advanced features like digital wallets and high-resolution touchscreens could delay deployment if not managed effectively.
Maintenance and Support: There needs to be consideration for regular updates, technical support, and troubleshooting to maintain functionality.

Self-Driving Car for Kids

You are a Product manager at Google, design a self-driving car for kids in India.

Clarification questions: I would want to clarify the meaning of self-driving car here so that we are on the same page.

A car that has sensors and has previous sensor data also live maps for driving on the road with human intellect or support.

Yes, you are right please proceed.

Does the business have a certain goal in mind?

No, you are free to define the goal

I would like to define the goal as adoption and engagement, as the use of self-driving is still in the growing stage. Is there a location that we are targeting in India?

No, you are free to choose a location

I would like to keep metro cities and some non-metro cities like Pune in the consideration set. I will proceed further to define the user persona, followed by identifying pain points and moving ahead by prioritizing pain points and defining a solution for them.

Sure, go ahead

To carve the scope, I would say that the service is essentially a ride service that starts once the identity of the rider is confirmed by adult and ends when the drop-off is confirmed at the designation by a designated adult.

I also think that this service can be of two types - personal cars and cabs. I would like to choose cabs so that impact of the service can be maximised by making it available to more users.

Sure.

Following are the basic use cases for which this service can be used

- A child needs to be transported from point A to B periodically

For example, home to school or sports center and vice versa

- Infrequent trips to grandparent's house
- One-time trips – for example visiting a friend's house

I can think of the following user personas -

1. Adults:
   - Primary caregivers
   - Responsible Adults
   - Institutions – schools, etc.
2. Children – Riders

I would like to focus on the persona of adult parents and caregivers to maximize impact and safety. The needs of this persona are:

Safety

1. They want to be aware of where the child is during travel
2. They want to monitor the child while he is in the cab
3. Safety procedures in case of an accident or car breakdown
4. Child handover: The child was dropped off and picked by an authorized person

Convenience

1. Easy accessibility of the car and hassle-free booking
2. Advanced booking should be allowed to plan the travel better
3. Different models like personal or call pools
4. Reliability after booking and should not get canceled right before the ride

Ride

1. Car should be hygienic and safe
2. Car should be able to entertain the kids
3. It should monitor that the kid is not engaging in any hazardous activity
4. Supplies of food and water
5. Content moderation on the car's infotainment system

Great. Can you prioritize the pain points?

I would like to prioritize the pain points. Prioritizing based on Value and Complexity (RICE/ ICE framework can also be followed).

Self-Driving Car for Kids

Pain Point	Value	Complexity
Registration of authorised user	High	Low
Booking a trip	High	Low
Content moderation	Medium	Medium
Safety and accidental fallback	High	High
Trip Monitoring	High	High

Sounds good please proceed

Registration and ride-booking are must-have features, this will allow caregivers to have peace of mind. Defining features for each of these pain points:

1. **Registration of authorized user**: Allowing the user to register for the service and
Registering for the person responsible for drop off and pick up for each trip. The person should be verified with legal ID proof, and the photo of the same should be uploaded during registration.
2. **Booking a trip**: Select start and endpoints. Mark of responsible adults on both points of the trip. Even allowing to keep a list of adults responsible, which can be reused in different trips
3. **Content Moderation**: Link to entertainment accounts that the rider is allowed to view, such as Netflix and YouTube. Allowed interactive games or select shows.
4. **Safety and Accidental fallback**: Create an interface or get off-the-shelf hardware with familiar interfaces
5. **Trip monitoring**: In-vehicle trip monitoring via monitors and trip tracking using analytics from the vehicle

Can you identify some metrics to evaluate the success of your product?

I will define success metrics as

1. Daily Active users
2. Number of trips completed in a day

Sounds good. Can you identify some potential risks?

I can think of the following risks. The inherent assumption is that being able to scale the service to pool and subscription models has been made, but there is no consideration and control mechanism for things going wrong with multiple riders, such as bullying on a school-bound car. These need to be thought out for long-term existence

Modernizing A Traditional Store

The CEO of a store has approached you to modernize their store. How would you proceed? (JPMC)

Thank you for the case. I would like to begin by asking a few clarification questions:
a. What kind of a store is this?
b. Where is this store located?
c. What kind of customers typically shop at the store?

Good questions! This is a typical store, similar to Spencer's or any other hypermarket that you can think of. The store is located at a posh locality in a metro city. Regarding the customers, you are free to make assumptions.

Thank you for the answers. Now that I have a basic understanding of the store, I would like to deep dive into the shoppers' personas. Since we are talking about adopting latest technology, I can think of 2 broad categories:

1. **Technology enthusiasts –** Typically, this would be people aged less than 35. They are early adopters of technology and are willing to try something new. They also prioritize convenience and speed over complexity of the product.
2. **Non-tech savvy shoppers** – Typically, this would include people aged over 35. They are late adopters of technology. They would love simple products and would not mind spending a few extra minutes at the store.

Sounds good. How would you go about designing features for these personas?

I would begin by drawing the user journey:



1. Pre-entry - Digital parking system: An automated screen would scan the car and suggest the optimal parking slot. It would also record the car number and the entry time automatically in a database.
2. Entry – Digital check-in system at the entrance: People can scan their shopping lists/select manually and it would generate a receipt with the bays at which those items are available. If some items are unavailable, an option of home delivery is also given to the customer. This is particularly useful for technology enthusiasts.
3. Entry – Smart carts: The shopping list would also be synced with the smart carts (similar to Amazon stores in US). In case the shopper has forgotten something and has reached the payment zone, the cart would display the missing items.
4. Shopping Experience – Digital Screens: A lot of food items might have fine prints which are difficult to read. Digital screens at convenient locations at each bay can help shoppers find out more about each product with a simple barcode scan. This would help them to make an informed choice.
5. Checkout – Express Checkout via Carts: Designated payment zones would be set up in each floor. When the carts are brought there, the users can opt for express checkout via automatic deduction from e-wallet or through UPI.
6. Customer service – Seva Desk: To encourage large purchases, a support desk for home delivery can be organized beyond a certain amount, say Rs. 1500. Also, loyalty cards can be issued at this counter to increase retention.

While these are features mainly for the technology enthusiasts, some manual counters and carts would also be kept for the non-tech savvy crowd. Gradually, as awareness increases, we can shift towards a completely automated model.

Great, you have touched upon all the aspects of the user journey. I am curious, how would you ensure that there are no duplicate entries during the login step?

I would be using the mobile numbers of the customers. Also, if there are no data privacy issues, after a one-time registration, even facial data can be used. That way, we can keep track without any duplicate entries.

We are running out of time, so briefly describe which features you would prioritize and the metrics you would use to judge the success of this transition?

I would prioritize the digital parking and check-in systems, smart carts and the seva desks. They have been implemented previously, all we need to do is assimilate them at a single place.
Regarding the metrics, I would look at the MAU, stickiness and retention rates at an overall level as well as across the 2 personas to see if any optimization is necessary.

That'd be all. Thank you for your time!

Car for the Deaf

How would you design a car for the deaf? (Flipkart)

Before we start, I want to ask a few clarifying questions. Are we building a new car from scratch or are we making changes to an existing one? Can we bundle other devices along with the car?

Let's assume we already have a car in the market. We need to modify it to suit the new segment. You can add any devices or software to the car.

Another clarification. Do the deaf include only people with full hearing loss or do we also think about partial hearing loss? A majority of the deaf people are also dumb. Do we have any information on this?

Just think about completely deaf users. Assume they are not dumb.

Got it! Let me think about a few deaf users who would be interested in such a car.

- **Young bachelors**: These are students or young professionals who need to commute to office or college and social gatherings with friends. For them cars are also a symbol of social status.
- **Middle aged professionals** : These are 35-50 year olds who use cars to commute to work everyday, for long distance travel and outings within the city. They tend to travel with their family who might not be deaf.
- **Elderly folks**: These are 60+ years old who need to use the car to travel to nearby locations for household chores. They would also be visiting hospitals and pharmacies.

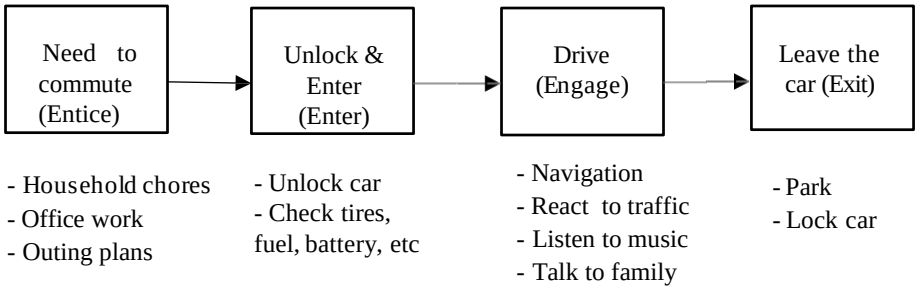
Among these I find **middle aged professionals** the most suitable target segment. Young bachelors lack the financial resources to purchase such a car. They mostly use public transport and cabs to commute from one place to another. Elderly folks too are not the right target segment since they would rarely travel alone. Their family takes care of them and ensures that they accompany them during their travels. Hence a younger person in the family would likely be driving and taking them to the nearby locations.

Middle aged professionals form a significant chunk of the population and have the buying power to purchase this car. Can I go ahead with this target segment?

Yeah, please go ahead!

Right. Let me go through the typical user journey of such a user and list down some needs and pain points.

Sure!



Here is the user journey of the usage of a car. For a deaf user in my segment, I can come up with the following pain points:

- While unlocking and locking, a major pain point for this segment is that they are unable to hear the beep sound for feedback. They need to look at the lights as a visual cue which could be a challenge especially during the day time.
- Deaf drivers have difficulty in conversations with passengers, requiring alternative communication methods.
- Drivers usually rely on the audio map directions for navigation. Deaf drivers would need to constantly look at their mobile screens during navigation which could be distracting and compromise on safety.

Car for the Deaf

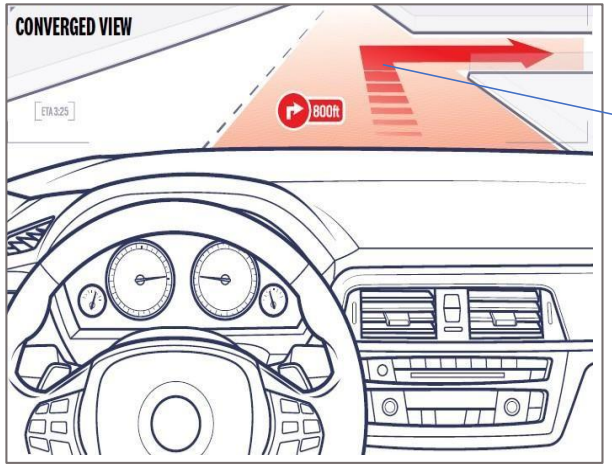
- A huge pain point for deaf drivers is that they are unable to hear the car horns. This could prove to be a problem at intersections, in overtaking lanes and when a police car or ambulance is nearby.

Among these pain points, what I want to prioritize is sorting out navigation and enabling reacting to car horns for deaf drivers. Solutions to other problems are good to have but these are must have since they would ensure a safe ride. Can I go ahead and think about the features for these?

I like your reasoning for all this. Please go ahead with the features.

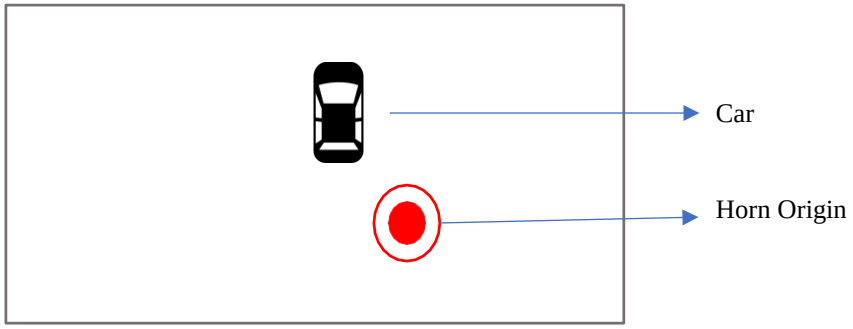
I have come up with two features:

AR Navigation: To make navigation easier, I would like to have directions on the windshield itself using Augmented Reality. The directions would be superimposed on actual roads to make it easier to navigate while driving without relying for audio cues. It would look something like this. (Candidate draws on a piece of paper)



Windshield Navigation

Horn Detection Dashboard: A sensor would detect when a sound greater than a particular decibel threshold happens around the car. This would generate a vibration on the steering wheel of the car and show the origin of the sound on a dashboard as a hotspot on a screen near the steering wheel. The driver can check the origin of the sound, look at the back view or sideview mirror and react accordingly. This is how it would look. (Candidate draws on a piece of paper)



These features look really cool. But how would you measure whether they are indeed making the car safer?

Ideally, it would be great if our analytics team could install a device in the car which could detect accidents. Through this we can compare the number of accident per car per month in this new car. However, it might be difficult to define the benchmark for this metric.

Another metric we could look at is NPS. We can send a survey to users after 6 months of their purchase and ask whether they would recommend this car to another deaf person keeping its safety in mind.

That makes sense. That would be all. Thank you.

Ecommerce for the Elderly

How would you design an e-commerce platform for senior citizens? (Flipkart)

Before we start, I want to ask a few clarifying questions. Firstly, are we building a new ecommerce platform from scratch or are we making changes to an existing one? Secondly, how do we define a 'senior citizen'?

We already have an e-commerce platform. We have a website as well as apps for Android and iOS. Senior citizens are folks above the age of 60.

Sure, let me list down the user personas who might have a need for using e-commerce.

- **Urban elites**: These are tech savvy senior citizens who order frequently through e-commerce platforms for their needs. They are comfortable with using computers and mobile phone apps.
- **Joint family elders**: These are senior citizens who live in a joint family, mostly in towns and villages. Whenever they need something, they mention it to their children or grandkids who either get the things needed from the nearby stores or order it online for them. The senior citizens rarely use the e-commerce platforms themselves.
- **Elders living away from kids**: These senior citizens live in cities or towns away from their children who are working in big cities. They don't get help for ordering online that easily. They sometimes try to order things themselves from online platforms but find it quite challenging. As a result, they often call or text their kids to buy a particular item for them. It is for this reason that I think we should target this segment.

I think that's a fair assessment. You can go ahead with this.

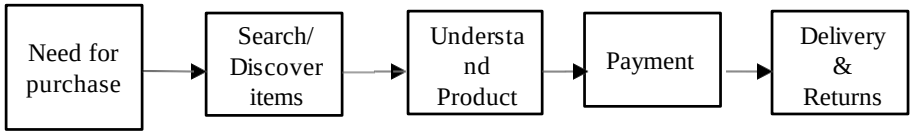
Do we have any platform specific usage data on senior citizens? In my opinion we should target the Android app. The users owning computers and iOS phone would likely be urban elites who would be much more comfortable in using e-commerce platforms.

Close to 90% of our users above 60 are on Android. Your logic sounds good! Let's go ahead with Android.

Great! Just give me a couple of minutes to gather my thoughts.

Sure.

This is the journey of a typical senior citizen using an e-commerce app.



Within this I have identified a few pain points in their journey.

1. **Trust**: Having shopped through physical stores all their lives, senior citizens look at e-commerce platforms with a lot of apprehension. They are scared about being duped and losing their money.
2. **Eyesight**: Senior citizens often have near sightedness which prevents them from reading small text in e-commerce apps. They are attracted to the product pages by the images on the home or browse page but read the product description thoroughly before completing the checkout.
3. **Difficulty with apps**: Senior citizens have great difficulty in discovering features which can be used. For example, they would rarely use filters or sort and instead just scroll through the list on the browse page.
4. **Payments**: Since senior citizens are not comfortable with online payments, Cash on delivery is the preferred option. However, having cash available after each delivery is a tough ask. They also have to heckle about having change.

Should I think about more pain points or should I go ahead with the ones I came up with?

I think these look good. You can go ahead with the solution to solve these.

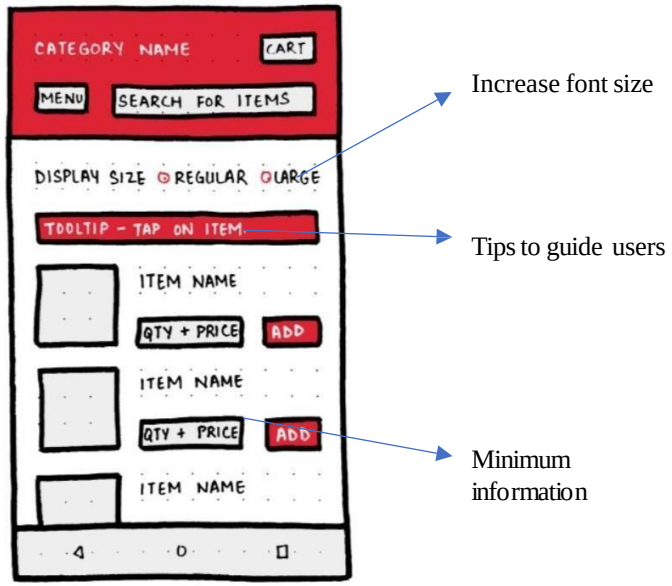
Ecommerce for the Elderly

Sure! Let me describe the features which can solve these pain points.

1. **UI Changes**: Before any new features, I would want to introduce basic UI tweaks to improve the experience for senior citizens. We should have larger texts, pictures, icons and buttons to ensure that users don't face any usability issues. Buttons should have colours contrasting with the background colour to attract the attention of users. We should also show tips on pages to guide the users.
2. **Minimal layout**: We should not overload the pages with features, hidden menus (hamburger menus, dialogue boxes which appear after tapping) and information (seller details, multitude of offers). Only the information which is absolutely necessary should be displayed.
3. **Send order updates on WhatsApp**: Senior citizens rarely open e-commerce apps after ordering but they are constantly going through their WhatsApp messages. Updates on their order (packed, shipped, out for delivery) should be sent on their mobile number on WhatsApp. This helps them trust the platform since they are able to view constant updates regarding their order.
4. **Bulk post-paid**: Instead of asking the users to pay after each delivery, senior citizens should only be asked to pay after a certain amount, let's say INR 5K is reached. This reduces the hassle of having money with change ready during each delivery. This also recreates their current relationship with the kirana store on the e-commerce platform. For this feature we would definitely need to collect details on users by making them upload pictures of their identification card. This process would be somewhat similar to KYC for banks since we are also extending credit.

Here is a wireframe of the browse page to visualize my UI and layout changes.

Although these are some interesting features, how do you plan to launch it? Will it be a new app?



No, this experience will be present on the existing app itself. No new app needs to be downloaded. We would show a toggle option on the homepage targeted to senior citizens. This toggle will change the UI and the layout. Senior citizens who are comfortable with the existing UI and layout can opt out of this experience. Similarly, users can opt out of receiving updates on their WhatsApp. Bulk post-paid will be a payment option during checkout.

That is a well thought solution. Thank you!



Root Cause Analysis (RCA)



RCA Questions – Approach

Metric ABC has seen a X% drop. What would you do?

- **Define the metric**: Be sure you understand what metric has been affected. Tell your interviewer what you think the metric means. Let your interviewer confirm before you proceed to solve the question.
- **Context**: Ask clarifying questions about the time duration in which the metric has been down, platform affected (mobile app/website), OS affected (Android/iOS), demographics affected & geographical region affected (international/India/regions within India).
- **Drop Rate**: Is this drop sudden or gradual? Both internal and external factors need to be evaluated either way, but the kind of suggestions you make should take this into account. For example, sudden declines are quite likely to be caused by new releases on our user facing platforms or servers which have introduced bugs, errors or crashes.
- **External factors:** These can be a new competitor entering the market, bad PR for our product, seasonality, a government regulation or legislation or a major temporary event (such as Covid-19, strikes).
- **Internal factors**: To evaluate these, create a user journey of your platform, ask for whether there has been any change in metrics at each stage of the user journey from the interviewer and understand the point at which the user behaviour has changed. This would help you zoom in on one stage of the user journey and understand the root cause of the change.
- **Solution**: After identifying the root cause you need to come up with a solution to address it. If the issue is within our platform or server, we should fix it immediately through a hotfix or new version release. If it is caused by some external factors, it will take more time to solve. We need to escalate it, involve more stakeholders in the organisation and devise a strategy to deal with it.

Swiggy Revenue

Swiggy has reported a 10% drop in revenue recently. How would you tackle this problem?

Sure. I would like to start with some clarifying questions in the beginning. By "revenue" are we referring to the total income generated from all orders placed on Swiggy's platform?
Also, could you provide details about the time frame over which this decline occurred? Is it a sudden drop or a gradual decline?

Yes, we are looking at the total revenue from all orders and the drop has been noticed over the past two months. So it is more sudden than gradual.

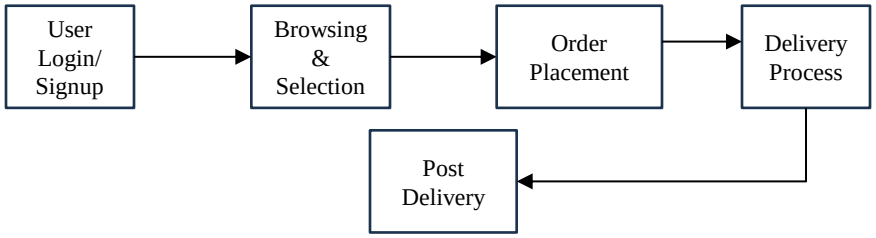
Understood. I'd also like to know if this drop is specific to any particular region, city, or type of service within Swiggy (e.g., food delivery, grocery delivery, etc.)?

The drop in revenue is across all regions, but it's most significant in the major metropolitan areas. It affects both food delivery and grocery delivery services

That's useful information. Now, I would like to know if there have been any recent market changes, such as new competitors, changes in consumer behavior, or regulatory impacts?

A few new food delivery startups have launched aggressive marketing campaigns, and there have been recent regulatory changes impacting delivery operations in some cities.

Okay, thank you. Now I would like to look at the internal factors and to do this I will come up with a user journey and the associated metrics for the Swiggy app which covers the different offerings in the app.



METRICS

User Login/Signup
- # New User Signups
- # App Downloads
- User Retention Rate

Browsing/Selection
- Session duration
- # Restaurants/items viewed
- Bounce Rate

Order Placement
- Conversion Rate
- Avg Order Value
- Cart Abandon rate

Delivery Process
- Avg Delivery Time
- Success Rate
- Cust. complaints

Post Delivery
- CSAT Score
- Repeat Order Rate
- User Reviews

I have listed the usual journey a customer would follow while ordering an item from Swiggy. Then I have associated few metrics that I believe would be important to look at to find the cause of the issue. From the listed metrics do you see any that you find to be impacted significantly in the recent months for Swiggy?

The metrics regarding User Login and Signup seem fine. We have however noticed a drop in CSAT Score and a decline in repeat orders. Also, delivery times have increased in certain areas, leading to more customer complaints

Based on this user journey analysis, it seems like the delivery inefficiencies are a significant pain point. The increased delivery times and subsequent delivery charges are likely affecting both new and repeat customers, leading to a drop in revenue.

That's a good insight. Let's focus on delivery inefficiencies as the primary root cause. Can you propose some potential solutions to address this issue?

The factors that we could improve could be optimizing the delivery routes by implementing a better algorithm and adjust our route based on real-time data better. Hiring of additional delivery riders in high-demand areas can be another solution to the same. Offering coupons to affected customers can help rebuild trust and also, we can reevaluate delivery charges based on dynamic pricing.

These seem like good recommendations. I believe we can conclude. Thank you!

Discrepancies in Google Maps ETA

Addressing Discrepancies in Google Maps ETA (Just Dial)

You are the PM for Google Maps. Google Maps is showing a shorter ETA than is actually required to travel between two points. Users are reporting that they are arriving at their destination later than expected, even when they follow the directions provided by Google Maps.

Sure. To better understand the issue, could you provide me with more details about when and where this problem is occurring? Is there any specific area where users have reported experiencing this problem more frequently? Is this problem present across all platforms, Android/iOS, Mobile/Web?

Yes, this is present across all platforms and has been only seen in the last couple of days. Additionally, the users are reporting from various regions and routes, so it doesn't appear to be isolated to a particular area. We have also noticed that the ETA inaccuracies are more prevalent during peak traffic hours.

That's definitely concerning. I would like to understand if there have been any recent updates to the algorithm or data source that could have caused this. Can you confirm if any updates were made to the routing algorithm or the data used for traffic predictions?

The data sources for traffic information remain the same as before. We did roll out a few updates to improve routing algorithm. It's worth noting that some users have mentioned that the ETA often remains the same regardless of the time of the day, leading to unrealistic expectations during heavy traffic periods.

Understood. It seems that the recent algorithm changes could be the likely cause of the ETA discrepancies. To address this, my next step would be to analyze the specific changes that were made to the routing algorithm. Could you provide more details about the nature of these updates and how they were expected to improve accuracy?

Certainly. The updates were intended to take into account more real-time traffic data and road closures due to construction. The idea was to provide users with more dynamic routing based on current traffic conditions and roadblocks.

Got it. While the intention behind these updates is commendable, it's possible that the algorithm is now overestimating the impact of road closures or underestimating the time required for specific traffic conditions.

To address this, I would propose conducting a thorough analysis of the algorithm's performance before and after the update. This analysis should involve comparing the ETA predictions with actual travel times for a wide range of routes and traffic scenarios.

That sounds like a reasonable approach. How would you proceed if the analysis confirms that the algorithm changes are indeed causing the ETA discrepancies?

If the analysis confirms that the algorithm changes are the root cause, I would recommend a two-fold approach. Firstly, I would work with the development team to fine-tune the algorithm parameters related to road closures and traffic data integration. This could involve adjusting the weightage given to different data sources.
Secondly, I would initiate a communication plan to inform users about the issue and the steps we are taking to resolve it. Transparency is crucial in maintaining user trust. We would provide regular updates through the app informing users about the progress and expected timeline for the fix.

Those are solid steps to take. One last question. How would you prevent similar issues from occurring in the future, considering the potential impact on user experience?

To prevent similar issues, I would recommend implementing a more robust testing and validation process for algorithm updates. Additionally, establishing a feedback loop from users, allowing them to report any discrepancies they encounter, would help us identify issues more quickly.

Great. We can stop it here. Thank you.

Amazon Return-Rate

There has been increase in the return-rate for Amazon products. What can be done?

Sure. I have some initial clarifying questions before proceeding. Is the decline sudden or gradual? Does the decline occur during any particular season?

The increase has been gradual. But, the return-rate peaks during the wedding/festive season.

Is there any specific product category for which the return-rate is increasing much?

Return rates of Clothes and Electronics have increased the most. You can focus on clothes for now.

Okay. Is this problem prevalent across India or is it specific to some particular geographic locations?

We are facing this issue across India.

Okay. I can think of 5 potential reasons for returning online ordered clothes–

1. **Wrong/defective products**: Sellers sometimes send defective or wrong item other than what the customer wants
2. **Wardrobing:** Sometimes customers buy clothes/footwear particularly for any occasion and then return the product after using it on that occasion.
3. **Size/fit issue:** It happens many times that the size of the product misfits the customer, so he/she has to return it.
4. **Change in mind**: Sometimes while searching for products, customers impulsively order items that they do not require. And after delivery, they realize that the product is not as appealing as they were anticipating and they have made an unnecessary purchase.
5. **Delay in delivery**: It happens sometimes that a customer requires something for a particular occasion, but the delivery gets delayed due to some reason and the customer no longer requires that product.

Is there anything else I need to add as a reason?

No, you have come up with good reasons. You can proceed.

Should I focus on any particular reason?

You can focus on the first three issues. Suggest ways by which we can address these issues and minimize the rate-rate.

Sure. Those 3 reasons of return can be tackled by–

1. Wrong/defective products:

1.1 Barcode authentication: Put a barcode not just on the shipping box, but on the product itself. And, scan the barcode at each stop during the transit and share the product details to the customer. This will alert the customer as soon as there is any change/defect in the product. It'll also help in tracking the package and minimize the chances of it getting lost/stolen or exchanged during transit.

2. **Wardrobing:**

2.1. Reduce the return window for clothes and perform stricter checking (tamper-evident tags) during pickup of returned products. We can implement 15-minute return policy as a pilot.

2.2. Flag customers with more than average return rate. Penalize the customer after a certain number of returns.

3. Size/fit issue:

3.1. Provide realistic product pictures, size-chart and insights from past purchasers. Incentivize reviews which help the costumers.

3.2. 360 degree view and virtual try-ons.

How will you prioritize among these?

For prioritizing features, I will look into the value added by the feature and the complexity of implementation for the company.

Feature	Value	Complexity
Barcode authentication	High	High
Reducing return window size	High	Medium - High
Flagging customers	High	Low
Realistic product pics and reviews	Medium	Medium
360 deg view and virtual try-on	Medium	Medium

Amazon Return-Rate

These recommendations sounds reasonable. Pick your top 3 solutions.

Sure. My top 3 recommendations would be:

1. Barcode authentication
2. Reducing return window size
3. Flagging customers

However, there can be dissatisfaction among customers due to reduction in return window size. We should implement it as a pilot for a few products and we have to be very vigilant about customers' reactions towards it.

Seems practical. That would be it. Thanks!

Prime Video Views

The video views on Prime Video have gone down. What should we do? (Microsoft)

Sure. Before I begin, I would like to ask some clarifying questions. What counts as a video view on Prime?

A video is counted as been viewed if at least 70% of the video is watched by the user. Views per day is the metric affected.

Okay. How much has the decline been? For how long have we seen this issue? Has the decrease been gradual or sudden?

We have seen a 20% decrease. This has been gradual, over the last month.

Is this issue prevalent across website and mobile apps on Android and iOS?

Yes, we are noticing this across all platforms, OS versions and devices.

Is this issue happening in a particular geographical region in India? Or to a specific demographic group?

No. This is happening across India for all demographic segments.

Understood. I will just reiterate the problem at hand. Video views on Prime Video is down by 20%. This has been gradual over the last month. We are seeing this issue on both website and mobile apps for all geographies and user segments in India. Can I proceed with this?

Yes. Please go ahead.

Alright. I would first analyse some external factors that might have impacted our users. For internal factors, I would understand the journey of our user while watching videos on our platform. Does this seem okay to you?

The structure looks good. Please go ahead and list all the external and internal factors which you can think of.

Sure! I will go over the external factors. Is there any major event going on currently which is being broadcasted? Something like the IPL or any football championships? This would have resulted in people shifting over there and spending less time on our application.

No. Nothing of that sort is happening.

What about a competitor related change? This change could be due to the entrance of a new competitor or some changes introduced by an existing one. If the new entrant is a big player, users could be moving to them due to their reputation or some exclusive content. For an existing competitor, they could have introduced some discounts, new features or content to lure users.

Can you think of some other factors?

Anything related to the government? Has any new legislation or regulation been introduced in India in this domain?

Nope. Nothing from the government side.

Has there been some bad PR? The company's name being dragged through a scam or scandal.

No, nothing like that.

Has this drop occurred earlier as well? Around the same time last year? This is just to make sure it is not a seasonal change.

No. This has happened for the first time.

Okay. Since none of the external factors have affected, I am shifting to internal factors. Has there been any new app release or server deployment around the time when the decline started?

If it were a new release or deployment, wouldn't the change then have been sudden and not a gradual one?

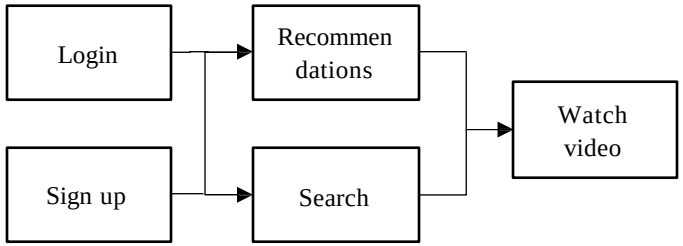
If this new release was rolled out gradually, its adverse impact would be gradual too.

Well pointed out. All our new releases and deployments are stable.

If all these do not seem to be the issue I would go through the user journey and look at metrics at each step of the journey to see where things start to go wrong.

Please go ahead.

Prime Video Views



Here is the user journey for a typical user using our platform. I would list a few metrics at each stage of the journey and ask whether they have deviated significantly from the baseline over the last month.

No need. I think the diagnosis has been quite comprehensive. In this case, as you had mentioned earlier, there is a new application which is offering free content. People are spending their time over there. Now give me some recommendations to address this issue.

Sure. The value proposition that people who migrated are looking for is value for money. This is understandable for the Indian market where people are price sensitive. So, we need to come up with an innovative way of pricing such as sachet pricing.

How would this sachet pricing work? Take the example of a series. How would it work?

Okay. Sachet pricing would be for the whole movie and per season for a TV show.

For illustration let us suppose we take the series Lupin. It has 5 episodes, and it has a new season which would get released next quarter. I would recommend that we price Lupin's entire season 1 and not do it per episode. Enabling payment per episode would mean that the users would need to pay again and again. This can result in a lot of dropouts in the user journey. Doing it season wise makes a lot of sense. Lupin season 1 ended on a cliff-hanger so people would be eager to comeback and pay for the next season when it arrives.

Coming to putting the actual number to the price, it would again depend on the type of content. If more people are interested in particular content, it could be priced high. But the price should not be so dynamic that it changes every few hours or days. This would discourage the viewers from paying if they would see the price rising for the content. Or they might be anticipating that the price might go down in the future.

Okay. This was quite interesting. The approach was very thorough. Thank you so much.

Flipkart UGC

Flipkart has seen a 20% drop in engagement on UGC. What could be done?

Okay. There are a couple of clarifications I want to make. From what I understand, UGC is User Generated Content. What all formats are present within Flipkart to support UGC?

UGC is supported in 3 formats – Reviews & Ratings, Questions & Answers and curated product collections by users.

Right. Is this drop across all formats?

No. Just Questions & Answers seem to have been impacted.

Another clarification. What exactly do you mean by engagement when it comes to Questions & Answers (Q&As)?

Engagement is defined by the metric – number of upvotes + downvotes per Q&A. Both are down.

Thanks. We know Flipkart works across mobile app & website. Is this issue platform specific?

No, we are seeing a drop across platforms.

Have we seen this drop pan India or is it specific to a particular region of users?

The metric is down for all users throughout India.

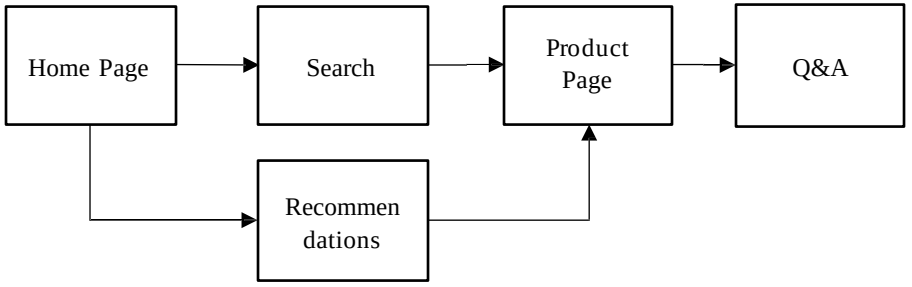
How long have we seen this issue? Was there a sudden drop or has there been a gradual decrease?

We have seen a gradual decrease over the last 2 months. It is likely due to some new release that we did. But Flipkart launches a multitude of features in each release. It is difficult to go through all of them to check what has happened.

I can also think of some external factors like some launch by competitors or some regulation changes which could have impacted e-commerce usage. But let's focus on our internal factors first.

That seems fair. You can go ahead with the analysis of our internal factors

Sure, let me build a user journey to understand how a user would typically use the Flipkart platform.



Homepage
- # sessions/day
- Bounce rate

Search
- # searches/session
- CTR on browse page

Recommendation
- CTR on recommendations on home page

Product Page
- # landings/session
- Bounce rate

Q&A
- # Q&A section impressions/session
- # Q&As viewed /session
- #upvotes/downvotes #upvotes/downvotes per Q&A

Here is the user journey of a typical user on the Flipkart platform. I have listed metrics I would check at each stage to ensure that things are alright at that stage of the journey. A drop in a metric at a stage before the Q&A stage might indicate that the root cause lies outside the Q&A construct. Have any of the listed metrics across the user journey been impacted over the last 2 months?

Interesting thought! None of the metrics before the user reaches the Q&A section have been impacted. Within the Q&A we have observed that the impressions per session and CTR of the overall Q&A section is stable. However, within the section, the number of Q&As viewed per session and total upvotes/downvotes has decreased but, upvotes/downvotes per Q&A has increased

Flipkart UGC

Since, impressions of the Q&A section hasn't changed, we can assume that the visibility and accessibility of the section is still the same. However, something must have changed within the Q&A construct in the release around 2 months ago. I can think of 3 factors – functionality which changed the process through which users browse the Q&As, new UI which changed the look and feel of the Q&As or some technical back-end change which might have impacted error rates, crashes or latency.

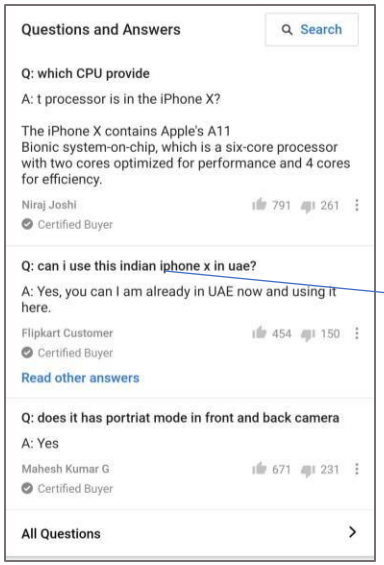
Now that I think of it, we did introduce the text search capability within the Q&A section. This allows to type in keywords and look for a question.

That makes a lot of sense. The fact that users can directly search for specific questions means that they no longer scroll through a multitude of questions like they used to earlier. This naturally means that the overall engagement goes down. It is more important to understand whether the users are able to locate the information they need. If the conversion rate has not dropped it essentially means we have made the purchase easier for users.

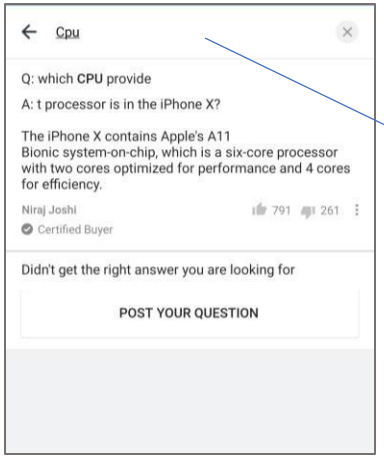
We ran an A/B with the Q&A search feature. The conversion rate in fact improved.

Great! I don't think we need to do anything then. The search feature has improved the user experience. We should move away from tracking engagement per Q&A and instead track engagement per Q&A viewed by the user.

That's a great solution! Thank you.



Q&ASection



Search Feature

Flipkart Notifications

The open rate of Flipkart notifications is down. What would you do?

I would like to ask a few clarifying questions before proceeding. Which notifications are we referring to? From my experience of using Flipkart, I have noticed the following different types:

- Push notifications on the notification panel
- In-app notifications for offers/cart updates
- Direct notifications sent to browsers

Is it one of them or am I missing something?

We are referring to push notifications.

Can you tell me what an open means for a push notification? Does this refer to the action of viewing the notification, clicking on the notification or actually going through with the purchase after opening the app?

Open rate is defined by the number of notifications opened per 100 notifications delivered. Opening a notification refers to the user seeing the notification and clicking on it to reach the intended landing page. That's where the job of our team ends. Rest of the funnel is handled by other teams.

Got it. Is this drop across all operating systems i.e. Android and iOS?

Yes, it is.

Is it limited to any geographical region or demographic segment?

Nope, we are seeing this drop all across India with all sorts of users.

How long have we seen this drop? Was this drop sudden or gradual?

We saw a sudden drop when the lockdown ended. Assume it was yesterday.

Understood. I will just reiterate everything we have discussed until now. We have seen a sudden drop in the open rate of push notifications. This drop occurred yesterday around the time lockdown ended and is seen across OS, geographies and demographics.

That's correct. You may proceed.

Sure! To debug this problem, I would go through both the external and internal factors might have caused this issue. Once we get to the root cause, I will ideate on what steps could be taken to rectify this issue. Does that seem okay?

That sounds fair.

I will go through the external factors first. The first factor to consider is a change to our competitive landscape. Did we have a new e-commerce platform emerging? Or did a competitor launch a major new feature? Or did local brick and mortar stores start offering free home delivery services? These events might have made Flipkart the less preferred shopping destination for users.

I don't think that's an issue. There was no major change in our competitors.

Ok. The next factor to consider is PR. Did we have any negative PR? Maybe some sort of news in the media that our warehouses are not safe? Any delivery guy who might have spread Covid? There was also a push towards boycotting Chinese goods, so something on those lines?

None at all. In fact, public sentiment towards Flipkart has improved. We are widely being recognized as a safe supply chain. More and more people want to start buying online rather than go to physical stores during the pandemic. These were some good points, but why don't you explore the internal factors now?

Sure! Let's brainstorm some internal factors. Since, this was a sudden drop, did we have a new app release or deployment which coincided with the end of the lockdown? There might be bugs in the release or deployment which might be causing this issue.

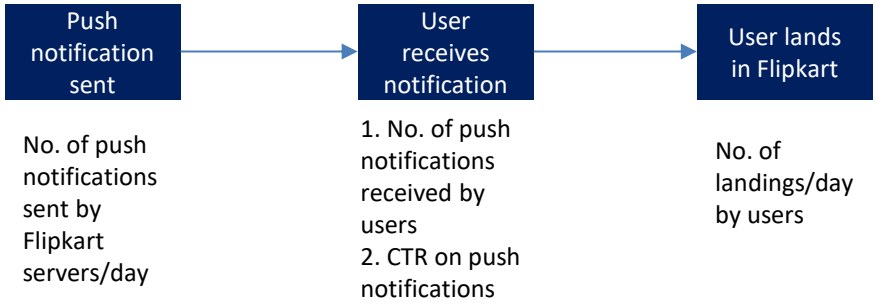
No, there were no recent roll-outs.

Alright. I would then go ahead and analyse the journey of our users receiving push notifications and acting on it. This will help us identify the root cause.

Sounds good. Please go ahead.

Here is the journey for a Push Notification (PN) sent to a user. At the first step, PNs are sent to users from our servers. Have we seen any change in the number of PNs sent per user per day?

Flipkart Notifications



Yes, we have seen a drastic increase in the number of push notifications being delivered to users.

Got it! The next stage is the user receiving the PN and clicking on it. Is the metric of number of PNs received per user per day consistent with PNs sent by our servers? This is just to make sure that we are not facing an issue in which the server is facing some errors and resulting in PNs not being received by the user.

We measure this through acknowledgement rate. The server gets an 'ack' when the user receives the PN. Acknowledgement rate's increase is inconsistent with the increase of PNs being pushed out. We don't see any errors or issues in our server bugs. You can assume the PNs are being delivered to users without any issues.

Have we seen a change in the CTR for PNs? Does it coincide with the time of the increase in PNs being pushed out from our servers?

Yes, CTR has decreased around the same time when our PNs increased.

The decrease in CTR could be as a result of the increase in PNs sent. The users might be feeling spammed or the notifications might not be relevant. Do we know why there has been an increase in PNs recently?

As you know, lockdown just ended. What could be the reason according to you?

With the lockdown ending, we might be informing users that we are resuming our operations. Since there is an immediate need for essentials, we might be sending PNs to let users know that we have all essentials available on our app. Also, users are really worried about their safety during the pandemic. So, we might be informing them about our safe practices and Covid protocols taken in the supply chain. Should I brainstorm further for more reasons?

You came up with some good reasons. However, the increase in PN has been due to wishlist additions. During the lockdown, products were not deliverable. Since people could not purchase them, they added it to their wishlists to buy it later when it's available. When the lockdown ended and the items in the wishlist became deliverable, our systems automatically generated and sent PNs to the users telling them the product is back in stock. This is the actual reason.

Got it ! And this sudden barrage of PNs must have felt like spam to the users. Or folks might have needed it earlier and added it to their wishlist but no longer found it relevant.

Correct. What can we do to rectify this?

One obvious way to do this is to limit the number of PNs sent to a user.

However, having an arbitrary upper cap would mean that we are unable to send across important in formation to the users. We can also limit the PNs by prioritizing products based on the history of shopping for the user. For example, if they have been historically more interested in electronics, we can send PNs for this category and hold other PNs for now.

A more measured approach would be to bundle the products for which we send PNs. These bundles can be created on the basis of product category. For example, the PN can be "X phone, Y TV and Z laptop are now back in stock". This PN combines all items in the electronics category. The trade-off would be that we won't be sending immediate PNs. However, this would be less of a nuisance to the users.

That would be all. Thank you!

Microsoft Word User Count

People are moving away from Microsoft Word. What do you do?

Before I proceed, I would like to ask a few clarifying questions. Are we focusing on a specific type of user? For example, I know that we provide Microsoft 365 bundle to both individuals and enterprises.

You can focus on individuals.

Okay. Are we looking at existing users, new users or both and is there a geographic area that I should focus on? Also, is this a recent trend?

For now, you can focus on existing users moving away from Word. You can consider all users of Word in any geography, and this has been a trend in recent years.

Got it, thanks. To begin with, we need to define the key metric we are examining. In this case, the metric is the drop rate – the rate at which users are discontinuing their use of Microsoft Word. To gauge the approximate drop rate, we would analyse usage data over time. We could look at:

- Active Users: Measure the decline in the number of active users on a monthly or yearly basis.
- Subscription Cancellations: Track the number of subscription cancellations or non-renewals.
- Usage Frequency: Monitor how often users are accessing and using Microsoft Word compared to previous periods.

For example, if we see a 20% decrease in active users over a year, that would be a significant drop rate indicating that users are moving away from the product.

What factors do you think could be involved in this drop?

Several external factors could contribute to the drop rate:

1. Increased Competition:
    1. Rise of free or more affordable alternatives like Google Docs, Zoho Writer, or Apple Pages.
    2. These alternatives often offer real-time collaboration and cloud storage integration, which can be more appealing to users.
2. Changing User Preferences:
    1. Growing preference for tools that offer seamless integration with other applications and platforms.
    2. Increasing use of mobile devices and the need for mobile-friendly applications.
3. Complexity and User Experience:
    1. Perceived complexity of Microsoft Word, especially for users who need only basic word processing features.
    2. User interface might be considered outdated or cluttered compared to competitors.
    3. Overwhelming number of features that are not relevant to the average user, leading to a steep learning curve.
4. Pricing and Accessibility:
    1. Higher cost of Microsoft Office subscriptions compared to free alternatives.
    2. Licensing and access restrictions that can be cumbersome for users with multiple devices or platforms.

Given these insights, what solutions would you propose to improve the situation?

To address the drop rate, I would propose the following solutions:

1. Simplify the User Experience:
    1. Redesign the user interface to be more intuitive and less cluttered.
    2. Introduce a “lite” version of Microsoft Word that offers essential features for basic users.
2. Adjust Pricing Strategy:
    1. Offer more flexible pricing plans, including a more affordable subscription tier.
    2. Consider freemium models where basic features are free, and advanced features are paid.
3. Focus on User Feedback:
    1. Regularly gather and act on user feedback to make necessary improvements.
    2. Conduct user surveys and usability testing to understand pain points and address them.

By implementing these solutions, we can aim to reduce the drop rate and retain more users, ensuring Microsoft Word remains competitive and user-friendly in the evolving market.

Excellent analysis. Thank you for walking me through your thought process.

Uber Drivers

Drivers on Uber have fallen down by 2%. What would you do?

Ok. I would like to ask a few clarifying questions before proceeding. What does 'drivers falling' mean? Is it the number of drivers signed up on the platform or is it the number of active drivers during a period?

Take the metric impacted to be active drivers per day. Active driver is someone who is logged in and willing to accept rides for at least 2 hours in a day.

Alright. For how long have we seen this adverse impact?

This drop started around 2 months back.

Was this drop sudden or has it been gradually going down for 2 months?

It has been gradual.

I am assuming we are debugging this for India. Has this metric dropped throughout India? Or has just a particular geographic region been affected?

We have just seen this drop in Hyderabad.

Has this affected drivers across different mobile devices and OS?

Yes.

Alright. I will state the facts of the case once. Active drivers on Uber have been gradually falling for the last 2 months. This drop is limited to Hyderabad. Drivers using all different mobile devices and OS have been impacted by this. Can I proceed with this?

Yes. Please go ahead.

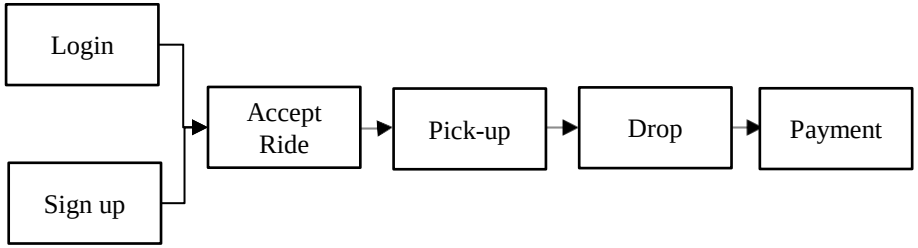
Alright. I am assuming that our releases are not localised to particular regions. Since this issue is limited to Hyderabad, I think that it is unlikely to be caused by any server deployments or new app releases.

That is correct. We don't have releases for specific regions.

One thing I can think of is some changes to our vernacular support but even that should have impacted all the Telugu speaking regions, not just Hyderabad. So, I would be constructing the user journey of drivers using our platform and see if any metric is down for the drivers operating from Hyderabad.

Sure. We can put a filter of the Hyderabad pin code on our analytics dashboard.

Here is the user journey for our drivers.



Login
- # Login failures per day
- Avg. time for login

Sign Up
-# Sign-up drop offs/day
- Avg. time for each sign up

Accept Ride
- # trip requests per day
- Avg. idle time
- Acceptance rate per driver

Pick-up
- Avg. distance to pick-up per trip
- Avg. time to pick-up per trip

Drop
- Avg. trip distance
- Avg. trip duration

Payment
- Avg. income per month per driver
- Avg. earning per trip
-Avg. base fare per trip
-Avg. tip received per fare
-Avg. surcharge per trip
- Avg. time to receive income from Uber

Starting off with logins and sign-ups, have we seen any issues there? An increase in failures, drop-offs or time taken to complete the process might frustrate drivers into leaving the platform.

All these are fine. We have observed no problems in our login or sign-up flow.

Uber Drivers

Next step in the journey is the driver accepting a ride. Is there any change in the number of trip requests per day? This could be due to a demand side issue. The reduced requests for trips could be forcing drivers away from the platform as second order effects.

The number of trips requested per day has not gone down in this period. It is just the drivers which have reduced.

Have we seen any change in driver behaviour? Increase in idle time or reduction in trip acceptance rates would suggest that drivers are slacking off or feel that the trip is not worth taking.

We have observed no significant change in these metrics.

The next stage in the journey for drivers is pick-up and drop. I am looking at metrics of distance and duration for pick-up as well as the trip. An increase in these metrics might make the trip more tedious for drivers.

This metrics are fine as well. What else could be the reason?

The last stage in my funnel is the payment to the drivers. Have we seen a decrease in average monthly driver income?

Yes, we have seen a decrease in this.

Since the number of requests and acceptance rate is unaffected, the total trips must still be the same for drivers. Hence this reduction must also have led to a decrease of earnings being received per trip.

Correct! Average earning per trip has indeed reduced.

Further, this can be attributed to earnings through the base fare or tip from customers. Have we seen a change in either?

Tips are unaffected. They have always been low. However, the fare given by Uber to drivers are come down.

Have we made any specific changes to our payment policy in Hyderabad?

Yes! Now that you mention it, we did change the incentive structure. We have a component called 'late night safety fee' for our drivers in many cities driving between 10PM – 6AM. As per our policies, Hyderabad has now been declared as a safe city. As a result, this incentive has been removed.

Got it! Just to validate this, is the drop in active drivers much more significant during late night (10PM – 6AM) as compared to the peak hours (8AM – 12PM & 5PM – 9PM)?

It indeed is. The drop is severe late at night. Other times of the day are relatively unaffected. How can we solve this?

Since the active drivers per day is the metric impacted by this, there must be drivers in Hyderabad who drive exclusively at night. Losing these drivers is an issue. I believe there should be some incentive for drivers to work late at night.

Firstly, instead of a binary classification of safety, I propose a safety index for cities. Fares late at night should be a function of this index. When cities become safer, there would be a relatively lesser reduction in revenue for drivers rather than the huge decrease which happened right now. We can run a few experiments with different incentive levels to get an estimate of price elasticity.

Secondly, we can change the incentive structure and provide this incentive bundled after rides rather than on each ride. This would encourage drivers to take more trips while limiting the costs for Uber.

Thirdly, we could pass on some of this cost to users by surging prices during late at night. These are my recommendations .

Sounds good! That would be all. Thank you.

Zepto Delivery Time

Zepto in Bangalore is observing breaches. Can you please diagnose the issue?

Sure. Before I begin, I would like to ask some clarifying questions. What do we mean by breaches here? How do we measure these breaches at Zepto?

For Zepto, we measure average delivery time per order. We are seeing a rise in the same.

Okay. How much is the rise observed in average delivery time per order?

We are seeing about 25% increase in the delivery times.

What categories are we observing these breaches in? Is it in a particular category like grocery, stationery, etc. or across all categories?

The rise is seen across all categories.

Is there a specific region of Bangalore in which we are seeing this increase or is it across Bangalore?

Good question. We are observing this in few parts of Bangalore. Other regions seem to be doing fine.

Alright. Lastly, for how long have we seen these breaches? Are these gradual or sudden?

We have seen this increase for a month now. Apparently, our dashboards show a step increase all of a sudden.

Okay. To begin with, I would first like to analyse some broad external factors that might impact our delivery commitments. I would then go ahead and understand the user journey of our delivery system to look for any issues at any of the steps. Does this seem okay to you?

Yes. The structure looks good. Please go ahead.

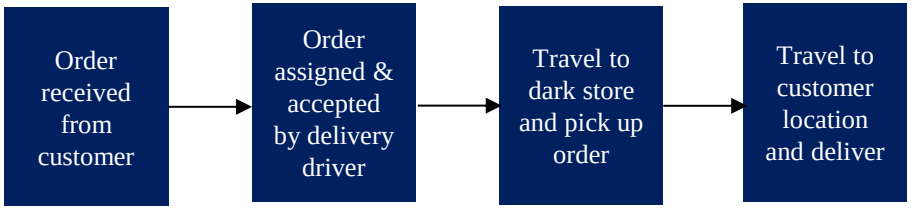
Thinking of seasonality as an external factor, we might see this rise because of rains or bad weather conditions in particularly our affected areas of Bangalore. Do we have any such observation?

No. The weather is good and there has been no drastic change compared to last month across Bangalore.

Okay. There are a few other factors that come to my mind. There can be some government interference affecting regional small stores that we serve from, impacting our supply chain. This rise could also be due to closing of some stores in those regions making our delivery guys take longer to procure items. Does any of this seem to be a probable reason?

These are interesting factors. But there are no such issues that have been observed and everything looks good.

Alright. I would then go ahead and analyse the user journey of our delivery system to get to the root cause of the issue.



Orders received
-Order Processing Time
-Order Error Rate
-Customer Order Confirmation Time
-System Uptime/Downtime

Order assignment & acceptance
- Avg. Order Assignment Time
-Acceptance Rate
-Average time to accept / order
-Reassignment Rate

Travel
- # stores covered /delivery guy/ day
-Avg. distance travelled / store
-Avg. time of travel / store

Pick up
- Dark Store Efficiency
Avg. time to confirm pickup / store

Travel & Deliver
-Avg. distance travelled / order
-Avg. time from pickup till delivery / order

Zepto Delivery Time

I have broken down the entire journey into 4 steps and listed a few metrics that I would check to see which part might be causing delays in delivery. Before we check the metrics, I would like to clarify the context of an order pickup. Delivery people might either go first to a dark store to pick up the order and then straight to the customer location or pick up orders for multiple customer by optimizing their route and then proceed towards end delivery. Should I consider multiple orders or a single order delivery for this problem?

Great question. For simplicity, let us assume the use case of a single order delivery.

Okay. So, I will keep the journey simplistic with respect to order pickup and analyse the metrics. Since the issue is observed in only a few parts of Bangalore, server-side error rates or any technical issues should not be the problem. Has any of the listed metrics across the journey been impacted for that particular region over the last month?

All the metrics look fine on the dashboard except the last one which you talked about. The average time from pick-up till delivery per order has gone up.

Okay. With this understanding, I would like to deep dive into just this part of the journey to find the root cause. Also, I would like to get an idea of the kind of increase in delivery time we are observing. From what I know, Zepto gives a 10 to 15 minute delivery assurance. So, a breach of 25% means the deliveries are delayed by 3-4 minutes on an average. Does my approach seem okay?

It's great that you brought out some numbers in the discussion. The approach is okay. Please go ahead.

Sure. Now, the time spent from pickup till delivery can be further split up into two parts – pickup till landmark of the location and landmark to exact address. For the first part, I can think of a few factors responsible. There might be some construction work in progress in these regions causing traffic to slow down. It might also be possible that drivers have started stopping enroute for some refuelling or refreshments. Does any of this seem to be the reason?

Good points. But, all these do not seem to be a problem in these regions. Think about how a typical end user location is in your campus at IIM Bangalore and then analyse.

Sure. I'll analyse the second part of this delivery time keeping a typical IIM Bangalore customer in mind. From my experience here, a significant amount of time gets consumed in the main gate in verifying the delivery guy details and directing the driver inside the campus, which increases the delivery time. Should I go ahead with to analyse it further?

Yes. This looks good. Please go ahead and extend it to a typical customer in Bangalore

Now, in Bangalore, a significant number of people live in mid-to-large-sized apartments with security guards at the main gate. Now, it is possible that these 3-4 minutes get wasted due to delivery guys talking to the guard, entering the apartment, and finally reaching the flat or house of the customer. Are we seeing any delays with respect to dealing with guards and reaching the customer?

Yes, we are. We came to know that the apartments in these regions do not have a smooth entry process for delivery agents. Hence, our guys waste about 5 minutes in the formalities with the guards and get delayed reaching the customer. That was a very thorough analysis from your side. Thank you so much.

Instagram App Opening Rate

There is a 15% drop in the Instagram app open rate. Tell us what happened.

Before moving ahead, I would like to ask some clarifying questions.

1. What do you mean by app open rate?
   - Does this imply that users are not opening the app at least once a day, indicating an increase in inactive users? Or are users simply not opening the app multiple times a day, leading to a reduction in the overall number of app opens?
2. Is it happening across geography? Is it across gender?
3. Is it on a specific OS - Android or iOS?
4. Is it happening gradually, or did it happen suddenly?

The Instagram app open rate refers to the number of times the app is opened daily through various means such as notifications, ads, and links. This trend is observed across different platforms, regions, and genders and occurs gradually over time.

Okay, I have a question regarding the metric measures:

1. Is the tool used for logging the data functioning correctly?

2. Have we changed the method or pattern used to measure the Instagram app open rate?

Yes, the tool used for logging data works correctly and we have not changed the method or pattern used to measure the app opening rate.

I will outline my approach to identify the root cause. I will first examine external factors such as competitors and new product launches. Then, I will consider internal factors, including tech releases, design changes, business decisions, and user journeys on the mobile app.

Okay, go ahead with the approach.

Promotional campaign: Was there a promotional or marketing campaign previously driving inorganic growth in the app open rate, which has now ended, resulting in a decline in the app open rate?

System Outage: Was there a system outage that prevented users from accessing the Instagram app at certain times of the day, leading to a drop in the app open rate?

Competitors: Has any new competitor entered the market?

Government Regulations: Did the government introduce any new compliance policy or law that Instagram failed to comply with, resulting in a temporary ban on the app and a drop in the app's open rate?

Bad PR: Was there any backlash on social media against Instagram, or was there some negative sentiment online that contributed to the drop in the app's open rate?

New OS: Was a new version released recently by Android or iOS causing compatibility issues with the Instagram app? If so, did this lead to an outage for users and a subsequent drop in the app's open rate?

App Store/Play Store update: Did Instagram fail to comply with the latest rules, regulations, or policy changes of the Android/iOS app stores? As a result, was the app banned for a certain period, leading to a drop in the open rate?

These are interesting factors. But no such things have happened. Please move to the internal factors.

App Crashes: Is the app crashing upon opening, preventing users from proceeding and using the app, leading to a drop in the open rate?

Bugs: Have we discovered any bugs in the app that are causing it to behave abnormally?

Features: Have we launched any new features that users may not have liked or removed/modified existing features that were popular? As a result, users are moving away from the app, leading to a drop in the Instagram app open rate.

Instagram App Opening Rate

UI or UX changes: Have we recently made any UI or UX changes that users may not have liked? As a result, are they moving away from the app, leading to a drop in the open rate?

Business strategies: Have we made any changes in the Business strategies that justify the drop?

This drop is due to a change in the behavior of users. Let's deep dive there.

Let's understand how users use the app. Users use the app for the following things-

- Feed interaction - checking feed or stories, likes, and comments
- DM - chat with friends
- Check friends - profile/photos
- Checking friends of friends - profile & follow them
- Posting a photo / video upload / take a photo apply filters adding captions (text, hashtag, location)
- Edit user bio or profile picture

We can say that the majority of users are consumers, with a much smaller percentage being posters/creators. Therefore, most user time is spent on feed interactions, while other activities are less frequent.

To check the above activities, we can check the following metrics:

- Number of posts scrolled
- Time spent on feed and stories
- Liking a post and stories
- Comments on post
- Replies on stories
- Stories opened
- DM - chat
- Number of user page opened in a session

Possible reasons for the lower open rate could be:

Case 1: It takes time to load new posts. In such cases, there would likely be reviews on the Play Store, App Store, or social media.

Case 2: The content in the feed/stories is no longer interesting or exciting, leading to fewer daily or repeat opens.

Okay, please dive down into case 2.

Feed/Stories feature posts from friends, social media influencers, and celebrities. Influencers and celebrities play a significant role in driving the app's open rate, as their frequent posts create a strong pull effect on users.

To evaluate this case, we can use the following metrics:

1. For less engaging content:
   - Trend in the number of likes and comments on influencer posts
   - Time spent on each post/story
2. To check if the Influencer hasn't been active
   - Number of posts per day for top influencers (top 1000)

Great. The number of posts generated by Instagram influencers is declining weekly. What do you think is the reason behind this reduction in posts by Instagram influencers?

Instagram influencers have reduced their posting frequency on Instagram, possibly because they find creating posts tedious and time-consuming compared to other platforms or have discovered easier platforms for content creation. To address this, we can develop an MVP incorporating Gen AI tools to help influencers create content more quickly.

These are some insightful suggestions. Your approach was very thorough. Thank you for sharing your thoughts.



Pricing



Pricing - Approach

How would you price ABC product?

- **Product Clarity**: What is the product? Don't hesitate to ask the interviewer clarifying questions. You should be sure about the product and its capabilities before answering the question.
- **Business Goal**: Your business goal determines the kind of pricing strategy you should use. Is the objective of setting a price to increase market share or profit or something else? A lower price increases market share, while a higher price increases profit. Having said that, do keep the characteristics of the product or company brand in mind. If ABC is a premium product, you shouldn't be slashing prices suddenly.
- **Market Evolution**: Is the market still forming (emerging), have dominant players already emerged (growing and mature), or is the market already in decline?
- **Pricing Strategy**: There are 3 main strategies for pricing – cost plus pricing, value-based pricing and competitor-based pricing. When the market is still forming, customers have not yet formed an opinion of the products, therefore companies can set prices based on cost and value perception. Once major players have formed, pricing will need to be benchmarked from the competitors.
- **Cost Advantage**: If you are looking to reduce the prices, be sure to think of whether the company has cost advantages over competitors? If no, is this a temporary measure to attract users? Without cost advantages, a company cannot compete by continuing with lower prices, because this would just continue to erode profits.
- **Product Differentiation**: If you are looking to increase prices, what are the additional features that the product already has, or you would offer in the future to justify the higher prices?
- **Risks**: How will all the stakeholders respond to your pricing? For analysing the impact of price changes on customers, knowing the elasticity of demand would be a great tool. Also think of what the reaction of your suppliers and competitors would be.

How would you price the family subscription pack for JioSaavn?

Just to clarify, JioSaavn is an already existing music streaming service in India. They already have individual subscriptions and are looking to introduce a family pack. Am I understanding this correctly?

Yes.

How many accounts would be present in the family pack?

We would adhere to 5 accounts in a family pack just like others in the industry.

Got it. We are playing in a mature market consisting of major players such as Spotify, Gaana and YouTube Music. Do we have a specific goal in mind? Do we want to wrestle users away from our competitors? Are we trying to increase our revenue? Or is there something else we are working on?

We want to increase our market share.

Got it ! Since we want to increase market share, one strategy could be to price our subscription lower than our competitors to entice users. This also synergizes with Jio's historical approach of cost leadership. We should keep a reduced price initially, attract a lot of users and then increase prices. Music streaming has some lock-in characteristics because users create their customized playlists on the app. Moving to a different app would mean doing a tedious job of creating these playlist again.

To come up with the numbers, can you help me with the prices of individual and family subscription packs of our competitors. I would also like to know the price of our own individual subscription pack.

JioSaavn's individual subscription price is Rs 100/month. Why don't you try to estimate the prices of our competitors.

Alright. I will take the competitor Spotify. Spotify should be priced at a premium as compare to JioSaavn due to its superior brand reputation as well as cutting edge recommendation algorithm. Let's assume that Spotify's individual pack is priced at Rs 120/ month. There are generally 4-5 people in the family. Hence, buying accounts for all of them would incur 4-5 times the cost. However, for the family pack to be appealing, its price would be significantly lower than that.

My estimate would be that the family pack price would be around 2 times the price of the individual pack. Does that make sense?

Your estimate of Spotify's individual pack is pretty close. I liked your reasoning for Spotify's family pack too. However, in reality, the family pack is priced around 1.5 times the individual pack.

Understood. Using this conversion factor of 1.5 times, are pricing for JioSaavn's family pack should be Rs 150/ month. However, since we want to wrestle away users from our competitors, we can also go for an aggressive strategy by pricing the family pack even lower.

At the same time, we wouldn't want to compromise too much on our potential revenues. Is there a prior research on the price elasticity in the market for music streaming? This would help us understand the price range in which we would still be able to earn a marginal profit from each additional user.

There has been no such analysis done prior to this.

In my opinion, before setting the exact price level, we should conduct a user survey to have a better understanding of how the market would react to our price.

Another thing we can look at is how we could reduce our costs to justify the lower prices. This can be done by optimising data storage and computing costs, making changes to human resources or by having partnerships with music series and record labels.

You followed a good approach for this question. Thank you!

Smart Speaker

How would you price Apple Home, the smart speaker from Apple?

Just to be clear, Apple Home is a smart speaker which in addition to its music playing capabilities, has an AI assistant which can take voice commands and fetch data over the internet. Am I correct with this?

Yes, you are correct. One thing I would like to point out – Apple Home can also connect with and control the smart home devices in the house..

Got it. From what I understand, these capabilities make it a competitor to smart speakers such as Alexa and Google Home. Is Apple Home a new launch by Apple or is it already present in the market?

We are looking to launch Apple Home in the next quarter. We hope to wrestle away at least 20-30% market share from other players in the market

Got it! Since this is a new product launch, one strategy could be to price Apple Home lower than our competitors to capture market share. However, historically, Apple's strategy in pricing has been to maintain high differentiation in their products in order to ask for premium prices.

Consumers associate Apple's brand with high-premium prices because they are used to Apple's innovative product designs and quality. Therefore, pricing a future Apple Home device should follow this same high-differentiation-premium price strategy to preserve the brand's leverage. To come up with the numbers, can you help with the products and prices our competitors Google and Amazon offer?

Sure, let me give you the prices of our competitor products. Google Nest is priced at Rs 7K while Google Nest Mini is priced at Rs 5K. For Alexa, we again have multiple versions ranging from at Rs 5.5K to Rs 10K.

Got it. First of all, just like our competitors, I would introduce 2 variants of Apple Home in the market. Offering a lower priced device would allow us to hit competitors both in the low end and high end of the market. This strategy is especially useful in India where most of the users are price sensitive. This would entice customers who want a premium Apple product at a lower price.

Secondly, I want to calculate the premium Apple charges for its products to arrive at the exact price point. Just to have an estimate of the kind of premium Apple charges, let's compare the Google Pixel 8 and Apple iPhone 15. Now they have comparable specs but from the prices I remember the iPhone price is 30% more. So, we can take around 30% as the premium for Apple products over similar Google products. Using this we come at the price level of Rs 6.5K for Apple Home Mini and Rs 13K for Apple Home.

I love the fact that you brought in so many numbers into the analysis. You mentioned that Apple justifies its higher price through product differentiation. What are the features you would offer as differentiation?

First let's talk about the key features of Alexa and Google Home. According to me, Alexa's key value proposition is its integration to Amazon's online store. This makes online shopping with Alexa seamless. For Google Home, the integration with Google's search engine is the key highlight. I would not want to compete with these devices on their strong points.

1) For Apple iPhone, the voice assistant Siri has been a huge success and hence becomes an obvious place where we can innovate. I would enhance personalisation in Apple Home. Usually, the device is used by multiple people in the house. Apple Home should store data mapped to different users and give personalised replies. These different users should be identified by their voice.

For example, let's take two users in a house who have had a historical interest in sports and politics respectively. When they get up in the morning and ask Apple Home to "Tell me the latest news", they should be given different answers.

2) Another of Apple's strong points has been its innovative design. There are numerous stories of Steve Jobs not going ahead with the design of iPhone and iPod until it was absolutely perfect. We should do something similar in Apple Home as well.

That was a thorough analysis. I really liked your approach. Thank you!

Amazon Prime

How would you price Amazon Prime?

Just to be clear, Amazon prime is a bundled service that offers multiple benefits like Amazon Prime Shipping Benefits, Prime Deals and offers, Prime Video, Prime Music, Credit card rewards and Prime Reading. Am I correct with this?

Yes, you are correct.

Are we focusing on any specific geography for this decision?

We are focusing on India.

Got it! Are we aiming to maximize our market share, or are we focusing on maximizing our profits with this pricing strategy?

Our goal is to attract more customers and boost our market share.

Got it! Then we should offer our service at a competitive price in order to penetrate the market and acquire customers. Can I go ahead with this assumption?

Yes, you can proceed.

There are primarily three methods to come up with a price point.

- First is the **Cost-based pricing** model. This is based on the costs incurred by Amazon to provide the services.
- Second is the **Value-based pricing** model. This is based on the customer's perceived value of the services provided.
- Third is the **competitor-based pricing** model. In this model, the price charged by the competitors for similar serviced is used to arrive at the price.

Since, Amazon Prime offers a bundle of services it is harder to find the cost incurred. So, I would like to use a combination of the Value-based and the Competitor-based models to arrive at the price. Can I go ahead with this method?

Sure, you can continue.

With the help of the models, we see that:

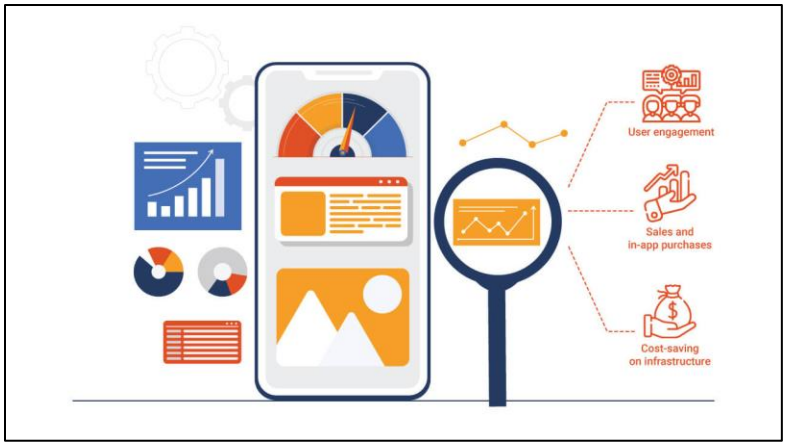
- Prime users get free delivery on all their orders. A typical Prime user orders at least once a month, saving ₹50 per delivery, totaling Rs.600 per year.
- The direct competitors for our OTT service are Netflix and Disney+ Hotstar. They charge around ₹900 per year, giving us the same value.
- The direct competitor to Prime Music is Spotify, which costs around ₹1100 per year, giving us the same value.
- Prime reading is similar to Amazon Kindle, which costs around ₹1800 per year, giving us the same value.
- In addition to these Prime members also get exclusive deals, discounts and offers on credit cards, providing additional value to our customers.

Thus, we arrive at a value of **₹4400**.

The industry standard is that the companies try to capture around 1/3rd of the customer's total perceived valued.

So, we should price our offering at around **₹1500,** this should allow us to improve penetration and acquire market share.

That was a thorough analysis. I love the fact that you brought in so many numbers into the analysis. Thank you!



App Improvement



Google Maps

How would you improve Google Maps? (VMock)

Let me start with a few clarifying questions. What do you mean when you say improve? Is the goal to enhance engagement, acquire more users, or to generate more revenue?

Let's go with engagement

Sure, and do we have any specific platform in mind? iOS app, Android app, or website? Or is it across all platforms?

Around 80% of our users are on Android so let's start with that. We can later expand it to iOS and web.

Sure. So first I'll start with listing down a few kinds of users for the app –

1. **Urban Millennials**: These are young folks in 20-35 years age group who are tech savvy, have high end phones, and good connectivity. They mostly use google maps for same city travel to schools, offices, new spots to visit on weekends etc.

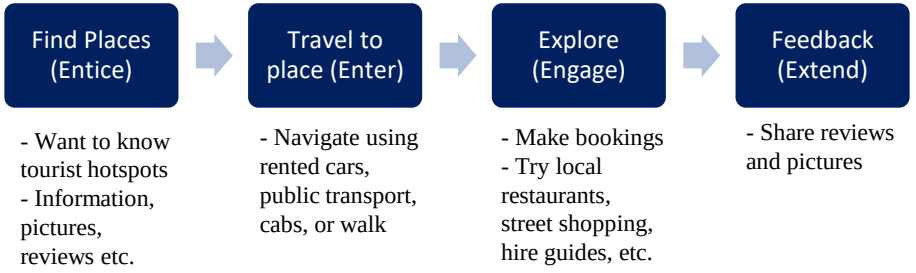
2. **Cab / Delivery Drivers**: The driver fleets for various cab services, e-commerce websites, food delivery platforms use Google Maps daily for their last mile connectivity.

3. **Tourists:** Tourists use google maps to find their way in a new city for finding vacation spots and restaurants.

Among these, I would like to proceed with the Tourists segment as I feel they rely a lot on maps when in a new city. Urban millennials mostly use maps for same city travel and have most frequently used routes memorized. Delivery drivers, although are heavy users, only use the app to reach from point A to point B in the shortest possible time and Google Maps is currently serving that need very well.

That sounds great, please go ahead.

The typical user journey through the app looks something like this -



In this journey, I can think of the following pain points -

1. Trip Planning: It is difficult to narrow down the places to visit while travelling. We can find the list of "top 10 places" online but those are not curated to interests, location, and period of stay of the user. For example, if a person is really interested in history and has a really short stay, they would ideally like to visit the museums.

2. Authentic Experience: People usually look for the authentic local experience when travelling. This may include finding the best places places to try the local food, or some local street shopping. One prominent issue is that they have to spend a lot of time researching the reviews and ratings of restaurants, and even local shopkeepers try to extort more money from foreigners.

3. Managing Bookings: Tourists need to buy tickets for museums, palaces, gardens, public transport, etc from different websites or stand in long queues at the counters. This can be very cumbersome.

4. Group Travel: Often tourists also rent cars to explore around the city. For a large group with multiple cars, it can get quite difficult to keep track of people in different cars.

Why don't you go ahead and tell me the solution for these?

I can club the first 3 pain points and address those using a '**Travel Planner**' feature. The 4th pain point I believe is faced by a small subset of travellers only and can be neglected for now. The Travel Planner will take the following inputs from the user -

- Location of the user: Can be entered manually (for pre-planning) or auto detected in Google Maps (for impromptu plans)
- Interests of the user: Explicitly asked through checkbox options.
- Duration of stay: Explicitly asked

Using these inputs the Travel Planner would create a personalized inventory for the user. It will include all tourist hotspots that align with the user's interests – museums, theme parks, galleries, monuments, street shopping locations, restaurants etc. The users are obviously free to rearrange the places and add/drop suggestions from the generated list. Once the users start their journey, they will be able to navigate to the next destination in the itinerary. After reaching the destination, Google Maps can also display some additional information about the places such as some historical facts, trivia, and reviews from other people who have visited before. There will also be an option to pre purchase tickets (or make reservations) for these destinations directly from the app itself.

Interesting. Do you think there is any way to make money out of this as well?

Currently all features on the app are free for end users so I don't want to add any paid aspect for them. The ticketing aspect however can help in generating additional revenue. Maps can take a listing fee for allowing businesses to sell tickets / make reservations on their platform. Alternatively, they can also charge a commission from the overall transaction value of tickets booked through Maps.

Okay, and how would you measure the success of your feature after the launch?

We could look at the number of users per month using the Travel Planner to generate an itinerary, the number of users who make bookings through maps itself. The revenue generated from commissions can also be noted.

Sounds good. Do you think this feature could impact any existing feature? What would the trade-offs be?

The app has different categories like hotels, restaurants etc. listed below the search bar that people use to find nearby destinations. The click through rate on these categories may drop for people who are using the Travel Planner. However, this shouldn't be a problem as we are still helping people find the places they are looking for.

That's great, thank you.

UPI Application - PhonePe

What is your favourite UPI application and how would you improve it? (AppDynamics)

I'll start with introducing PhonePe so that we are on the same page.
PhonePe is a payments app that meets all payment, investment, mutual funds, insurance and banking needs. They have moved from being a simple payments app to being a complete financial services platform, recently adding wealth management solutions to their portfolio. It has a super app platform called Switch with integration across multiple merchants related to food, travel, lifestyle, etc.
Another interesting product is an interactive website called PhonePe Pulse showing data, insights, and trends on digital payments done through its app.
Their major competitors would be GPay, PayTM, and Cred.

What would be the main differentiators for PhonePe?

PhonePe has not restricted themselves to only UPI and are making financial services products accessible to people, be it insurance, mutual funds, or gold.
For investing needs, they have interesting features like SIP and wealth calculator, Super Funds and multiple other educative resources to help investors.
The UI is very intuitive and clean. The buttons and icons are also well-spaced out with a big font, making them readable and accessible even for elderly.
In terms of the home page, I noticed a difference between GPay and PhonePe. For GPay, the focus seems to be on sending and receiving money and the frequent people we've transacted with being on the top of the home page. On PhonePe, the home page has multiple sections - Recharge, Insurance, Switch, Investment.
Also, compared to GPay, PhonePe has a higher payment success rate.
Can I now proceed with analyzing how they can improve their product?

Yes, please proceed.

I would like to know what is the goal of the app improvement? Improving user engagement, increasing market share, increasing user retention or increase revenue?

What is the most pressing issue in the payments space according to you?

The digital payments market is mature but has less customer loyalty and stickiness.
Considering this, I would like to consider user retention as the goal of improvement, which would ultimately translate to an increase in revenue.

That sounds fair. Go ahead.

I'll now move on to identifying the user personas for the product. The personas I can think of are:

1. Students

- Age group 15-21, tech savvy, usually transact using online payment channels, use the app to send money or split expenses among friends.
- Saving money is their primary objective and they are attracted to cashbacks and discounts.
- **Pain point:** They are most affected when they face payment failures if bank servers are down, since they don't usually have cash on hand and are in a hurry to complete transactions. They don't find the current offers on PhonePe lucrative.

2. Young Curious Investors

- Age group 21-25 years, have disposable income which they are looking to invest, are new to the investment world and would like to know how and where to invest.
- **Pain Point:** They don't know which funds to invest in. Also, trust on making investments using PhonePe is low.

3. Working Professionals

- Age group 25-40, comparatively loyal to the app, use the app to pay bills, for recharges, buying insurance, booking travel tickets, etc.
- **Pain Point:** Amidst their schedule, they forget due dates for utility bills. They have to use multiple apps for recharge, bill payment and booking needs.

4. Parents/Grandparents

- Aged 55+, not comfortable with using mobile apps, use the app to send and receive money.
- **Pain Point:** They tend to forget the sequence of steps to make a transaction. Also, trust on online payments is low.

Do these personas look fine to you or do you want me to consider anything else?

UPI Application - PhonePe

This seems good, you can go ahead and prioritize a persona.

Sure. Though the parents' segment has a differentiated need and might present as a lucrative segment, they have trust issues. This requires a different approach and the app features wouldn't be able to contribute much to this. I would not prioritize this segment for now. Based on my understanding of the needs of the other users, I would prioritize students and young working professionals, as there is no technological barrier for them. Also, 66% of the Indians fall under this bracket and forms the majority of the market, hence prioritizing makes more sense.

Are you fine with me proceeding with this or would you like me to prioritize a different persona?

The target personas seem fine. How would you solve for the problems?

I will now suggest 2 features for these 2 target segments, plus an additional delight feature that can cater to all our customers and act as a differentiator.

1. Reward Points/Cashback: Assured points after a certain transaction amount and vouchers across more categories (like Cred). The points can be redeemed at a certain conversion rate. However, this might burn some capital and collaborations with brands need to be sorted out.

2. Reminders & Auto-Payments: Payment reminder notifications can be sent to users. If somehow the users ignore this, then payment mandates to top-up PhonePe wallet and make the payments on behalf of the user can be done to ensure that the credit scores are not affected.

3. (Delight) Fail-proof transactions: Sometimes, UPI transactions fail in crunch moments. Many users end up paying double the amount and have to haggle for refunds later. This is particularly problematic in situations where the receiver is a person with whom we won't be transacting again (like public transport/roadside stalls).

For such situations, PhonePe can create a **suspense wallet** in the backend, which will act as a middle layer between the sender and the receiver. When a transaction is initiated, the amount is credited to this wallet first and then sent to the receiver's bank. In case the transaction fails, the suspense wallet would retry the payment after some time, however, the receiver would receive a notification stating that the transaction has been made. This is similar to durable execution seen in workflow engines like Temporal.

This seems innovative, but difficult to implement.

The efforts would be high initially, but once implemented, this can result in nearly 100% transaction success rate. We can also gatekeep this technology and keep it exclusive for PhonePe users. Later on, as the competition catches up, the ecosystem becomes better and it's a win-win for all customers.

Fair, what all metrics do you plan to track?

I would look at the user retention rate, the daily and monthly active users and the number of transactions on an average per customer.

For the delight feature, I would look at transaction success rate and number of complaints related to payment failures.

That sounds alright. Thanks, we are done with the case.

Microsoft Powerpoint

You are tasked with improving Microsoft Powerpoint. How will you go about it?

I have some initial clarifying questions. To ensure we're aligned, could you please clarify what you mean by "improvement"?

By "improvement," we mean enhancing the overall user experience, addressing pain points, and possibly introducing new features that align with user needs and market trends

Before we begin, I'd like to start by understanding and clarifying the context and main use cases of Microsoft PowerPoint. PowerPoint is a widely used presentation software that helps users create, edit, and present slideshows. The primary users include professionals, educators, and students. They use PowerPoint for business presentations, academic lectures, project reports, and more. Does this overview sound accurate?

Yes, that's a good overview. What would be your next step?

Next, I would define the different user personas to understand their goals and pain points.

Persona 1: Business Professional - John

John is a 35-year-old marketing manager who frequently creates presentations for internal meetings, client pitches, and marketing campaigns. John seeks efficient tools to create visually appealing and persuasive presentations quickly. John often struggles with time-consuming design tasks, lack of advanced graphics options, and difficulty collaborating with team members in real-time.

Persona 2: Educator - Dr. Smith

Dr. Smith is a 45-year-old university professor who uses PowerPoint to create lectures and academic presentations. Dr. Smith needs to create engaging and interactive presentations to keep students interested. Dr. Smith finds it challenging to incorporate interactive elements and multimedia seamlessly. He also faces issues with version control when updating slides for different classes.

Persona 3: Student - Emily

Emily is a 21-year-old college student who uses PowerPoint for class projects and presentations. Emily aims to create professional-looking presentations that can impress her professors and peers. Emily often struggles with design consistency, finding suitable templates, and integrating multimedia content.

Those personas are well-defined. Which one of these personas would you prioritize?

I'd like to prioritize solving for John, the Business Professional, as his needs seem to align closely with maximizing productivity and collaboration, which are critical for business success. John as a user needs more flexibility in customizing presentations with unique, non-template-based designs and would also benefit from better tools for gathering and incorporating feedback from team members and stakeholders.

Proposed Features

1. **Custom Design Studio**
   - **Feature**: An advanced design studio with tools for creating bespoke designs, including custom shapes, animations, and transitions.
   - **Benefit**: Provides John with the flexibility to create unique, task-specific presentations without relying on pre-made templates.

2. **Enhanced Integration with Business Tools**
   - **Feature:** Deeper integration with CRM systems (like Salesforce), project management tools (like Trello or Asana), and data visualization platforms (like Power BI).
   - **Benefit:** Allows John to pull data and insights directly into his presentations, keeping them up-to-date and relevant.

3. **Advanced Feedback and Review System**
   - **Feature**: A feedback system that allows team members to leave comments, suggest changes, and rate different sections of the presentation.
   - **Benefit**: Streamlines the review process and ensures that all feedback is consolidated and easy to implement.

Great! These seem like some interesting features. Please go ahead with how you would go about prioritizing them and what kind of metrics will you track to judge their performance?

I would like to then prioritize the listed features based on the value they would generate for a business professional and the complexity required to build it.

Feature	Value	Complexity
Custom Studio	Medium	High
Business Tool Integration	High	High
Adv Feedback and Review System	High	Medium

Based on the Value-Complexity prioritization, I would prioritize the features in the following order :

1. Advanced Feedback and Review System
2. Enhanced Integration with Business Tools
3. Custom Design Studio

These features I have decided based on what would generate the most benefit for our user persona John. Regarding the Metrics for the features, I would go for the following ones for each of them

1. Advanced Feedback and Review System
    - # Feedback instances per presentation
    - User satisfaction scores with the review process
    - Reduction in Revision Cycles
2. Enhanced Integration with Business Tools
    - # Integrations used per user
    - Increase in Productivity Metrics (e.g time saved, data accuracy)
3. Custom Design Studio
    - Usage rates of custom design tools
    - # New design tools created per user

That was a really detailed breakdown of your prioritization and metrics. Thank you for such an elaborate discussion!

Music Streaming - Spotify

Critique and Improve an application you like (Atlassian)

I would like to choose Spotify which is a music streaming app. Elaborating a little more on the problems Spotify addresses. As a music lover, I wish for unlimited song access without buying, discovering new music aligned with my tastes, and match my mood with the music I listen to.

Sure. Please go ahead. Can you start by mentioning some use cases of Spotify?

When I have a particular thing in mind, like a specific genre, artist, podcast, or my favorite songs, I simply open Spotify and go straight to it. On the other hand, when I'm in the mood to explore and discover new music, I use Spotify to find something fresh or to come across a new favorite artist.

That sounds good enough. Now please move ahead

Spotify has many different kind of users and before I go ahead and critique it, I would first like to define different users and their goals and pain points. After that I would prioritize one of the personas and eventually solve for it. Does this sound fair or should I include something else?

That is a fair approach. Please proceed.

Sure. I would like to define 3 personas along with their goals and pain points

Persona 1: Music Enthusiast - Emily

Background: Emily is a 24-year-old music enthusiast who loves discovering new artists and genres. She spends several hours a day listening to music on Spotify during her commute, work breaks, and leisure time.

Goals: Emily seeks a seamless and personalized music experience. She wants to explore a wide variety of music and discover new tracks and playlists tailored to her taste.

Pain Points: Emily finds the ads on the free-tier version disruptive and considers upgrading to Premium for an uninterrupted listening experience. She also wants more effective ways to discover lesser-known artists and emerging music trends.

Persona 2: Casual Listener - Alex

Background: Alex is a 30-year-old casual listener who enjoys music but doesn't have much time to explore extensively. He mostly uses Spotify during workouts and while doing household chores.

Goals: Alex wants a simple and user-friendly app that allows him to quickly access his favorite playlists and play music without much hassle.

Pain Points: Alex occasionally finds the app overwhelming with its extensive features. He sometimes struggles to find specific playlists or songs he added previously and desires an intuitive search and navigation system.

Persona 3: Independent Artist – Sarah

Background: Sarah is a 28-year-old independent musician who uses Spotify to share her original music with a wider audience. She relies on the platform to gain exposure and connect with potential fans.

Goals: Sarah aims for greater visibility and engagement for her music. She wants fair compensation for her streams and opportunities to feature in Spotify's playlists to reach a broader audience.

Pain Points: Sarah finds it challenging to get her music noticed among millions of tracks on the platform. She's uncertain about how to promote her music effectively and desires more insights into her audience's preferences.

That was a thorough approach. Please proceed to the next step.

I would like to prioritize Sarah as a user for recommending features as there as very less apps that focuses on content creators. The budding artist as she could gain a lot by Spotify providing her a platform to collaborate, engage and promote her art journey thereby increasing her reach among people.

That is fair.

Now I will be defining features to solve Sarah's issues. I would like to define 3 features.

Promotion & Discovery Opportunities: Introduce a new program called "Spotify Spotlight" that showcases hand-picked independent artists to a wider audience. Spotify's editorial team and AI-driven algorithms will curate these selections based on the quality of music and listener engagement. Artists like Sarah, who demonstrate promising talent and engagement, will have a chance to be featured in Spotify's official playlists and in-app banners.

Music Streaming - Spotify

Personalized Artist Support and Resources: A dedicated support hub offering resources and guidance on how to maximize their presence on Spotify. This could include tutorials, best practices for music promotion, and personalized advice based on their performance data. This will help artists like Sarah navigate the platform more effectively and optimize their strategies for success

Fair Compensation and Monetization Options: Enhanced monetization options, including tiered subscription models for exclusive content, concert ticket draws, and a "tip jar" feature where fans can support artists directly. This will help provide artists with additional revenue streams, ensuring fair compensation and more control over their income.

Those were interesting features. Thank you for your time.

Fashion App - Myntra

Take any fashion ecommerce website / app of your choice and improve it

Alright, before I begin solving the question, I have some clarifying questions

Sure, please ask your questions

What is the primary target demographic of the app? Are there any areas of the app that have received negative feedback recently? What are the business goals you hope to achieve with these improvements? Are there any budget or technical constraints we need to be aware of?

Our primary target demographic is millennials and Gen Z. Recently, we've received feedback about the browsing experience and checkout process. Our key goals are to improve user retention, increase conversion rates, and enhance overall user satisfaction. We do have some budget constraints but are open to significant changes if they promise a good ROI.

Thank you for the information. Based on my experience, I would like to focus on improving the user experience of Myntra, one of my favourite fashion e-commerce apps.

Sounds good, please go ahead.

I'll first start by defining user personas, then I'll prioritize one of the personas, define its pain points, come up with certain features and use RICE score to prioritize those features. And finally, I'll define success metrics.

That sounds like a good approach, you may start

Sure, I'll start by defining the personas:

1. Priya, 21, is a **fashion enthusiast** who loves staying updated with the latest trends, shops frequently, values high-quality images and detailed product descriptions.
2. Namrata, 30, is a **busy professional** who shops occasionally, prefers quick and efficient shopping experiences, values fast delivery and easy returns.
3. Akash, 35, is a **bargain hunter** and looks for the best deals and discounts, shops during sales, values price comparisons and reviews.

That's a decent list of personas, if I ask you to come up with 2 more, what would you say?

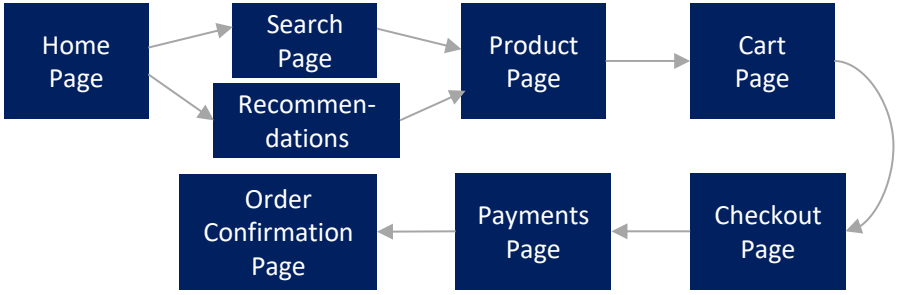
4. Rishabh, 34, is **eco conscious** and prioritizes sustainable and ethical fashion, looks for eco-friendly brands and products, values transparent supply chains.
5. Tanvi, 27, is **tech-savvy** and loves integrating technology into her shopping experience, values innovative features like AR, VR, and AI-driven recommendations.

Great, please continue

I'd prioritize Namrata, the busy professional as she shops occasionally so there's scope to increase her shopping frequency and this segment also denotes a significant proportion. Fashion Enthusiast Priya already has a high shopping frequency and eco conscious and tech savvy customers may represent a minor segment. Due to the above reasons, I'll go ahead with Busy Professional Namrata

Alright, I've understood your logic, Busy Professional Namrata sounds good, please proceed

Sure, I'll create a user journey to identify the pain points in the entire funnel



That user journey looks quite complete, could you now list some pain points?

Sure, following are the pain points:

1. On the **Home Page**, there's an overwhelming number of recommendations, lack of context for personalized recommendations and limited visibility of ongoing offers and deals.

Fashion App - Myntra

2. On the **Browse/Search Page**, filters are not intuitive or responsive enough, difficulty in finding specific styles or brands, slow load times for product images and lack of inspiration for users unsure of what they want.
3. On the **Product Page**, there is inconsistent quality of images, limited interactive features, lack of user-generated content and difficulty in understanding the fit and size.
4. On the **Cart Page**, lack of dynamic pricing alerts, no collaboration features for shared shopping experiences and limited cart management tools can lead to bad experience for the user

Alright, I think these many pain points would be enough for now, you may move ahead to creating solutions to solve these pain points

Sure.

- **Brand and Style Shortcuts**: Introduce quick access buttons for popular brands and styles.
- **Progressive Image Loading**: Use technology that loads low-resolution images first and then high-resolution ones to speed up browsing.
- **Contextual Recommendations**: Show recommendations with a brief explanation of why they are being shown, e.g., "Based on your recent search for formal wear."
- **Inspiration Boards**: Create curated boards based on current trends, seasons, or events.
- **Frequent Purchases Shortcut**: Add a "Frequently Purchased" button for quick access to commonly bought items.
- **360-Degree Product Views**: Offer a full 360-degree view of products.
- **Collaborative Cart**: Allow users to share their cart with friends/family for opinions or joint purchases.
- **Smart Cart Insights**: Provide insights such as "Frequently Bought Together" or "Complete the Look" suggestions directly in the cart.

Awesome! Can you now prioritize these features and select the ones you'd implement first and why?

Sure

Feature	Reach	Impact	Confidence	Effort
Brand & Style Shortcuts	M	M	H	L
Progressive Image Holding	H	H	H	L
Contextual Recommendations	H	H	H	M
Inspiration Boards	M	L	L	M
Frequent Purchases Shortcut	M	L	M	L
360-Degree Product Views	H	H	M	M
Collaborative Cart	M	M	M	H
Smart Cart Insights	M	M	M	M

I have used basic tshirt sizing for RICE scores above based on which I'd prioritize **Brand & Style Shortcuts**, **Progressive Image Loading** and **Contextual Recommendations** as all these three features have high confidence and low-medium effort.

Great, what would be some success metrics that you'd look at to see how well your improvement features are doing?

- Conversion Rate
- Customer Satisfaction
- Checkout Abandonment Rate
- User Retention (DAU / MAU)
- Return Rate

Great, we can close the case now

Dating App - Tinder

We would like you to pick any dating app and improve it.

I would like to choose Tinder, a popular dating app. Elaborating a little more on the problems Tinder addresses: As someone seeking romantic connections, they desire an efficient way to meet new people, discover compatible matches, and have a fun and engaging dating experience

Sure. Please go ahead. Can you start by mentioning some use cases of Tinder?

Yes! When I want to meet new people, whether for a relationship or a casual date, I open Tinder and start swiping through profiles. When I'm in the mood to chat and connect, I use Tinder to message my matches and get to know them better.

That sounds good enough. Now please move ahead..

Tinder has many different kinds of users. Before I go ahead, I would like to define different users, their goals, and pain points. After that, I would prioritize one of the personas and eventually solve for it. Does this sound fair, or should I include something else?

Fair approach. Please proceed

Persona 1: The Desperate and Single
Background: Individuals in their 20s, divorcees, or LGBTQ members seeking relationships or casual companionship. They may feel pressure to enter a relationship due to age, social, or peer influences.
Goals: They aim to find a relationship while exploring and trying various options.
Behaviour Patterns: They frequently swipe right, exploring numerous options.
Pain Point: These individuals have not had much success with relationships in the past and are often driven by a desire for instant gratification.

Persona 2: The Casual Dater
Background: Young adults, often recent high school graduates, not interested in serious relationships or matching deeply with companions. They seek casual companionship.
Goals: They wish to meet numerous people before considering a potential relationship.
Behaviour Patterns: They are curious, often use multiple dating apps, and casually swipe right to explore options.
Pain Point: They struggle to find others who seek the same casual, no-strings-attached relationships.

Persona 3: The Relationship Seekers
Background: Adults in their mid-20s to early 30s who are financially stable with fewer family or social responsibilities. They are mature and capable of making responsible decisions.
Behaviour Patterns: They are sociable, willing to face complex situations, and often rely on their intuition while swiping.
Pain Point: They often find it challenging to meet individuals and who share similar interests and personalities.
Goals: To meet as many people as possible until they find the “ONE”

I would like to choose "The Desperate and Single" as the focus because they represent users looking for serious relationships but who struggle due to past experiences and instant gratification tendencies.
Now I will define features to solve "The Desperate and Single" persona's issues.

1. Dating Advice and Guidance:
- Implement a feature that offers personalized dating advice and emotional support. This could include tips on creating a more appealing profile, advice on engaging in meaningful conversations.
- Provide access to articles, videos, and resources related to building healthy relationships

2. Date Planner:
- Feature that allow users to search for unique and interesting date options based on location, category, date, and pricing. Users can browse through a curated list of date ideas, including activities, restaurants, events, and experiences, and filter them according to their preferences and budget.
- Simplifies and enhances the date planning process, leading to more enjoyable and successful dates.

3. Icebreaker Challenges
- A series of fun and interactive challenges designed to help users break the ice and get to know each other better. These challenges can range from trivia quizzes to collaborative tasks, with rewards such as profile boosts or special badges.
- Provides a playful and engaging way for users to initiate conversations and build connections, making the dating process less awkward and more enjoyable

Dating App - Tinder

How will you prioritize among these?

For prioritizing features, I will look into the value added by the feature and the complexity of implementation for Tinder. The order :
Value: How beneficial the feature is to the user, including its impact on user engagement and satisfaction.
Complexity: How difficult and resource-intensive the feature is to develop and implement.

Feature	Value	Complexity
Dating Advice and Guidance	High	Medium
Date Planner	High	High
Icebreaker Challenges	Medium	Medium

How would you measure the effectiveness of the "Dating Advice and Guidance" feature?

For effectiveness, I would look into 5 stages:

1. **Acquisition:**
   - Number of Users Accessing Advice: To gauge initial interest and adoption of the feature.
2. **Activation:**
   - User Engagement with Advice Content (e.g., reading articles, watching videos): To measure how effectively the feature captures user attention and prompts them to interact with the content
3. **Retention:**
   - Repeat Usage Rate / month: To assess how valuable and engaging the advice is in encouraging users to return and continue using the feature.\
4. **Referral:**
   - Shares and Recommendations
5. **Revenue:**
   - Conversion Rate to Premium Features

Those were interesting features. Thank you for your time

Food Delivery App - Swiggy

We would like you to pick a food delivery app and focus on improving it. (Infoedge)

I would like to choose *Swiggy*. It is a food delivery app which focuses on elevating the quality of life of Urban consumers by providing them convenience through food delivery. It has recently started its grocery delivery service too. It is considered to have a duopoly in the market and covid has led to changing habits of people which caused an overall increase in the size of the pie. This phenomena along with regulations has led to new players trying to enter the market.

That was comprehensive. Please proceed further.

Is there a specific goal I should be able to achieve through the improvement features?

No, you are free to define the goal.

Swiggy is a three way marketplace with 3 types of users - restaurants/grocery stores, end consumers and delivery partners. I would want to focus on end consumers, will that be okay?

Sure. Please proceed.

It has verticals of food delivery, instamart and Swiggy Genie, are we specifically looking for one of these?

You can focus on one or more, it is solely your choice.

Thank you for answering all the clarifying questions. I would like to define the scope of the problem before I proceed. I would like to focus on customer retention metrics and revenue. Since it is a popular name in the market with increased demand for delivery, customer acquisition doesn't seem to be a problem anymore. While defining improvements, I would also like to focus on enhancements which would serve as a competitive advantage. Will that be okay?

Sure. Go for it

Since we have defined the goal, I would like to define user personas with their needs, goals and pain points. I would also like to focus on how and when do they really use our product. This is important to create value according to the context.

Sounds fair.

User Personas can be defined as follows:

Families at Home: These are families with grandparents, parents, and kids – usually around 4 to 6 people. They usually cook their own meals and rarely order food, except for special occasions or when they want a break once a week. They like food that's good for both kids and older family members. They can afford to spend a bit more for high-quality and healthy food.

College Students: These are students who live away from home in college hostels. They get a fixed amount of money each month. The hostel food isn't the best, so they often try different food options. They're influenced by their friends and tend to eat at cheap stalls near the college to save money and hang out with friends.

Busy Professionals: These are people with busy jobs and big goals. They either don't like cooking or don't have time for it. They earn good salaries and don't have many responsibilities. They often order food about three-four times a week and go out to eat on weekends. They rely on recommendations from friends and colleagues for food choices.

Great! Please proceed.

According to the context I would like to prioritize the user according to customer life time value.

Households don't frequently order food, but their orders are usually larger. This presents a chance to offer them additional grocery items, increasing their long-term value. On the other hand, working individuals tend to order food frequently and prefer the convenience of not going to local stores for daily essentials. Given these insights into user preferences and behaviors, I would like to prioritize **households and working individuals** as the target audience. Is that fine?

Sure. Please proceed

Defining the user journey in various paths to identify the pain points



Food Delivery App - Swiggy



Here are some **challenges** users face and certain beliefs they currently hold, which are affecting their optimal use of our product. Our aim is to transform these existing beliefs into more favorable ones:

- **Perception of Grocery Quality and Cost**: Some users consider buying groceries through Swiggy to be pricey and worry that the freshness might not be up to the mark.

- **Discovering New Places**: Users feel that Swiggy lacks information about recently opened restaurants in their vicinity, especially those offering the types of dishes they usually enjoy.

- **Hesitation due to Hunger**: Users are reluctant to place orders when they're very hungry, as browsing through options and delivery takes a significant amount of time.

- **Emotional Connection with Food**: The platform is perceived as lacking the emotional connection that fancy restaurants provide, appearing more transactional in nature.

- **Complex Ordering for Households**: In households, the process of taking everyone's food preferences becomes a hassle, leading to longer ordering times. This problem escalates when hosting parties or having friends over.

- **Reliability of Ratings**: Users find it hard to trust restaurant ratings since nearly all establishments seem to have high ratings, making the authenticity of these ratings questionable.

You can focus on one or more, it is solely your choice.

First, I would like to define solution to these problems and then prioritize using RICE framework

- **Sampling and Grocery Incentives**: To engage households who order from restaurants, we could offer a complimentary Instamart sample. Coupled with coupons, this could encourage them to explore Swiggy's grocery offerings. After delivery, a push notification can inquire about their experience with the sample and nudge them to visit Instamart for groceries. This might require delivery partners to pick up samples during downtimes, with extra compensation.

- **Restaurant Collaborations and Emotional Content**: Collaborating with new or recently opened restaurants presents an opportunity. Short videos could provide insights into the food, chefs' stories, and even delivery partner experiences. Such content, along with brief restaurant descriptions in menus, fosters an emotional connection. Users could also share this content with their social circles.

- **Voice Ordering**: Integration with virtual assistants like Alexa, Google Assistant, or Siri for hands-free ordering. This would be extremely helpful for adding multiple groceries to instamart cart in an instant.

- **Group Ordering and Scheduled Deliveries**: Introducing a group order button on menu pages, generating a time-limited link for shared orders, adds a sense of urgency. Implementing scheduling features, although complex, would enhance convenience. Balancing item availability, restaurant timing, and delivery partners' schedules is crucial.

- **Variable Rewards for Engagement**: Utilizing a variable reward concept post-order could maintain user engagement. Random coupon or scratch card rewards provide personalized discounts and offers, deviating from generic promotions.

In the next step I would like to prioritize these features.

Prioritization using the **RICE framework** is a logical step. Ensuring win-win-win situations, where consumers, restaurants, and delivery partners all benefit, aligns with Swiggy's approach. Each proposed project or feature will be assessed based on its potential positive impact on all stakeholders.

Sure. Sounds good. Please proceed with the roadmap.

Food Delivery App - Swiggy

Objectives:

3 Months: Product Roadmap and Immediate Enhancements

In the shorter term, spanning 3 months, our product roadmap will come into sharper focus.

6 Months: Business Metrics and Objectives in Focus

Within 6 months, we will have clear visibility on our business metrics and set goals. This step is pivotal in aligning our efforts towards achieving growth and sustainability. By monitoring these metrics closely, we can gauge our progress and make informed decisions.

18 Months: Vision for Evolving Key Levers in Food Delivery Business Model

Over the span of 18 months, we will be focusing on understanding and shaping the fundamental drivers of our food delivery business model. These key levers will evolve to adapt to changing market dynamics and customer needs. During this period, we will also strategize around competition and navigate regulatory considerations related to commission charges and customer data sharing.

Actions

3 Months: This will include introducing integration with virtual assistants and providing concise yet impactful descriptions of restaurants. This effort will be complemented by curating engaging content showcasing newly opened restaurants, sharing chef and delivery partner stories.

6 Months: Boosting User Engagement and Revenue: As we approach the 6-month mark, our attention will shift towards enhancing user engagement and revenue generation. Exciting features such as scheduled orders, complimentary items, and scratch cards will be integrated. These additions are aligned with our core objective of retaining users and driving increased revenue.

18 Months: In-depth Exploration and Potential Group Orders Feature: At the 18-month juncture, we will delve deeper into understanding customer needs. An exploration phase, involving customer feedback and possibly a landing page, will help us determine the demand for a Group Orders feature. Given its potential impact, this feature will be a long-term consideration.

Throughout this roadmap, we are committed to maintaining a balanced approach. We will incorporate quick wins, strategies to move key metrics, significant initiatives, delightful user experiences, well-considered experiments, and the ongoing evolution of our technological capabilities. By following this comprehensive approach, we will ensure our roadmap caters to a holistic range of objectives and aspirations.

That was an elaborate and comprehensive solution. Thank you for your time

Note: These kind of questions might require success metrics as well. Please ask the interviewer if he/she wants you to define success metrics. Keep the success metrics both a mix of negative and positive metrics.

Ride Hailing App - Ola

We would like you to take a lead on improving ride hailing apps like Ola.

Of course, but before diving into the solutions, I'd like to clarify what exactly you mean by "improve"? Could you please elaborate on the specific aspects of user engagement you're looking to enhance?

Sure, by improvement, we mean increasing user interactions, app usage, and overall satisfaction with the platform.

Thank you for the clarification. In that case, I would use the CIRCLES method to brainstorm and produce some potential solutions. One idea that comes to mind is to enhance accessibility for specially-abled persons. For instance, we could introduce a product viability feature for the visually impaired, making it easier for them to use OLA's services independently.

That sounds interesting! Can you explain how this feature would work and how it fits into the customer journey?

Absolutely! I would like to use AI for this feature. The AI-driven product viability feature could use image recognition technology to identify nearby vehicles and read out relevant details like car type, license plate number, and driver information. This feature would be particularly helpful during the booking process, enabling visually impaired users to locate and identify the right vehicle, improving their overall experience with OLA. By ensuring inclusivity and accessibility, this feature would create a positive impact on the customer journey, fostering customer satisfaction and loyalty.

That's a great initiative. How do you plan to implement it, and what other features do you have in mind

To implement this AI feature, we would collaborate with accessibility experts and organizations for the visually impaired to ensure the technology meets their specific needs. Additionally, we would conduct user testing and feedback sessions to refine and optimize the feature before rolling it out to a wider audience.

As for other features, I would focus on enhancing the overall user experience throughout the customer journey. For example, we can introduce eco-friendly rides for creating a positive perception of the app and in-ride entertainment options so that it enhances passenger experience.

Those sound like excellent additions. How would you prioritize these features, and what metrics would you use to measure their impact?

Prioritizing these features would involve considering both their potential impact on user engagement and the resources required for implementation. I would suggest starting with the AI-driven product viability feature, given its significant positive impact on inclusivity and accessibility, aligning with OLA's commitment to a customer-centric approach.

As for eco-friendly rides, we can first offer a differential pricing strategy, which could be something similar implemented by Swiggy and Zomato and check the number of rides booked and increase in the app usage by having incentive schemes for customers using eco-friendly rides. For in-ride entertainment, we can track the percentage number of users and average time spent on them to assess customer satisfaction. We can continuously monitor which features are used the most and update the features regularly to get more traction.

Your approach seems well-thought-out and customer-focused. We appreciate your insights and look forward to seeing these improvements in action. Thank you for your thoughtful response

Thank you! I'm excited about the opportunity to contribute to the improvement of OLA and enhance user engagement. I believe that by focusing on the customer journey and incorporating AI-driven features, we can make a meaningful impact on the overall user experience.

Amazon Kindle

How would you improve Kindle?

Before I start, I would like to ask a few clarifying questions. What do you mean when you say 'improve'? Do we want to enhance engagement by making users read more, acquire more users or increase sales of Kindle books?

Target engagement. We have noticed that a lot of users make the purchase and start reading books on Kindle. However, we see a large number of drop-offs. People stop reading and never finish their books.

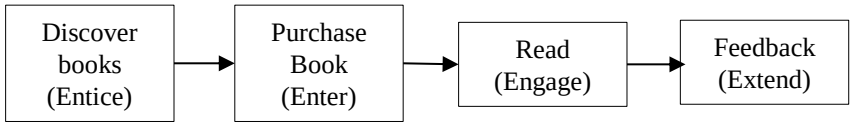
Great! Let me list down a few users who could be using Kindle.

- **Passionate readers**: These people have been avid readers for a long time. They actively look for new books to read by looking online lists and reviews and take suggestions from their friends.
- **Educational readers**: These are school or university students who have a digital copy of their course book on Kindle. They go through specific chapters as per their syllabus, highlight important portions, and make notes in a physical copy.
- **Occasional readers**: People who have lost the habit of reading books and want to start again or new readers altogether. They might lose concentration or interest while reading.

Among these, occasional readers are the most appropriate target segment. They would find reading for a long duration difficult and would need a push from us to go on. Passionate readers don't need any external motivation to continue to read. They are constantly looking for new books, but this is solved largely through Goodreads, which is already integrated with Kindle. Goodreads provides a social network where people can read and post reviews and ratings for books and look for what others are reading. Although Kindle can potentially have a lot of new features for educational readers, I feel the market is really small. Hence, I would proceed with occasional readers as my target segment. Does that seem okay, or should I think more about user segments?

These segments and your logic for the target segment sound good to me. Please go ahead!

Sure. Just give me a moment to think through the user journey for this segment.



- Search for popular books in favourite genre - Ask friends	- Read summary - Purchase through Amazon or Goodreads	- Track progress - Explore meaning of difficult words - Bookmark for reading later	- Share reviews, ratings with fellow readers - Get recommendation

Within this journey, for an occasional reader, I have identified a couple of pain points

1. **Unsure about the purchase**: Since this segment rarely reads books, the users think a lot before making a purchase. Oftentimes, ratings, reviews and a summaries might not be enough to convince them to purchase the book. We could provide a preview of all books on Kindle, which has the first a couple of chapters of the book.
2. **Lack of motivation**: New readers can find reading books cumbersome. Books can have portions where the author is setting up the story and hence a lot of action might not be happening. This can cause users to drop off since they are not used to reading books.

Sorry to interrupt you but the first pain point you mentioned is already being solved. If you go to Amazon, you can download a "sample" for your Kindle. Can you continue to brainstorm and come up with some other pain point?

Sure. Another issue is the fact that these users might have gaps in between reading. When they start reading the book after such a gap they would need a summary of what happened up until now before they continue reading. This would be similar to the recaps we have before TV episodes. However, the content creation for this would be really tricky.

Fair enough. Let's focus on motivation. How can we encourage users to comeback to read?

I propose a form of **gamification** to make it interesting for the readers. We can have badges that users need to earn. These badges could reward users for their reading speed, streaks such as the number of consecutive days read or the number of novels read by an author or in a genre, reading at a particular time of the day (early riser for morning readers or night owls for those reading late at night). We can use the integration with Goodreads to create a leaderboard where users can see where they stand with their friends or globally.

We can also add other features like:

- AI-powered notifications
    - Summary Notifications: Send AI-generated summaries or intriguing excerpts from the last chapter to remind users of the story and create curiosity for the next chapter. Example: "Last time on your Kindle: Detective Jane found a mysterious clue. What's next? Dive back in to uncover the truth!"
    - Reading Reminders: Personalized reminders based on users' reading habits and preferred reading times.
- Daily Reading Goals: Allow users to set personal daily reading goals.
- Integrated Audiobook Option in the Kindle. Offer seamless integration with Audible and Ensure users' progress is synced across text and audio formats.

Interesting. Do you think this could encourage users to not actually read and just go through books for the sake of appearing higher on the leaderboards?

That is something which I believe would not be a widespread problem. These scores and streaks are just for the personal satisfaction of users; they don't bring any monetary or other benefits. So, I think most users would not be gaming the system. Additionally, we would be tracking the metrics such as reading speed and hours taken to complete the book. If they differ significantly from the usual data (let's say beyond 2 or 3 standard deviations from the mean) we would flag this and not give any points or streaks to the user.

Fair enough. What other metrics would you look at to measure the success of the gamification feature?

The most important thing to measure is engagement. For this I would look at the time spent per user per month on Kindle. For ensuring that this increase in time spent is due to the gamification feature, I would also track the number of badges and points earned per user.

I noticed that you are using time spent as the north star metric for engagement. Why not metrics such as words read per user per month or pages read per user per month?

Since we are encouraging a lot of new readers to read more on Kindle, metrics such as words or pages read per month per user might actually go down since these new readers would be relatively slower readers.

That's a great analysis! Thank you.

Youtube

We would like you to take the lead in improving YouTube. How would you approach this task?

Before I begin with the solution, I have some clarifying questions

Yes, go ahead

When you say “Improve Youtube”, is there any specific aspect of “improvement” you’re looking at? Or are there any areas of user engagement I should focus on?

For the scope of this question, you can choose how you want to define improvement, you can consider any product metric and work towards enhancing it

Alright, thanks! I would work on improving user engagement and retention since Youtube is a well-known app and currently focusing on top of the funnel metrics such as awareness and discoverability may be deprioritized.

Sounds good, please go ahead.

In context of improvement. Would you like me improve the platform as a whole or I should focus on some specific features. Do you want me to solve for specific users? Are there any constraints on resources?

You should focus on YouTube’s free website platform, you can solve for any user group. There are no other constraints.

Thanks for your inputs. I would start by defining the user personas.

1. Content Creators

These are users who upload videos regularly or occasionally, driven by either monetary gains or personal enjoyment. They form the foundation of the platform's value. However, due to YouTube's exceptional success as a content-sharing platform, creators might find limited alternatives.

2. Content Viewers

These individuals engage with YouTube for diverse purposes, including education, entertainment, news, and gaming. Their presence generates revenue, which serves as a crucial incentive for creators to continue using YouTube as a platform to share their content.

3. Live Streamers

This group comprises both individuals and groups who live-stream activities like gaming, news, sports, and national events. The popularity of this media form has surged recently. While YouTube faces competition from platforms like Twitch & Mixer, which have gained prominence in the live streaming realm, my primary goal is to prioritize user retention over new user acquisition.

That seems like a good list, please proceed

Prioritizing user groups against two criteria on a scale of 1 (lowest) to 3 (highest)
User Impact — How beneficial is it going to be for the user group? Can we make their experience significantly better?
Potential Monetization/Size — Looking at the broader scope to think about which segment has more users. For example: if we had 100 users on our platform and segment A had 95 users, and B had 5, we would probably target the 95 users first.

User Group	User Impact	Potential Size
Content Creator	2	2
Content Viewer	3	3
Live Streamer	1	1

Based on the above analysis, I’d prioritise *Content Viewers*

Can you list down the pain points of these viewers and your solutions?

Pain Points:

- Users often find YouTube's recommendation algorithm to be a "black box" and feel they have little control over the content suggested to them.
- Users with niche interests struggle to find relevant content amidst the more popular mainstream videos.
- The educational content on YouTube lacks interactivity and structure, making it difficult for users to follow a learning path.
- Users experience inconsistent video quality due to fluctuating internet speeds.
- Inadequate and slow response to inappropriate content and comments.

Youtube

I'd prioritise **algorithm transparency and control** and **content curation for niche interests** as other pain points are sort of related to these ones. Addressing these two pain points would lead to the alleviation of other pain points as well

Great, can you quickly come up with solutions to address this challenge?

Following are a few solutions:

- **Recommendation Weight Dashboard**: Allow users to access a dashboard where they can see the factors influencing their recommendations, such as watch history, liked videos, and subscribed channels. Include sliders or toggle buttons for users to adjust the weight of different factors (e.g., more educational content, less entertainment).
- **Social Sharing**: Allow users to create and share their own playlists and recommendation sets. Integrate a social component where users can follow playlists from friends, influencers, or topic experts, enhancing content discovery based on trusted sources.
- **Niche Interest Column**: A dedicated section for niche interests where users can explore curated playlists and channels specific to their chosen topics.
- **Creator Tagging**: Tagging for Creators: Creators can create tag for themselves such as "Travel & Lifestyle", "Tech", "Design", etc and users can filter based on their preferences.
- **Personalized Notifications**: Personalized notifications for new content in niche interest areas. Users can subscribe to specific tags or keywords related to their interests. Whenever new content matching these criteria is uploaded, they receive a notification or see it highlighted in their feed.

How would you prioritize among these solutions?

I'd prioritise the solutions based on user impact & the effort required (technical, business) to build the feature.

Feature	Effort Required	User Impact
Recommendation Weight Dashboard	H	H
Social Sharing	H	M
Niche Interest Column	L	H
Creator Tagging	L	M
Personalized Notifications	L	L

Based on the above analysis, I'd firstly prioritise "Recommendation Weight Dashboard" even though it has a high effort, but I believe its impact will also be considerably high which would make up for the high effort. By building this feature, we'll be able to improve the overall recommendations we give to our audience which seems like a good effort for a long-term investment.
The second feature I'd prioritise is "Niche Interest Column" as it would cater to the audience that isn't able to find unique and specific content in the flood of oversaturated content by the more famous youtubers. This would also require low effort, which means this could be a quick win for us.

Alright, could you quickly illustrate success metrics for this feature?

Watch Time: Total time users spend watching videos. This helps gauge content popularity.
Average View Duration: Average length of time users spend watching a video. Longer durations indicate higher engagement.
Click-Through Rate (CTR): Percentage of impressions that turn into views.
Views per User: Average number of videos watched per user. Higher numbers suggest greater engagement.
Subscriber Growth Rate: Rate at which new users subscribe to channels.
Frequency of Visits: Average number of times a user visits the app in a given period.
Return Rate: Percentage of users who return to watch more videos after their initial visit.

Thanks for an insightful discussion! We can close the case now

Travel Booking App - MMT

We would like you to improve the MMT Travel Booking application

Great!, I will start by clarifying the understanding of Make My Trip, it is an application where users can go to book flights, domestic or international; it also offers hotel, cab, train and bus booking. You can also book home stays; travel insurance service is also present.

I would like to move forward to goal setting if this description is alright.

Yeah, that seems alright. Let's move on

Okay, can you please tell me if this improvement has any goal and are we targeting to improve any module of the application ?

The goal is to improve engagement and retention, and the module we are looking to improve the flight booking

I would like to go ahead and define some user personas of the application

User Personas	Pain Points
Business Traveler	• Lack of flexibility in last-minute booking changes • Need for quick and reliable customer support • Difficulty in managing and storing receipts for expense claims
Family Traveler	• Difficulty in finding seats together for the family • Lack of child-friendly services and amenities • Complicated booking processes for group travel
Elderly Traveler	• Difficulty navigating complex booking platforms • Limited mobility and need for special assistance during trave • Anxiety about traveling alone or managing logistics

Seems alright to me, what persona are you prioritizing? Also, reasons for your choice

Sure, I would like to prioritize the family traveler persona the reason is because firstly, for business traveler persona MMT already has MMT Biz which covers most scenarios, also the elderly travelers are quite few in numbers to make a huge impact - the de-facto method for them is rely on 3rd party. The family traveler persona brings the element of group bookings, a large average order value, and a larger customer base in numbers.

That sounds reasonable enough, lets move forward to the features

Yes, for the Family Traveler persona we can release a **Family Mode** on MMT that has the following features to address their pain points -

1. Group Booking Support:
- Simplified booking process for families, allowing users to book multiple seats in a single transaction.
- Option to book seats together, ensuring that family members are seated near each other.

2. Sit Together feature:
- Highlight the flights where it is possible for the group booking to sit nearby

3. Child and Infant Ticketing:
- Dedicated sections for booking child and infant tickets with age-specific pricing and requirements.

4. Family Travel Hub:
- A centralized platform for families to collaboratively plan, book, and manage travel itineraries with real-time alerts, expense tracking, and seamless communication for a coordinated travel experience.

5. Family Loyalty Program:
- Develop a loyalty program that rewards families for booking together, with points redeemable for discounts, upgrades, and exclusive offers.

Looks great, can you finalize what feature will you bring to family mode?

I will prioritize the listed features now, using Impact, Confidence and Effort framework

Travel Booking App - MMT

Features	Impact	Confidence	Effort	Score
Group Booking Support	9	8	5	14.4
Sit Together feature	8	7	4	14
Child and Infant Ticketing	7	8	5	11.2
Family Travel Hub	10	9	9	10
Family Loyalty Program	7	7	6	8.2

I would like to prioritize Group Booking support, sit together and child and infant ticketing feature as the first release in the MVP of **Family Mode** on MMT, they will be coherent features which support each other and have a higher reach and impact.

This looks great, can you think of metric for this feature?

- **Feature Adoption Rate:** Percentage of total users who use the Family Mode features. **MAU (Monthly Active Users)**
- **Retention Rate:** Increase in repeat bookings by families using the Family Mode.
- **CRR (Customer Retention Rate)**
- **Average Transaction Value:** Comparison of transaction values before and after the implementation of Family Mode.

Sounds great! That was a comprehensive analysis, we can end the case here.



Guesstimates



Bike Taxis

You are starting a bike taxi company, how many bikes would you require? (Google PM)

At first, we need to determine the scope of operations. Are we starting in a single city or planning to expand to multiple cities simultaneously, like pan-India?

We are just starting out in Bangalore for now.

Okay, so the first step would be to forecast the demand in the city. We can start with the population of the city which is approx. 1.4 crore.

What other factors will you need to get to a number?

We can consider other factors like population density, age and income segmentation of the population, existing public transport options, and even the competitor's market share in the city and how much market we can expect to capture.

How will you use these factors?

We can start with segmenting by age. In India roughly 25% of the population falls in the 0-15 years age group, 40% in the 15-35 range, 25% in the 35-60 range, and the remaining 10% of the people are 60+ years old. Bike taxis will majorly be used by the 15-35 year old age group which includes college going students and working professional. So, the number of potential riders comes out to be 1.4cr x 40% = **56 lakh people.**

Then we can look at income segmentation. In a metro city like Bangalore the distribution of lower, middle, and upper-class people will be approx. 20%, 70%, and 10% respectively. I believe bike taxis will be a viable choice for the middle-class population which comes out to be 56 lakh x 70% ~ **40 lakh people**.

That seems awfully high. Will all 40 lakh people really use bike taxis for conveyance?

No, we can further drill down by looking at the alternative transportation options that people have. Common ones being personal cars/bikes, cabs, autos, bike taxis, metro, and even bicycles. People generally prefer bike taxis for travelling over short distances (upto 10kms maximum) and considering that these bikes can carry only one passenger at a time, a person would prefer a bike taxi out of all available options maybe 10% of times whenever they want to travel. So, the number of rides comes out to be 40 lakh x 10% = **4 lakh.**

Finally, we also need to consider the existing competition. There are many prominent companies already providing bike taxi services in Bangalore. Initially it will be difficult to capture a sizeable share of the market, but with the right kind of marketing and promotional offers, we can aim to capture 10% of the market. That comes out 4 lakh x 10% = **40,000 rides daily**.

How will you get to the number of bikes required from this?

Considering that bike taxis will be used for distances upto 10kms, which might take 20-30 minutes on average to complete depending upon the traffic conditions, it is safe to say that a rider may only complete 2 rides in an hour. Also considering that a rider works 10 hours a day on average, the total number of trips completed by a rider in a day comes out to be 20.

So, to get the total number of bikes required, we can simply divide 40,000 / 20 to get **2000 bikes**.

That seems alike a fair estimate. But what about maintenance, repairs, and having some bikes as backups?

Yes, these are important considerations, and I would recommend adding around 10% to the estimated number of bikes to account for repairs and maintenance of some bikes. The total number of bikes required in that case will come out to be **2,200 bikes**.

Seem practical. That would be all.

Age Group	Split
0-15	25%
15-35	**40%**
35-60	25%
60+	10%

Income Group	Split
Lower Class	20%
Middle Class	**70%**
Upper Class	10%

Transportation Mode	Preference
Walking	5%
Bicycle	5%
Bike Taxi	**10%**
Autos	20%
Cab Services	20%
Metro	15%
Personal Car/Bike	25%

Note: Segmentation breakup followed

Storage for Google Photos

Can you give a quick estimate on how much storage is needed by Google Photos? (Google PM)

Sure. I would like to clarify a few things before I begin. Are we focusing on just photos, or are we considering videos as well?

You can consider only photos.

Are we considering only clicked photos or we are looking at received photos too?

Consider both.

Should I consider Gmail users or only Android user? Some Gmail users might not be using Android.

You can consider only Android.

The approach I will be following is, I will be estimating the number of users based on segment. I will multiply that by number of photos. Then multiply it with total population, multiplied number of Android users.

That's a good approach.

Total users = 2.4 Billion
Average photos per user : 3 segments
- Heavy User = 5 clicks/week
- Medium User = 2 clicks/weeks
- Low end user = 1 click/week

Assuming Heavy users are 20%, medium users are 50 % and low users are 30%

Seem practical, go ahead.

Now let's assume 1:3 ratio for click to receive photos
Hence for clicked pictures = (.2*5+.5*2+.3*1)*52 = 260
No of photos received = 260*3 = 780
Average photos per person = 780+260 = 1040
No of photos = 2.4 Billion * 1040
we can take 1:3 ratio above to estimate photos received are smaller
Smaller = 200Kb, average photos clicked = 2Mb

Sure, go ahead.

No of photos = 2.4 Billion * 1040 ~ 2.5 Trillion
we can take 1:3 ratio above to estimate photos received are smaller
Smaller = 200Kb, average photos clicked = 2MB
Dividing 2.5 Trillion in ratio 1: 3 and multiplying by .2MB and 2MB
= 2.5*.25*.2 + 2.5*.75*2 ~ 3750 petabytes

Seems practical. That would be it. Thanks!

Note : If historical photos is considered it will be a lot higher , this is only for this year

Number of Uber Drivers

Can you quick estimate on how many uber drivers are there in Bangalore? (Samsung R&D)

Sure. I'm assuming we are estimating the total number of uber drivers that serve Bangalore in a day. Is that correct?

Yes you are right please proceed.

I'll calculate the number of drivers during the peak demand hours, let's say office hours. People are coming back from work on a Friday night.

At a high level, we need to determine how many rides are demanded per peak hour and how many rides a driver can complete within that hour. We can estimate the number of drivers needed by dividing the total demand by the rides per driver per hour.

Uber Drivers = (Drivers / Ride/Hours) / (Rides/Hour)

Sounds about right.

The total population of Bangalore is close to 1.4 Cr.
More than 50% of the population is between 20 and 60.

People in the age group of 20-60 years = 50% * 1.4Cr. = 70L
Out of the assuming 60% are working professional = 60%* of 70L = 42L

Sounds like a fair estimate, go ahead.

Not every one of them will use the cab assuming the following number-

1. No transport — 5%
2. Bus — 30%
3. Personal Vehicles — 20%
4. **Cabs — 40% = 40% * (42L) = 17L approx.**
5. Others — 5%

There are other cab services like Ola, Rapido, and Meru since Uber and Ola are the top 2 players.

Assuming Uber Market share = 40%
Uber Customer = 40% * 17L = 6.8L

Some people might be sharing cabs with friends and colleagues.
On an avg. we can assume 2 people per ride.

No. of rides = 6.8L/ 2 = 3.4L approx.

Sounds good.

Let's say it takes 30 min to complete a ride.
So, a rider makes 2 trips in an hour.

Therefore, the number of Uber drivers = 3.4L/2 = 1.7L drivers.

Seems practical. That would be it. Thanks!

Google Drive Storage for Singapore

Estimate the total storage needed if 200 GB free storage is to be given to people using Google Drive in Singapore

Sure, I have a few clarifications before I begin. Are we considering all residents of Singapore or a specific segment like adults or internet users?

Consider the Internet users

Okay. Should we assume that every eligible person will use the full 200 GB of free storage?

You can make reasonable estimates.

Alright. Should we also consider any potential limitations, such as a maximum number of users or an uptake rate?

No specific limitations. Assume maximum possible uptake.

Great. Here are my initial assumptions:

Population of Singapore: Approximately 5.5 million.
Adoption Rate: Considering Singapore being a developed nation, internet penetration is ~80%.
Storage Utilization: Assuming each user will utilize approx. 60% of the total 200 GB of free storage.
Buffer for Variations: Include an additional 10% buffer to account for variations in usage and potential new users over time.

Sounds good. Go ahead with the calculations.

Okay. Based on my assumptions:
Total population: 5.5 million
Internet users: 80% of 5.5 million = 0.8 * 5.5 million = 4.4 million users
Average user utilizes ~60% of the storage
Storage needed = 4.4 million * 0.6 * 200 GB = 528 million GB = 528,000 TB
Buffer storage = 10% of 528000 TB = 52800 TB
So, the total storage needed = 528000 + 52800 = 580,800 TB

Seems reasonable. That would be it. Thanks!

Number of Google Searches

Estimate the number of queries answered by Google per second.

Surely. I have a few clarifying questions before we begin. Is there any particular geography that I should focus on? And also any particular mode of search like via the mobile or desktop?

You can consider the global population, and both modes of search are acceptable.

Thank you. I'll go with a top-down approach starting with the global population, incorporate for certain factors like internet penetration, the market share for google, and frequency of search for different types of users. Does that sound fair?

Sure, can you assign some numbers to these factors?

The global population is close to 8 Bn people but as far as I'm aware, Google search is blocked in China, so that takes away a big chunk of 1.4 Bn away leaving 6.6 Bn people.

Internet penetration globally is ~ 60% and that gives the total number of internet users as 6.6 Bn x 60% = 4 Bn (approx.)

I know there are a lot of other search engines available, but Google holds a big chunk of that market. Is it fair to assume that Google's market share in search is 90%?

Yes, you can proceed with that.

Thank you. So, now I have the total number of users of Google search as 4 Bn x 90% = 3.6 Bn.

Next, I would like to incorporate some user personas because different people will have differing usage. For instance, certain segments like students at a higher level of education, or professionals in fields like research, coding, etc. would use Google search extensively for their day-to-day tasks.

I can segment the population into the following 3 groups -

Description	Split	Number	Queries / day	Total number of queries / day
Power User	15%	0.54 Bn	10	5.4 Bn
Casual User	70%	2.52 Bn	5	12.6 Bn
Occasional User	15%	0.54 Bn	2	1.08Bn
Total		**3.6 Bn**		**19.08 Bn**

From this I can get the total number of queries being searched on Google as ~19 Bn.

To get the number of queries per second we can simply divide this by 24x60x60.

Therefore, number of queries answered by Google per second

= 19 Bn / (24x60x60) = **220K** (approx.)

You've used a good structured approach, but can you think of anything that you may have missed?

Recently we've seen the rise of many AI powered search engines like ChatGPT and Perplexity.AI that people are now shifting to for getting answers to many of their queries. So, the number of queries being answered by Google would have taken a slight hit.

That's right, we have actually observed an average drop of 10% per user for our search engine in recent times because of many more AI tools in the market.

Got it, so the total number of queries answered would come down to 220K x 90% = **200K** (approx.)

Thank you, we can close the case.

Number of Swiggy Orders

Can you estimate the number of orders delivered by Swiggy per hour in India?

Sure. I would like to clarify a few things before I begin. Should I consider both peak and off-peak hours in my estimation, or focus on an average over the entire day?"

You can decide.

First, I'll estimate the number of food delivery app users in India, then determine the average number of orders per user per month, and finally apply Swiggy's market share. Does that sound good?

That is a good approach. Go ahead.

I want to divide the population into 3 segments. Segment 1 having Tier-1 cities of population over 10 million, segment 2 having Tier-2 cities of population over 5 million and segment 3 consisting of Tier-3 cities and towns

Sure, please proceed.

For the top 1-5 cities, I am assuming a penetration rate of 40% for people who order food online. Therefore, 10 million x 5 x 0.4 = **20 million users**

For the top 6-10 cities, I am assuming a penetration rate of 30%. Therefore:
5 million x 5 x 0.3 = **7.5 million users**

Adding another 5 million users from smaller cities, the total is:
20 + 7.5 + 5 = **32.5 million app users**

To estimate the average number of orders per month, I want to divide the usage patterns into four categories: monthly, bi-monthly, weekly and alternate day. Is it okay?

Yes, that is a fair segmentation. You can proceed.

Average users of food delivery apps:
Monthly : 40%
Bi-monthly: 30%
Weekly: 20%
Alternate day: 10%

Average number of orders per customer per month:
0.4 * 1 + 0.3 * 2 + 0.2 * 4 + 0.1 * 15 = **3.3**

Total number of orders per month:
32.5 million * 3.3 ~ **105 million orders**

Number of orders per day:
105 million / 30 ~ **3.5 million orders**

Assuming Swiggy has a market share of 50%, number of Swiggy orders per day:
0.5 * 3.5 = **1.75 million orders per day**

Number of Swiggy orders per hour in India:
1.75 million / 24 = **70,00 orders per hour (approx.)**

Seems practical. That would be it. Thank you.

Number of Google Meets

Estimate the total number of Google meets that happen daily

Before I start with the solution, I have some clarifying questions

Yes, go ahead

Is there any region or country we're focusing on, or should we look at it globally? And are we considering only professional meetings or personal as well?

To estimate the total google meets accurately, it would be better if we focus on a global level. Whether to consider professional & personal, that's a choice you can make

Alright, thanks! I believe that Google meets are used more professionally, in corporate settings, whereas for personal use, apps such as Whatsapp, FB Messenger and FaceTime are preferred. For the above reason, I'd consider only professional use of Google meets to do an estimation.

That sounds like a decent logic, please go ahead

I'll solve the problem in 4 steps which are as follows:
Step 1: Estimate the number of Google Meet users
Step 2: Estimate the total number of professional Google Meet users
Step 3: Estimate the frequency of use of the app
Step 4: Calculate the total number of Google Meets per day

If the approach seems okay, then I'll start with the calculations

Yes, the approach looks good. Please proceed

Step 1: Total number of Google Meet users

- Total World Population ~ **8 billion**
- Internet Penetration Rate ~ **60%**

Therefore, Internet Users: **8 * 0.6** = **4.8 billion**

- % of Internet users with a Google account ~ **70%** (Combining all Google accounts such as Gmail, Youtube, etc)

Google Account Users: **4.8 * 0.7** = **3.36 billion**

Step 2: Estimating the total number of professional Google Meet users

Assuming ~**25%** of Google account users use Google Meet for professional purposes. This includes employees, businesses, educational institutions, and organizations.

Professional Google Meet users = **3.36 billion*0.25** = **0.84 billion** or **840 million**

Step 3: Estimating the frequency of use of the app

- Assume an average professional user joins **10 meetings per week** in a 5-day work week
- Average number of professional Google Meets per day per user:

Professional Google Meets per user per day = **10** meetings per week / **5** days = **2 meetings / day**

Step 4: Calculate the total number of Google Meets per day

Total professional Google Meets per day = **840 million users** * **2 meetings per day** = **1.68 billion**

Therefore, the total number of **Google Meets** that happen in a **professional** setting **daily** is **1.68 billion**

Great, we can close the case now.

Number of WhatsApp Chats

Estimate the number of WhatsApp chats occurring in India in a day?

I have some initial clarifying questions. We are focusing on chats and not on messages? Also chats includes individual chats, group chats or both?

Yes, we are focusing on the number of chats and it covers both individual and group chats.

I would like to start by categorizing the Indian population based on urban and rural segments. Then, I will estimate the number of WhatsApp users in each segment. Finally, I will estimate the average number of chats per day for each group, considering both individual and group chats. Is this approach alright?

Sounds like a comprehensive approach, please proceed.

Population of India is approximately 1.4 billion people. Considering the urban and rural split to be 40% and 60%. The urban and the rural population will be around 560 and 840 million. Now, assuming smartphone penetration rates in urban and rural India to be 80% and 60%, we get 448 million urban smartphone users and 504 million rural smartphone users. Does this seem reasonable?

Yes, you may continue.

Considering the popularity of WhatsApp, I will estimate that around 90% of urban smartphone users and 70% of rural smartphone users have WhatsApp, resulting in around 403 million urban WhatsApp users and 352 million rural WhatsApp users. If this sound reasonable, I would like to further categorize urban and rural WhatsApp users into three age group: under 18, 18-45, and over 45. Should I proceed with this approach?

Yes, please continue.

Considering the population distribution to be approximately 40% under 18, 40% between 18-45, and 20% over 45, we get:

1. Number of urban WhatsApp user aged <18: 161 million
2. Number of urban WhatsApp user aged between 18-45: 161 million
3. Number of urban WhatsApp user aged >45: 81 million
4. Number of rural WhatsApp user aged <18: 141 million
5. Number of rural WhatsApp user aged between 18-45: 141 million
6. Number of rural WhatsApp user aged >45: 70 million

Does this breakdown seem reasonable?

Yes, that is a fair segmentation.

Now, I will estimate the average number of chats per day for each segment.

1**. Under 18:**
   Individual chats: 15 per day
   Group chats: 5 per day
   Total chats: 20 per day

2. **Between 18-45:**
   Individual chats: 10 per day
   Group chats: 3 per day
   Total chats: 13 per day

3. **Over 45:**
   Individual chats: 5 per day
   Group chats: 2 per day
   Total chats: 7 per day

Does this seem reasonable?

Yes.

Now, calculating the total chats per day for each segment, we get:

1. Chats in age group, under 18:
   $(161+141)\times 20 = 6.04$ billion
2. Chats in age group, 18-45:
   $(161+141)\times 13 = 3.93$ billion
3. Chats in age group, over 45:
   $(81+70)\times 7 = 1.06$ billion

Total Number of WhatsApp chats in a day:
$= 6.04+3.93+1.06$
$=$ **11.03** billion WhatsApp chats per day

This was a comprehensive estimate. Thank you.



Miscellaneous



Fake Products on Instagram

How to stop fake product selling on Instagram?

I have some initial clarifying questions. First, What types of fake products are most commonly sold on Instagram, and is there any existing system in place to combat fake product selling?

Let's say clothing accessories and there is no existing system.

One more clarifying question, What is the primary goal for Instagram in addressing this issue? Is it user trust, legal compliance, or revenue protection?

You are free to decide.

I believe the primary goal should be to enhance user trust, as this will indirectly lead to improved legal compliance and revenue protection. Before proposing specific solutions, I'd like to identify different stakeholder groups and their pain points. Is that acceptable?

Sure, please proceed.

I can identify 4 potential stakeholder groups:–

1. **Consumers:** Risk purchasing fake products, leading to disappointment, financial loss, and potential health/safety risks.
2. **Legitimate Sellers:** Face unfair competition and brand damage from fake sellers.
3. **Instagram:** Risks reputation damage, legal issues, and loss of advertising revenue from legitimate businesses.
4. **Brands:** Lose sales and brand value due to counterfeiting.

I'd like to focus on Consumers and Instagram as the primary stakeholders, as addressing their needs will have the most significant impact. If that's agreeable, I can proceed with listing pain points

Yes, that is a reasonable approach. You can proceed.

Pain points for Consumers:

1. Difficulty in distinguishing between authentic and fake products
2. Financial loss from purchasing counterfeit items
3. Potential health and safety risks from substandard fake products
4. Loss of trust in Instagram as a shopping platform

Pain points for Instagram:

1. Reputation damage as a trustworthy platform for e-commerce
2. Legal liability for facilitating the sale of counterfeit goods
3. Loss of advertising revenue from legitimate businesses
4. Increased moderation costs to combat fake product listings

Coming to the features for these pain points –

1. **AI-powered Product Verification**: Implement machine learning algorithms to analyze product images and descriptions, flagging potential counterfeits for review.
2. **Seller Verification Program**: Create a rigorous verification process for sellers, including identity checks and proof of authentic product sourcing.
3. **User Reporting System**: Develop an easy-to-use reporting system for users to flag suspicious listings, with rewards for accurate reports.
4. **Blockchain-based Product Tracking**: Partner with brands to implement a blockchain solution for tracking authentic products from manufacture to sale.
5. **Educational Content**: Create in-app tutorials and warnings about fake products, helping users identify potential counterfeits.

How will you prioritize among these?

For prioritizing features, I will look into the value added by the feature and the complexity of implementation for the company:

Feature	Impact	Complexity
AI-Powered Product Verification	High	High
Seller Verification Program	High	Medium
User Reporting System	Medium	Low
Blockchain-based Product Tracking	High	High
Educational Content	Medium	Low

How will you measure the success of these features implemented?

1. **User Trust Score:** Track changes in user trust through surveys and feedback.
2. **Report Resolution Time:** Measure the average time to resolve reported fake product listings.
3. **Decrease in Scam Incidents:** Monitor the reduction in reported scams and fake product incidents

Thank you, we can close the case now.

Digital Transformation of a Pizza Store

How will you help in digital transformation of a small pizza store?

Thank you for the question. I have some initial clarifying questions. First, How many employees does the shop have?; What's their current order volume (eat-in, take-away, delivery if any)?; What are their main pain points with the current offline system?; Do they have any existing technology infrastructure (POS, computers, etc.)?; What's their budget for this digital transformation?

Great questions. The shop has 10 employees, handles about 100 orders per day (60% eat-in, 40% take-away), and their main pain points are order management, inventory tracking, and staff scheduling. They have a basic POS system but no other technology. Their budget is flexible but they prefer cost-effective solutions. Now, what would be your approach to suggesting solutions?

Thank you for the information. I'd like to start by identifying the key areas where digital solutions could make the most impact, and then propose specific tools or systems for each. Is that approach okay?

Yes, that sounds good. Please proceed.

Based on the information provided, I'd identify four key areas for digital transformation:

1. Order Management System
2. Inventory Management
3. Staff Scheduling and Management
4. Customer Relationship Management (CRM)

For each of these areas, I can suggest specific digital solutions that would help streamline operations. Should I go ahead and outline these?

Yes, please outline your suggestions for each area.

Certainly. Here are my suggestions for each area:

1. Order Management System:

a) Implement a digital POS system that can handle both eat-in and take-away orders

b) Include features like order tracking, digital receipts, and sales reporting

2. Inventory Management:

a) Use a digital inventory tracking system integrated with the POS

b) Automate reordering processes and provide real-time stock levels

3. Staff Scheduling and Management:

a) Implement a digital scheduling tool for shift management

b) Include features for time tracking and payroll integration

4. Customer Relationship Management (CRM):

a) Develop a simple CRM system to track customer preferences and order history

b) Implement a loyalty program to encourage repeat business

These seem like solid suggestions. Can you elaborate on one of these areas in more detail? Let's focus on the Order Management System.

Definitely, here's a more detailed specification for the Order Management System:

1. Digital POS System:

- User-friendly touchscreen interface for easy order input
- Customizable menu items with modifiers (e.g., toppings, sizes)
- Split bill functionality for eat-in orders
- Integration with kitchen display system for order preparation
- Support for multiple payment methods (cash, card, mobile payments)

2. Order Tracking:

- Real-time status updates (received, preparing, ready for pickup/service)
- Estimated preparation times for better customer communication
- Ability to modify or cancel orders

3. Digital Receipts:

- Option to email receipts to customers
- Store digital copies of all transactions for easy reference and accounting

4. Sales Reporting:

- Daily, weekly, and monthly sales reports
- Breakdown of sales by item, time of day, and order type (eat-in vs. take-away)
- Ability to export data for further analysis

5. Integration Capabilities:

- API for future integration with online ordering platforms
- Compatibility with the inventory management system for real-time stock updates

6. Data Security:

- Secure user authentication for staff access
- Regular data backups to prevent loss of information
- Compliance with data protection regulations

This system would significantly streamline the order process, reduce errors, and provide valuable insights for business decision-making.

That's a comprehensive breakdown. How would you prioritize the implementation of these features?

To prioritize the implementation, I'd consider both the impact on efficiency and the complexity of implementation. Here's how I'd rank them:

Feature	Impact	Complexity
Digital POS System	High	Medium
Order Tracking	High	Low
Sales Reporting	Medium	Low
Digital Receipts	Medium	Low
Integration Capabilities	High	High
Data Security	High	Medium

I'd recommend implementing these in phases, starting with the core POS functionality and gradually adding more advanced features. This approach allows the staff to adapt to the new system gradually and provides immediate benefits in terms of order management efficiency.

That's a sensible approach. Lastly, how would you measure the success of this digital transformation?

To measure the success of the digital transformation, I'd focus on the following key performance indicators (KPIs):

1. Order processing time: Measure the reduction in time taken to process orders
2. Order accuracy: Track the decrease in order errors
3. Staff productivity: Monitor the increase in orders handled per staff hour
4. Inventory wastage: Measure the reduction in food waste due to better inventory management
5. Customer satisfaction: Conduct surveys to gauge improvement in customer experience
6. Revenue growth: Track any increase in sales or average order value
7. Cost savings: Calculate the reduction in operational costs (e.g., paper, time spent on manual processes)

What would be the North Star metric among these?

The North Star Metric for this digital transformation will be **Customer Satisfaction**. This metric reflects the overall success of the implemented changes in improving the customer experience, which in turn drives repeat business and growth. Customer satisfaction can be measured through surveys, feedback, and repeat purchase rates.

Great! We can close the case nw.

Thank you.

Users moving away from WhatsApp

How to stop people moving from WhatsApp to Telegram?

I have some initial clarifying questions. First, What are the primary reasons users are switching to Telegram, unique features Telegram has, and any recent triggers (past 6 months) causing the migration, including trust factors and user experience issues?

Users are switching due to recent updates by Telegram focused on privacy and user control

What is WhatsApp's main goal in addressing this issue? Is it user retention, feature parity, or maintaining market dominance?

You are free to decide.

I believe the main goal should be user retention through feature enhancement and improved user experience, as this will naturally lead to maintaining market dominance. Before proposing specific solutions, I'd like to identify different user segments and their pain points. Is that acceptable?

Sure, please proceed.

I can identify 4 potential customer segments –

1. Privacy-conscious users: Concerned about data security and encryption.
2. Power users: Require advanced features and higher file-sharing limits.
3. Group communication users: Extensively use groups and community features.
4. WhatsApp Business users: Businesses leveraging WhatsApp for customer communication

I'd like to focus on Privacy-conscious users and Power users as the primary segments, as addressing their needs will likely have the most significant impact on preventing migration to Telegram. If that's agreeable, I can proceed with listing pain points?

Yes, that is a reasonable approach. You can proceed.

Pain points for Privacy-conscious users:
1. Concerns about data sharing with Facebook (now Meta)
2. Lack of anonymity options
3. Limited control over personal data
4. Concerns about government surveillance features.

Pain points for Power users:
1. File size limitations for sharing media and documents
2. Lack of advanced customization options
3. Limited multi-device support
4. Absence of features like message scheduling and self-destructing messages

The following can be the proposed features to address these pain points:

1. **Enhanced Encryption:** Implement end-to-end encryption for all types of communication, including backups, with user-controlled encryption keys.
2. **Anonymous Chat Mode:** Introduce an option for users to chat without sharing their phone numbers, using temporary usernames instead.
3. **Increased File Sharing Limits:** Raise the file size limit for sharing media and documents to match or exceed Telegram's limits.
4. **Advanced Customization:** Provide more options for app themes, chat backgrounds, and notification settings.

How will you prioritize among these?

For prioritizing features, I will consider their potential impact on user trust and implementation complexity:

Feature	Impact	Complexity
Enhanced Encryption	High	High
Anonymous Chat Mode	High	Medium
Increased File sharing limits	Medium	Low
Advanced Communication	Medium	Medium

How will you measure the success of these features implemented?

1. **User Retention Rate:** Measure the percentage of users who remain active on WhatsApp over time to assess the effectiveness of the new features
2. **Feature Adoption Rate:** Track the usage of newly introduced features to determine user engagement and acceptance.
3. **Churn Rate:** Monitor the rate at which users are leaving WhatsApp for Telegram

Users moving away from WhatsApp

What would be the North Star metric among these?

User Retention Rate is a direct measure of the percentage of users who remain active on WhatsApp over time. This metric provides a clear indication of how effectively WhatsApp is maintaining its user base and preventing churn. By focusing on retention, WhatsApp can ensure long-term user engagement and loyalty, which are critical for sustaining growth and countering the shift of users to Telegram. High retention rates typically reflect positive user experiences and satisfaction, which are essential for keeping users from switching to alternative platforms.

Great! We can close the case now.

Thank you.

Market Entry - Make My Trip

MMT, is considering entering the Dubai market. What should their market entry strategy be?

Before starting with the strategy, I have some initial clarifying questions. By MMT, are we referring to the online travel platform, MakeMyTrip.

Yes !

First, I would like to evaluate the desirability of entering the Dubai market. So, what are MMT's specific objectives and expectations for entering the Dubai market? Do we have any specific metrics like revenue or profit targets?

The firm's main objective is to maximize its revenue and ROI over the next five years. Dubai's travel market is rapidly growing, driven by increasing tourism and luxury travel. There is no specific revenue target, but they aim for significant growth.

That sounds promising. What about the competitive landscape? Who are the key players, and what are their strengths and weaknesses?

The market is competitive with established players like Booking.com and local agencies.

Are there any desirable customer demographics that I should focus on (e.g., luxury travel, budget travel, business travel) that MMT is targeting?

MMT should target a mix of high-income residents and tourists looking for comprehensive travel services.

Understood. To build a comprehensive strategy, I would want to evaluate the business from 3 perspectives:

1. **Economic Analysis** – To determine market size and potential market share
2. **Internal Capabilities** – To assess MMT's feasibility in the market.
3. **Execution Plan**: To outline the steps for market entry

I would want to start with economic analysis first.
I would like to assume that the fixed costs are essentially zero because of an already established technical platform . So, the initial costs include marketing, partnerships, and regulatory compliance.
Do let me know if my assumption is fine and are there any estimated costs that I should think of ?

The cost drivers are fine. Also, the expected revenue is substantial, with a forecast payback period of 3-4 years and a strong ROI.

Understood. Given these investments, what are the main risks we should be aware of when entering the Dubai market?

What do you think could be some of the potential risks?

1. Well, one potential risk could be **regulatory** changes. Given that Dubai's regulatory environment can be quite dynamic, changes in laws or regulations could impact our operations or increase our compliance costs.
2. Another risk could be intense **competition**. Considering that Dubai is an attractive market, there are likely several well-established local and international players. Competing against them might require significant effort and resources.
3. Besides regulatory changes and competition, there could also be risks related to **cultural differences and customer preferences**. MMT might need to invest in understanding the local market better to tailor its offerings effectively.

The risks are relevant. Proceed next

Given the promising revenue prospects and manageable risks, the market entry seems economically viable. Now, I'll like to evaluate the feasibility
Next, we need to assess if MMT has the internal capabilities to enter the market successfully. Are MMT's current product offerings suitable for the Dubai market?

Yes, MMT's offerings, are well-suited for Dubai. However, they might need some localization.

Localization is indeed crucial. Could you specify what kind of localization might be required? For instance, would we need to customize the language, payment options, or perhaps the types of travel packages offered?
I could think of customizing the language to Arabic, offering region-specific payment options like local credit cards and digital wallets, and tailoring travel packages to include popular local destinations and experiences would be necessary.

All of those are relevant

What about the technological infrastructure? Does MMT's current platform support these localization needs ?

Market Entry - Make My Trip

The platform is robust but will need some adjustments to integrate local payment gateways and to handle multi-language support efficiently.

It's good to know the platform is strong. Moving on to organizational structure, do we have the right teams in place to manage operations in Dubai? For example, do we need a local office with a dedicated team for marketing, customer service, and partnerships?

Yes, MMT can think of a local team, but you can skip this aspect for now

Finally, we need to develop an execution plan. How should MMT enter the market? Should we expand organically, enter a joint venture, partner with local firms, or acquire a competitor?

What would you recommend?

I recommend a partnership strategy. By collaborating with local travel agencies, hotels, and airlines, MMT can leverage their market knowledge and networks. Additionally, launching a targeted digital marketing campaign can help build brand awareness and attract customers. Developing localized travel packages will also cater to the preferences of Dubai's residents and tourists.

What timeline would you propose for this market entry?

It can be divided into 3 phase:

1. **Short-Term (0-6 months):**
   Conduct market research, establish partnerships, and adapt the platform for local use.
2. **Mid-Term (6-12 months):**
   Launch marketing campaigns, start offering localized travel packages, and monitor market response.
3. **Long-Term (1-2 years):**
   Expand service offerings, optimize operations based on feedback, and aim for market share growth.

That sounds comprehensive. What are the next steps?

The next steps are:

1. **Establish Local Partnerships**: Engage with local travel agencies, hotels, and airlines.
2. **Digital Marketing Campaign**: Initiate targeted campaigns to build brand presence.
3. **Regulatory Compliance**: Ensure compliance with local regulations and adapt services to meet cultural norms.

That was good. Thanks for your time

Success Metrics – Swiggy IPO

You are the CPO of Swiggy, what key product metrics would you monitor in preparation for an IPO?

To ensure a successful IPO, I would focus on product metrics that provide valuable insights into user engagement, satisfaction, and overall platform performance, these factors would help convince investors. Does this sound reasonable, should I proceed and list the metrics?

Yes, please proceed. Also, include the rationale behind selecting each metric, particularly in the context of the upcoming IPO.

1. **Monthly Active Users (MAUs):** MAU indicates overall user engagement and growth. It is important for IPO as investors look for strong user growth and engagement as a sign of a successful and scalable product.
2. Daily Active Users (DAUs): DAU measures daily user engagement and platform stickiness. It becomes important for IPO because a high DAU to MAU ratio can signal strong user retention and satisfaction, a key factors that can help attract investors.
3. Customer Retention Rate: It shows how well the platform can keep its users over time. A strong retention rates can help demonstrate long-term user loyalty and the product's potential for sustained revenue.
4. Order Volume and Growth: Order volume tracks the total number of orders placed, reflecting market penetration and user activity. An increasing order volume showcases the platform's growth trajectory and can help appeal to investors looking for scalable businesses.
5. Average Order Value (AOV): AOV helps measure average spending per order. A high AOV contributes to overall revenue and will help highlight the platform's potential for profitability.
6. Average Delivery Time: It helps track the efficiency of the delivery process. Hence, critical for IPO as efficient delivery operations enhance user satisfaction and operational excellence, key considerations for investors.
7. Feature Usage and Engagement: Feature-wise insights into user engagement. Important for IPO because identifying popular features helps enhance user experience and engagement, attracting investors.
8. Customer Satisfaction (Net Promoter Score - NPS): NPS measures likelihood of users recommending the platform. A high NPS indicates strong user satisfaction and loyalty, vital for growth. Therefore, vital for IPO
9. Churn Rate: It measures the percentage of users who stop using the platform over time. Important for IPO because managing churn is essential for long-term user base stability and revenue predictability, factors closely monitored by investors.
10. Operational Efficiency Metrics: Assessing how efficiently the platform operates, including fulfillment and staffing costs is again critical. Important for IPO as investors value companies that manage costs effectively while scaling, indicating a sustainable business model.

It was a comprehensive list. How would you use these metrics to prepare for the IPO?

I would closely monitor these metrics and utilize them to:
1. Demonstrate Growth: Use positive trends in MAUs, DAUs, and order volume to showcase the platform's growth and user engagement.
2. Highlight Efficiency: I will showcase improvements in delivery times and operational efficiency to underline our ability to scale effectively.
3. Emphasize User Satisfaction: I will use high NPS and retention rates to reflect strong user satisfaction and loyalty.
4. Provide Financial Insights: Finally, linking metrics like AOV and feature usage to revenue potential and profitability should be insightful.

This was comprehensive solution.

Thank you.

Success Metrics – Gmail

What would be the success metrics for Gmail?

First, I will start by going over Gmail as a product, the tasks that can be done and then define the success metric for the same,

Sure, go ahead

Gmail is an application for sending emails. Using this Gmail ID, users can also use other suits of products provided by Google like *Chrome, Maps, Youtube, Calendar, Drive, Photos, etc.* The users can be of two types the consumers who use gmail for personal use, these users utilize the application for free for upto 15 GB of storage space. The second type of users are the G-suite or corporate users who pay for the whole package of the G-suite and hence for the Gmail corporate service as well.

Some tasks that a user over Gmail can do are:

1. Send/Receive emails.
2. Delete/Archive emails.
3. Snooze or set a Reminder for emails.
4. Categorize, Organize and Prioritize emails using Tags/Folders.
5. Integrate with other Google products/services.
6. Send invites for a virtual meetup or meeting via integrating Google Calendar and Google Meet links in the email.
7. Send files(audio/image/video) attachments in the emails via Google Drive.
8. Chat with other Google users using text, voice, and video via Google Chat.

Should I be focusing on only email feature of the other features should be considered as well?

You can focus only on the mailing feature

I will be considering both web as well as app user on both Android and iOS. Will that be okay ?

Yes, that is the right approach please proceed

Is there a specific business goal on which I should focus.

No, please feel free to define the business goal

I would like to focus on Engagement and retention out the listed business goals (Reasons for choosing or not choosing certain metrics is given below)

1. Acquisition — Google has reached good user penetration. Gmail being an entry-level product, may have excellent acquisition metrics.
2. Revenue — As we are looking at the free version of the product, revenue is indirectly through advertising. Also, there is a paid option to get more storage, but that does not apply to large sections of the user base.
3. Engagement and Retention — The most critical goal for Gmail is to keep Google's users engaged so that they will stay with Google's app and services ecosystem and reoccurring engagement shows Retention.

Thus, Engagement and Retention are essential for this product. I am right or shall I consider some other parameters too.

Yes, that is correct please proceed.

I would like to define the User journey to identify the metrics for retention

***User Journey* (Desktop)**

- **Acquisition**

1. Visit mail.google.com and click on "Create account".
2. Fill in personal details and set a strong password to sign up.
3. Send/Receive the first email.
4. Repeat the previous step multiple times (Loop)

- **Engagement and Retention**

1. Open Gmail on the browser organically.
2. Send emails.
3. Read emails from the Primary Inbox.

Success Metrics – Gmail

1. Read emails from Social, Promotions, Updates, Forums, and other categories.
2. Reply to emails.
3. Other actions — forward, delete, archive, sort, snooze.
4. Close tab/browser.

Mobile (Android/iOS)

- **Acquisition**

1. Download the Gmail App from Play Store/App Store.
2. Sign up and create a Gmail account.
3. Send/Receive the first email.
4. Repeat the previous step multiple times (Loop).

- **Engagement and Retention**

1. Open Gmail app Organically or via push notification.
2. Send emails.
3. Read emails from the Primary Inbox.
4. Read emails from Social, Promotions, Updates, Forums, and other categories.
5. Reply to emails.
6. Other actions — forward, delete, archive, sort, snooze.
7. Exit app.

You have defined the user journey correctly now tell me how exactly would you measure engagement and retention.

Broadly engagement and retention can be measured on the basis of mail sent , mails received , storage space utilised, number of emails opened, emails deleted without opening

Specify metrics for engagement and retention

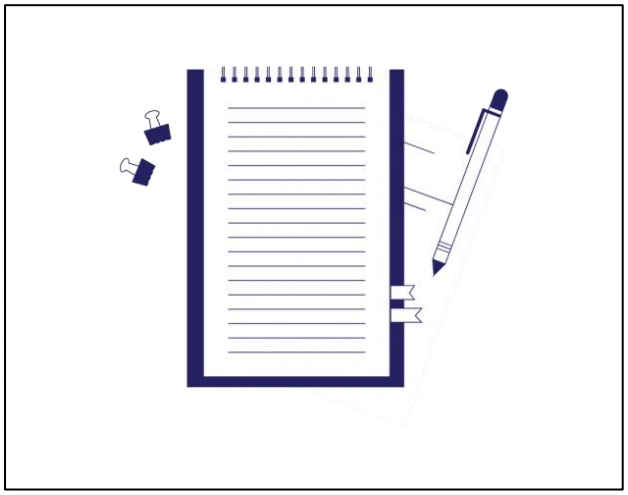
For engagement:

1. # emails sent/read per session, per day, per month.
2. # of searches in the mailbox per user.
3. Avg Time Spent/Session Duration per day.
4. % of users who checked their inbox organically or via notifications on *Day 1, Day 7, Day 15, Day 30.*
5. # of days the user was active in *30 days.*
6. # of users who read emails from Primary VS other categories *(Social, Promotions, Updates, Forums, etc.)*
7. % Split of total users who read emails from Primary and from different categories *(Social, Promotions, Updates, Forums, etc.)*
8. # of users who forward, delete, archive, sort, snooze their mails.
9. % Split of the total users who forward, delete, archive, sort, snooze their mails.
10. % of users who used other Google Services integrated inside Gmail.
11. # of other Google Services also used.

For Retention

1. *1 Day, 7 Day, 15 Day, 30 Day* Retention rate.
2. DAU, MAU.

That was a comprehensive answer. we can stop here it was nice talking to you



Appendices



Appendix 1 – North Star Metrics

List of potential North Star Metrics for some popular apps -

Airbnb	Number of Nights Booked
Amazon	Number of Purchases per Month
BookMyShow	Number of Tickets Booked
Facebook	Monthly Active Users
Google	Searches per User
Instagram	Daily Active Users
LinkedIn	Monthly Active Users
Netflix	Total Hours Viewed
Spotify	Time Spent Listening
Swiggy	Number of Orders Completed
Quora	Number of Questions Answered
Uber	Weekly Active Riders
WhatsApp	Number of messages Sent
YouTube	Total Watch Time
Zoom	Weekly Hosted Meetings

Note: These are just potential metrics obtained from company interviews and various online sources. The actual North Star Metrics internal to the company may differ. This list is intended to give a tentative idea of the possible key performance indicators.

Appendix 2 - Frameworks at a Glance

CIRCLES Method

Comprehend Situation – What? Who? Why? How?
Identify customer – User Personas
Report customer needs – As a [user], I want [task], so that [goal].
Cut through prioritization
List solutions – Features, wireframes, metrics
Evaluate trade-offs
Summarize your recommendation

5Es Framework for User Journey

Entice – What event triggers a user to enter into the product?
Enter – What are the first few steps while using the product?
Engage – What task(s) is the user trying to accomplish?
Exit – How does the user complete the task?
Extend – What follow-up actions occur after the user completes the task?

RICE Prioritization Framework

Reach – How many users will use your feature?
Impact – Measure of user need or potential delight from the feature.
Confidence – Technical & financial feasibility, ability to meet deadlines
Effort – Estimate of human resources needed, measured in man-hours or person-months

HEART Framework for Metrics

Happiness – How users feel about the product? (NPS, App Ratings, CSAT scores)
Engagement – How often are people using your product? (DAU, session length, activity (# comments on Instagram))
Adoption – How easy is it to complete the onboarding process? (New users, drop-offs during onboarding)
Retention – How many users are returning to your product? (Churn rate, customer lifetime value)
Task Success – Are users achieving their goal? (Conversion rate in Flipkart, time taken to book a ride in Uber)

Appendix 3 - Datasheet for Guesstimates

India's Digital Context

Population	1.4 Billion
Internet Penetration	55%
Smartphone Penetration	70%
% internet users on mobile	90%
Urban internet users	400 million
Rural internet users	450 million

India's Internet Usage Pattern

Internet Users	800 million
Chatting and Social Media Users	450 million
Video Content Consumers	400 million
Financial Transactors	300 million
Online Shoppers	250 million
Avg. time spent online per day	6 hrs

Advertising Benchmarks

Avg. CTR for a display ad	0.5%
Avg. CTR for a search ad	3%
Avg. Cost per Click(CPC) for search ads	$2
Avg. CPC for display ads	$0.5

E-commerce Benchmarks

Avg. conversion rate	2.5%
Avg. order value	Rs 1500
Avg. cart abandonment rate	70%

Additional resources for common data points

https://www.mypminterview.com/p/guess-estimation-data-to-remember
https://www.tryexponent.com/courses/estimation/product-management-estimation
https://economictimes.indiatimes.com/tech/technology/how-india-is-using-the-internet/articleshow/108354854.cms?from=mdr

Appendix 4 - Resources

Books

Cracking the PM Interview by Gayle Laakmann McDowell
Decode & Conquer by Lewis C. Lin
The Product Manager Interview – 167 Actual Questions & Answers by Lewis C. Lin

Online Resources

Interview Questions - **https://www.productmanagementexercises.com/interview-questions**
Interview Preparation - **https://zapupp.com/**
Exponent - **https://www.tryexponent.com/** Exponent YouTube Channel - **https://www.youtube.com/channel/UCjm_qVkCPjOVDz9BWjNqO9A**